

浙江大学

本科生毕业论文

文献综述和开题报告



姓名与学号	张琨皓 3220105256
指导教师	赵成成
年级与专业	2022级自动化（控制）
所在学院	控制科学与工程学院

一、题目：一类融合地形感知的水下机器人非对称追逃博弈多智能体深度强化学习策略研究

二、指导教师对文献综述、开题报告、外文翻译的具体要求：

文献综述和开题报告的要求：

1. 查阅国内外相关文献资料 10 篇以上（其中外文文献不少于 5 篇）；完成至少一篇 3000 字以上外文翻译。
2. 完成文献综述（3000 字以上）和开题报告（3500 字以上）。

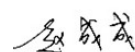
对文献综述的具体要求：

- （1）了解非对称追逃博弈与水下环境的基本理论与发展现状。
- （2）综述水下多机器人协同与强化学习策略的研究现状。
- （3）比较现有环境建模与地形感知方法，阐述地形与战术协同的研究意义。
- （4）做好毕业论文相关知识的准备工作。

对开题报告的具体要求：

- （1）分析当前追逃博弈与多智能体协同策略的算法类型与特点。
- （2）提出融合地形感知的协同策略设计方案，明确核心算法选择。
- （3）对论文要完成的工作有清晰的认识，给出具体的研究方案，预见可能遇到的问题 and 困难。
- （4）提出论文的研究成果或预期达到的目标。
- （5）安排好论文的进度，对于每个阶段的工作有详细的规划。

指导教师（签名）



2026 年 1 月 3 日

目录

一、 文献综述	1
1 研究背景	1
1.1 非对称追逃博弈问题的理论基础	1
1.2 水下环境特性与多机器人协同需求	2
1.3 深度强化学习的发展及其在机器人领域的应用	3
2 国内外研究现状	5
2.1 非对称追逃博弈理论与方法研究	5
2.2 水下多机器人协同与强化学习策略研究	6
2.2.1 水下多机器人协同控制	6
2.2.2 水下机器人强化学习控制	6
2.2.3 分层强化学习与层次化决策框架	7
2.2.4 水下机器人非线性动力学建模与控制	9
2.3 环境建模与地形感知方法比较	10
2.4 地形与战术协同的研究意义	11
2.5 存在问题	12
2.5.1 非对称博弈建模与策略设计的系统性不足	12
2.5.2 水下场景的针对性研究匮乏	12
2.5.3 动力学建模非线性与学习控制的融合不足	13
2.5.4 地形感知与博弈策略的耦合深度不足	13
2.5.5 综合性研究空白	14
3 研究展望	14
3.1 融合地形感知的非对称追逃多智能体强化学习框架	14
3.2 分层多智能体深度强化学习架构设计	15
3.3 高保真水下仿真与系统验证路径	16
3.4 技术挑战与未来方向	16
4 参考文献	18

二、 开题报告	23
1 问题提出的背景	23
1.1 背景介绍	23
1.1.1 项目提出的原因	24
1.2 本研究的意义和目的	24
2 项目的主要研究内容和技术路线	25
2.1 主要研究内容	25
2.1.1 阶段一：水下机器人动力学建模与分析	26
2.1.2 阶段二：低层运动控制策略学习	26
2.1.3 阶段三：融合地形感知的一追一追逃博弈策略	27
2.1.4 阶段四：融合协同地形利用与目标预测的多追一策略	28
2.2 技术路线	29
2.2.1 水下机器人六自由度动力学建模	30
2.2.2 分层强化学习框架设计	33
2.2.3 基于 PPO-LSTM 的低层运动控制策略学习	35
2.2.4 融合地形感知的一追一非对称追逃博弈策略学习	37
2.2.5 融合地形利用与目标预测的多追一协同追逃策略学习	41
2.3 可行性分析	44
2.3.1 理论可行性	44
2.3.2 技术可行性	45
2.3.3 研究基础	45
3 研究计划进度安排及预期目标	46
3.1 进度安排	46
3.2 预期目标	46
3.2.1 理论与方法目标	46
3.2.2 系统实现目标	46
3.2.3 验证与成果目标	47
4 参考文献	49

三、 外文翻译	51
摘要	53
1 引言	53
2 问题描述	56
2.1 围捕控制场景	56
2.2 基于 ROS2 的仿真平台	57
3 预备知识	58
3.1 多智能体强化学习	58
3.2 课程学习	59
3.3 跳跃启动强化学习	60
4 面向策略的围捕控制学习	60
4.1 JSRL 的引导策略	61
4.2 通过多课程调节器训练	63
4.3 MARL 框架设计	64
4.3.1 奖励函数	64
4.3.2 状态空间和通信协议	67
4.3.3 网络结构	68
5 仿真与结果	68
5.1 仿真设置	69
5.2 MRA-RLEC 的可扩展性和鲁棒性	70
5.3 消融测试	73
5.4 与其他围捕算法的比较	74
5.5 性能分析	76
5.6 ROS2 中不同逃跑策略下的实验	78
6 结论与未来工作	80
7 外文翻译参考文献	81
四、 外文原文	85
毕业论文（设计）文献综述和开题报告考核	99

一、文献综述

1 研究背景

1.1 非对称追逃博弈问题的理论基础

追逃博弈 (Pursuit-Evasion Game) 是博弈论与控制理论交叉领域的经典问题, 其研究历史可追溯至 20 世纪中叶。该问题的核心在于: 在给定的空间几何与动力学约束下, 追击方 (Pursuer) 如何设计最优运动策略以捕获逃逸方 (Evader), 而逃逸方则致力于最大化逃脱概率或延长被捕获的时间^[1]。追逃博弈的数学理论基础由 Isaacs 在其 1965 年出版的经典著作《Differential Games》中系统建立, 该书奠定了微分博弈理论的框架, 并给出了诸如“Homicidal Chauffeur”等经典追逃问题的解析解^[2]。Weintraub 等人在 2020 年发表的综述中系统梳理了追逃微分博弈的理论发展, 指出该领域的核心在于求解 Hamilton-Jacobi-Isaacs (HJI) 偏微分方程以获得鞍点均衡策略^[3]。国内学者王宏健等人^[4]较早地对多机器人追捕问题进行了系统研究, 分析了多追击者协同围捕的策略框架。

在实际应用场景中, 追逃双方的性能往往呈现显著的非对称性 (Asymmetry)。这种非对称性可体现在多个维度: 最大速度差异、加速度与机动能力差异、感知与通信能力差异、以及数量上的差异^[5]。典型的非对称场景包括: (1) 追击方在绝对速度上处于劣势但在加速性能或转弯能力上占优; (2) 追击方数量众多但单体能力较弱; (3) 逃逸方具备更优的态势感知但机动空间受限。在速度非对称的双人追逃中, 经典的 Apollonius 圆方法提供了分析可截获域的几何工具, 当追击者速度劣于逃逸者时, 单个追击者无法在开放空间中保证捕获, 这一结论为非对称博弈的协同策略研究奠定了理论基础^[6]。

非对称追逃博弈在众多领域具有重要应用价值。在军事领域, 导弹拦截、无人机编队对抗、海上缉私与反潜作战等场景均可抽象为追逃博弈问题^[7]。Chen 等人研究了多追击者对单逃逸者的 reach-avoid 博弈问题, 证明了在特定数量优

势下即使追击者存在速度劣势仍可实现成功围捕^[8]。Sun 等人将追逃博弈扩展至动态流场环境，研究了洋流等环境因素对博弈结果的影响^[9]。Li 等人提出了层次化多玩家追逃博弈框架，通过分层决策结构降低大规模博弈的计算复杂度^[10]。

传统追逃博弈研究主要依赖解析方法，包括微分博弈理论、最优控制、以及图论方法等^[11]。然而，当博弈场景扩展至高维连续状态空间、复杂几何约束（如障碍物、地形）以及多智能体协同决策时，传统解析方法面临严峻挑战：HJI 方程的数值求解遭遇“维度灾难”，计算复杂度随状态维度呈指数级增长；启发式规则难以应对对手策略的动态变化；预设战术缺乏对复杂环境的适应性。这些局限性催生了数据驱动方法特别是深度强化学习在追逃博弈中的应用研究^[12]。

1.2 水下环境特性与多机器人协同需求

水下机器人（Underwater Vehicle）已成为海洋探测、资源勘探、水下搜救、海洋国防等领域不可或缺的技术装备。Sahoo 等人在其综述中系统回顾了自主水下航行器（AUV）的发展历程与技术现状，指出 AUV 已从早期的单体遥控作业演进为具备自主决策能力的智能平台^[13]。我国在 AUV 研发领域也取得了显著进展，Yu 等人报道的“海翼”系列水下滑翔机在深海科考与海洋观测中发挥了重要作用^[14]。徐德民等人^[15]系统分析了水下无人航行器的发展趋势及关键技术，指出智能化、协同化是未来发展的重要方向。

随着海洋任务复杂度的提升，单一水下机器人的能力已难以满足大范围协同搜索、动态目标跟踪、协同围捕等任务需求，多水下机器人协同系统应运而生^[16]。Zeng 等人针对 AUV 持续自主任务系统梳理了路径规划方法的研究进展，强调了多 AUV 协同在提升任务效率与覆盖范围方面的优势^[17]。Hadi 等人的综述则聚焦于多 AUV 路径规划与编队控制，分析了集中式与分布式协同架构的适用场景^[18]。朱大奇等人^[19]对多水下机器人协调控制的研究进展进行了全面综述，探讨了任务分配、编队控制、协同定位等关键问题。严卫生等人^[20]则针对自主水下航行器的路径规划技术进行了系统梳理。

水下环境具有区别于陆地与空中的显著特殊性，这些特性对多机器人协同

控制提出了独特挑战。Akyildiz 等人在其开创性工作中系统分析了水下声学传感器网络面临的挑战,指出水下通信主要依赖声学信道,其带宽受限(通常为 kbps 量级)、传输时延大(声速约 1500m/s,远低于电磁波)、且易受多径效应与环境噪声干扰^[21]。Yan 等人的综述进一步讨论了 AUV 导航定位技术,强调了 GPS 信号在水下不可用带来的定位难题,以及惯性导航漂移、声学定位受限等问题^[22]。

水下环境的另一显著特性是复杂的海底地形。海底的沟壑、山脊、礁石群、沉船等地形特征既构成导航与避障的严峻挑战,也蕴含潜在的战术价值。Wöfl 等人的研究指出,全球海底地形测绘覆盖率仍然较低,高分辨率地形数据的获取与利用是海洋机器人领域的重要课题^[23]。在追逃博弈场景中,追击方可利用地形遮蔽实现隐蔽接近,利用狭窄通道构建围堵陷阱,或借助地形限制逃逸方的机动空间。Bhattacharya 和 Hutchinson 研究了存在视线约束(Visibility Constraints)的追逃博弈,证明了障碍物的存在可显著改变博弈的纳什均衡结构^[24]。因此,如何将高分辨率地形信息有效融入多机器人协同决策,实现从”地形作为障碍”向”地形作为战术资源”的认知转变,是本研究关注的核心问题之一。

1.3 深度强化学习的发展及其在机器人领域的应用

强化学习(Reinforcement Learning, RL)是一种通过与环境交互、基于试错反馈学习最优行为策略的机器学习范式。Kober 等人在其经典综述中系统梳理了强化学习在机器人领域的早期应用,涵盖了运动控制、抓取操作、自主导航等任务^[25]。深度强化学习(Deep Reinforcement Learning, DRL)通过将深度神经网络的强大函数逼近能力与强化学习的序贯决策框架相结合,突破了传统 RL 在高维状态空间中的应用瓶颈。2015 年, Mnih 等人提出的深度 Q 网络(DQN)在 Atari 游戏中达到人类水平的表现,标志着 DRL 的重大突破^[26]。张铭钧等人^[27]对水下机器人智能控制技术的研究进展进行了系统总结,分析了深度学习与强化学习方法在水下机器人控制中的应用前景。

Tang 等人在其 2025 年发表于《Annual Review of Control, Robotics, and Autonomous Systems》的综述中,全面审视了深度强化学习在机器人领域的真实世

界应用成就^[28]。该综述提出了一套系统的分类框架，按机器人能力（Competencies）将应用划分为：（1）移动能力（Mobility），涵盖运动控制（Locomotion）与导航（Navigation）；（2）操作能力（Manipulation），包括抓取、装配、灵巧手操作等；（3）交互能力，包括人机交互（HRI）与多机器人协同。更重要的是，该综述提出了“真实世界成功等级”（Level of Real-World Success）评估框架，将 DRL 应用的成熟度划分为六个等级（0-5 级），从仅仿真验证到商业化产品部署。这一框架为评估 DRL 在不同机器人任务中的应用前景提供了清晰的参照。在运动控制领域，Tang 等人指出四足机器人是 DRL 应用最为成功的领域之一，已有多家公司（如 ANYbotics、Swiss-Mile、Boston Dynamics）将 DRL 集成到商业产品中^[28]。这些成功案例的共同特征包括：（1）采用“仿真训练-真实部署”（Sim-to-Real）的零样本迁移方案；（2）使用稳健的在线策略优化算法（如 PPO）；（3）通过域随机化（Domain Randomization）和学习执行器模型来弥合仿真与现实的差距。这些经验对水下机器人 DRL 控制具有重要借鉴意义。

多智能体深度强化学习（Multi-Agent Deep Reinforcement Learning, MADRL）将 DRL 扩展至多智能体系统，旨在解决多智能体间的协调、竞争与混合博弈问题。Nguyen 等人^[12]在其综述中系统分析了 MADRL 面临的核心挑战，包括环境非平稳性、信用分配困难、联合动作空间指数增长等。为应对这些挑战，研究者提出了“集中训练、分散执行”（Centralized Training with Decentralized Execution, CTDE）范式。Lowe 等人提出的 MADDPG 算法是 CTDE 的代表性工作，在训练阶段利用全局信息学习评价函数，执行阶段各智能体仅依赖局部观测独立决策^[29]。Rashid 等人提出的 QMIX 算法则通过单调值分解实现了更好的可扩展性^[30]。在追逃博弈领域，MADRL 的引入为解决高维非对称博弈问题提供了新途径。研究表明，基于 MADRL 的追逃策略能够涌现出包抄、驱赶、诱捕等复杂协同行为，且无需人工设计繁琐的启发式规则^[31]。刘明雍等人^[32]针对多 AUV 协同围捕问题，提出了基于深度强化学习的策略学习方法，验证了学习型方法在水下协同任务中的有效性。然而，Tang 等人同时指出，多机器人协同是 DRL 应用中最具挑战性的领域之一，当前最先进的方法主要限于合作场景，在通信受限、部分可观测的分布式对抗决策场景中仍面临诸多开放问题^[28]。

2 国内外研究现状

2.1 非对称追逃博弈理论与方法研究

非对称追逃博弈的研究经历了从理论分析到计算求解、从双人博弈到多人博弈的发展历程。在理论层面，Zhou 等人提出了 reach-avoid 博弈的一般性开环表述，通过将追逃博弈转化为可达性问题，利用水平集方法求解 HJI 方程获得最优策略^[6]。Huang 等人研究了多追击者分散追逃的保证捕获条件，证明了在特定几何构型下追击者协同可弥补个体性能劣势^[5]。多追一（Multi-Pursuer Single-Evader）场景是非对称追逃博弈的典型形式。Pierson 和 Rus 提出了基于 Voronoi 划分的分布式围捕策略，追击者根据其 Voronoi 胞元边界动态调整位置以压缩逃逸者的机动空间^[33]。该方法的核心思想是将空间划分为各追击者的“责任区域”，通过协同移动逐步缩小逃逸者的可达区域。Chen 等人研究了追击者速度劣势条件下的多人 reach-avoid 博弈，证明了通过数量优势可以补偿个体速度的不足^[8]。Vidal 等人的早期工作将概率方法引入追逃博弈，为处理感知与运动不确定性提供了理论框架^[7]。

近年来，深度强化学习方法在追逃博弈中的应用日益广泛。Yu 等人在 IJCNN 上发表的工作^[31]提出了多智能体对抗逆强化学习方法，通过在训练中引入对抗机制提升策略鲁棒性，该方法在多无人机追逃任务中展现出优于传统方法的性能。Pinto 等人提出的鲁棒对抗强化学习（RARL）框架^[34]通过训练对抗性扰动器来增强策略的抗干扰能力，该思想在追逃博弈中可用于应对对手策略的不确定性。在非对称性能建模方面，现有研究主要关注速度差异对博弈结果的影响。Sun 等人将追逃博弈扩展至动态流场环境，发现洋流等环境因素可显著改变追逃双方的相对优势^[9]。然而，综合考虑“速度劣势但加速度/机动优势”以及“利用地形弥补性能差距”的系统性研究仍相对缺乏，这正是本研究拟重点突破的方向。表1.1对追逃博弈策略研究的主要方法进行了分类总结。

表 1.1 追逃博弈策略研究方法总结

方法类别	具体方法	核心思想	特点
传统解析方法	微分博弈理论	求解 HJI 方程获得鞍点均衡	理论严谨，存在维度灾难
	Apollonius 圆	几何分析可截获域	直观明确，仅适用双人场景
	Voronoi 划分 Reach-Avoid 博弈	空间责任区域分配 转化为可达性问题	分布式协同，逐步压缩 通用性强，计算复杂度高
多智能体强化学习	MADDPG QMIX 对抗强化学习	集中训练分散执行 单调值函数分解 引入对抗机制提升鲁棒性	连续动作空间，稳定性好 可扩展性强，协调性好 策略鲁棒，训练较复杂
	Options 框架 Feudal 架构 HIRO	时间抽象与技能封装 管理者-工人层次结构 高层子目标低层条件策略	长时序决策，可迁移 子目标分解，信用分配 样本高效，适合稀疏奖励

2.2 水下多机器人协同与强化学习策略研究

2.2.1 水下多机器人协同控制

水下多机器人协同是海洋机器人领域的前沿课题，涉及通信、感知、决策、控制等多个层面的技术挑战。Cao 等人研究了多 AUV 协同目标搜索与跟踪问题，提出了基于行为的分布式控制架构，各 AUV 根据局部感知信息独立决策并通过简单通信实现协调^[35]。Wu 等人针对水下目标搜索-跟踪任务设计了异构无人系统协同路径规划方法，考虑了空中与水下平台的能力互补^[36]。Cai 等人利用深度强化学习方法研究了多 AUV 协同围捕任务，证明了学习型方法在复杂协同任务中相较于传统规则方法的优势^[37]。

2.2.2 水下机器人强化学习控制

在水下机器人运动控制层面，深度强化学习展现出突破传统控制方法局限性的潜力。水下机器人的动力学模型具有强非线性和参数不确定性，包括附加质量效应、非线性阻尼、科氏力耦合以及洋流干扰等复杂因素，传统基于模型的控制方法对模型精度要求较高^[39]。传统方法如计算流体动力学（CFD）虽可获得精确的水动力系数，但计算成本高昂且对几何模型敏感^[40]；经验公式在非设

表 1.2 深度强化学习在水下机器人领域的应用进展^[38]

年份	任务类型	算法	仿真验证	实海验证	开源
2005	目标跟踪（仅距离）	Q-learning	○	○	○
2015	路径优化	Q-learning	●	○	○
2018	低层运动控制	DDPG	●	○	○
2019	路径跟踪	DQN/DDPG	●	○	○
2019	目标搜索	A3C	●	○	○
2019	对接控制	DQN	●	○	○
2020	碰撞避免	DDQN	●	●*	○
2021	涡流场导航	V-RACER	●	○	●
2021	动态定位	PPO	●	○	●
2022	路径跟踪/避障	Actor-Critic	●	○	○
2022	拦截任务	SLDDPG	●	○	○
2022	动态定位	RL-NMPC	●	●*	○
2023	水下目标跟踪	H-LSTM-SAC	●	●	●

注：●表示是，○表示否；*表示受限条件下的实海测试（如湖泊、短时间等）

计工况下精度下降；代理模型方法需要大量离线训练数据^[41]。这些困难使得直接采用强化学习绕开显式建模成为一种有吸引力的技术路线。

Carlucho 等人发表于 IEEE Transactions on Robotics 的工作^[42]是该方向的代表性研究，提出了端到端深度强化学习 AUV 位置跟踪控制方法，无需精确动力学模型即可学习有效的控制策略，在轨迹跟踪与姿态稳定任务中取得了优于传统 PID 控制器的性能。该研究表明，DRL 能够隐式学习复杂的非线性动力学关系，为水下机器人控制提供了“模型无关”的替代方案。Chu 等人针对 ROV 轨迹跟踪问题提出了基于观测器的自适应神经网络控制方法，通过在线学习应对模型不确定性^[43]。Sun 等人将生物启发神经网络算法应用于 AUV 完全覆盖路径规划^[44]。Wang 等人的最新研究将强化学习应用于 AUV 自主导航与目标跟踪的联合学习^[45]。Yan 等人利用分层强化学习解决 AUV 辅助水下传感器网络的能效数据采集问题，展示了分层架构在复杂水下任务中的有效性^[46]。表1.2总结了深度强化学习在水下机器人领域的应用进展。

2.2.3 分层强化学习与层次化决策框架

分层强化学习（Hierarchical Reinforcement Learning, HRL）通过将复杂任务分解为多个层次来应对长时序决策与稀疏奖励问题。Pateria 等人的综述^[47]系统

梳理了 HRL 的理论基础与算法进展，主要范式包括：（1）Options 框架：将时间抽象引入强化学习，定义具有起止条件和内部策略的”选项”；（2）Feudal 架构：高层管理者（Manager）设定子目标，低层工人（Worker）执行达成子目标的动作；（3）子目标条件策略：高层策略输出子目标状态，低层策略以子目标为条件生成动作序列。

在机器人控制领域，分层结构的优势体现在多个层面。Nachum 等人提出的 HIRO 算法^[48]通过高层策略输出子目标、低层策略以子目标为条件学习的方式，实现了高效的数据利用与策略学习，在稀疏奖励的导航任务中表现优异。Jeon 等人在 ICRA 上发表的工作^[49]针对多机器人协同任务设计了层次化学习框架，高层负责任务分配与协调，低层负责单体运动控制，这种解耦设计显著降低了联合学习的复杂度，使得大规模多机器人系统的训练成为可能。分层结构在水下机器人控制中具有特殊价值。考虑到水下环境的复杂性与通信限制，分层架构可以有效分离关注点：低层策略专注于应对复杂的流体动力学与运动控制，高层策略专注于任务规划与协同决策。这种结构既可充分利用仿真环境训练低层技能，又可保持高层策略的灵活性以适应多变的任务需求^[50]。

在安全性方面，Zhang 等人发表于 IEEE Robotics and Automation Letters 的工作^[51]提出了事件触发的分层安全强化学习框架。该框架在高层策略中嵌入安全监督模块，当检测到潜在风险时触发安全恢复策略，通过 Lyapunov 约束确保系统稳定性。这种安全感知的分层设计对于水下机器人等安全关键应用具有重要参考价值，可在追求任务性能的同时保障系统运行安全。Tang 等人指出^[28]，分层架构的一个关键优势在于可有效隔离底层动力学不确定性对高层决策的影响。在仿真到真实迁移中，低层策略可针对性地进行域随机化训练以适应真实环境的不确定性，而高层策略则可在更抽象的状态空间中学习任务策略，从而提升整体系统的迁移鲁棒性。这一特性对于水下机器人尤为重要，因为水下环境的流体动力学效应难以在仿真中完全复现。

2.2.4 水下机器人非线性动力学建模与控制

水下机器人动力学建模面临强非线性与参数不确定性挑战，包括附加质量效应、非线性阻尼、科氏力耦合以及洋流干扰等。针对这一问题，研究者发展了多种建模与控制方法，可归纳为四大类：传统物理建模、数据驱动代理模型、端到端强化学习以及混合控制架构。表1.3对这些方法进行了系统比较。

传统物理建模方法中，CFD 虽可获得精确的水动力系数，但计算成本高昂且对几何模型敏感^[40]；经验公式在非设计工况下精度下降；系统辨识方法依赖实验条件，泛化性有限。数据驱动代理模型方法中，高斯过程回归可提供不确定性量化但面临大数据计算瓶颈^[52]；神经网络代理模型拟合能力强但需要大量离线训练数据；混合物理-数据模型通过将物理约束嵌入神经网络提升数据效率，但设计较为复杂^[41]。端到端强化学习方法直接学习从状态到动作的映射，规避了显式建模困难，其中 SAC 等最大熵方法在探索性与样本效率方面表现较好^[42]。混合控制架构将强化学习与传统控制相结合，在保证安全性的同时利用学习能力应对不确定性，是当前研究的重要方向^[50]。

表 1.3 水下机器人非线性动力学建模与控制方法比较

方法类别	具体方法	核心理念	特点与局限
传统物理建模	CFD 方法 经验公式法	数值求解 N-S 方程 基于实验数据拟合系数	精度高，计算成本极高 简单快速，非设计工况精度差
	系统辨识	实验激励获取模型参数	依赖实验条件，泛化性有限
数据驱动代理模型	高斯过程回归	贝叶斯非参数建模	不确定性量化，大数据计算慢
	神经网络代理	学习输入输出映射	拟合能力强，需大量离线数据
	混合物理-数据	物理约束嵌入网络	数据效率高，设计较复杂
端到端强化学习	DQN/DDQN	值函数逼近离散控制	离散动作，探索效率低
	DDPG/TD3	确定性策略连续控制	连续动作，超参数敏感
	SAC	最大熵框架	探索性强，样本效率较高
	PPO	近端策略优化	稳健性好，样本效率较低
混合控制架构	RL + PID 底层	RL 输出参考，PID 跟踪	安全性好，性能受限于底层
	分层 DRL	高层决策 + 低层技能	模块化，接口设计是关键
	RL + MPC	RL 学习代价函数/约束	兼顾优化与学习，计算复杂

2.3 环境建模与地形感知方法比较

机器人环境建模是实现自主导航与决策的基础。根据表示形式的不同，主流环境建模方法可分为以下几类：

（1）栅格地图（Grid Map）方法：栅格地图将连续环境离散化为规则网格单元，每个单元存储占据概率或语义标签。Hornung 等人提出的 OctoMap^[53]采用八叉树数据结构实现了高效的三维概率栅格地图表示，支持动态更新与多分辨率查询，已成为机器人三维环境建模的标准工具之一。栅格地图的优势在于表示直观、查询高效，适合与规划算法（如 A*、Dijkstra）直接结合；局限性在于存储开销随分辨率提升而快速增长，且在大规模场景中面临可扩展性挑战。

（2）距离场（Distance Field）方法：Oleynikova 等人提出的 Voxblox^[54]采用截断符号距离场（TSDF）与欧氏距离场（ESDF）表示三维环境，支持在线增量更新。距离场表示的优势在于可直接提供到最近障碍物的距离信息，便于梯度规划与碰撞检测；同时支持高效的 GPU 并行计算。该方法在无人机实时规划中得到广泛应用，对水下机器人同样具有参考价值。

（3）同时定位与建图（SLAM）：SLAM 旨在机器人运动过程中同时完成环境地图构建与自身定位。Cadena 等人的综述^[55]系统回顾了视觉 SLAM、激光 SLAM、语义 SLAM 等技术进展，指出 SLAM 正向着更加鲁棒、语义化、长期自主的方向发展。Palomeras 等人将主动 SLAM 应用于水下机器人自主探索，在不确定环境中规划信息增益最大的探索路径^[56]。水下 SLAM 面临的特殊挑战包括声呐数据稀疏与噪声大、光学视觉受能见度限制、以及缺乏全局定位基准等。

（4）地形可通行性分析：Papadakis 的综述^[57]系统梳理了面向地面机器人的地形可通行性分析方法，包括基于几何特征（坡度、粗糙度、台阶高度）、基于外观（颜色、纹理）、以及基于学习的方法。Fankhauser 等人^[58]提出了考虑定位不确定性的概率地形建图方法，通过不确定性传播提升地形估计的鲁棒性。Frey 等人发表于 IEEE TNNLS 的工作^[59]将地形语义信息与深度强化学习相结合，设计了地形感知导航框架，通过将地形特征编码为强化学习的状态表示，引导机器人学习适应不同地形的运动策略，实验表明融合地形感知的策略在复杂户外环

境中的导航成功率显著优于传统方法。

(5) 水下环境的特殊性: 水下地形感知主要依赖声学传感器。Meng 等人^[52]采用高斯过程回归对多波束声呐数据进行地形插值与不确定性量化, 为 AUV 地形辅助导航提供了可靠的地形估计基础。Medagoda 等人^[60]研究了时变洋流场中的 AUV 定位问题, 提出了融合地形与流场信息的定位滤波器。这些工作为水下地形感知提供了技术基础, 但如何将地形信息有效融入多机器人博弈决策仍是开放问题。

2.4 地形与战术协同的研究意义

将地形信息融入追逃博弈决策具有深远的理论与应用意义。从理论角度, Bhattacharya 和 Hutchinson^[24]的研究表明, 障碍物(地形)的存在可根本性地改变追逃博弈的均衡结构, 使得在开放空间中不可能的捕获在有限空间中成为可能。这一理论结果揭示了“地形作为战术资源”的潜在价值, 为非对称博弈中的劣势方提供了“扳回一城”的可能途径。

在多机器人协同围捕领域, Yang 等人^[61]改进了人工势场方法用于多 AUV 协同狩猎, 通过设计吸引势场与排斥势场实现对目标的协同围捕, 但该工作未显式考虑地形的战术利用。Huang 等人^[62]在多 AUV 协同目标搜索中引入地形引导的势场方法, 通过地形特征调节搜索优先级, 展现了地形信息在任务规划中的价值, 但仍停留在将地形作为搜索辅助信息的层面, 未深入探讨地形在对抗博弈中的战术作用。

地形与战术协同研究的核心意义体现在以下方面:**(1) 弥补性能劣势:** 在追击方速度劣势的非对称博弈中, 地形提供了“扳回劣势”的战术手段。通过利用狭窄通道、死角、遮蔽区等地形特征, 追击者群体可限制逃逸者的机动空间, 将开放空间中的不可能捕获转化为可能。例如, 追击者可利用海底沟壑的瓶颈区域设置“埋伏”, 或利用礁石群的遮蔽实现隐蔽接近。**(2) 增强策略多样性:** 地形感知为强化学习策略提供了更丰富的状态信息, 有助于涌现多样化的协同战术(包抄、驱赶、诱捕、封锁等), 超越简单的距离最小化追击。地形特征的引入使

得策略空间更加丰富，有助于学习到更具创造性的协同行为。**(3) 提升仿真-现实迁移能力：**地形是环境的相对稳定特征，将地形信息显式纳入策略学习，有助于提升策略在不同场景间的泛化能力。相比于依赖绝对位置信息的策略，基于地形相对关系的策略具有更好的场景迁移性。**(4) 连接规划与决策：**地形感知架起了底层路径规划与高层战术决策之间的桥梁，使”哪里可走”与”如何利用”形成有机整体，实现从导航层面到战术层面的信息贯通。

尽管地形感知在陆地与空中机器人领域已有较多研究，但在水下多机器人追逃博弈场景中的应用仍处于起步阶段。水下环境的三维性、声学传感的特殊性（噪声大、分辨率低、存在阴影区）、以及通信受限条件下的信息融合问题，均需要针对性的研究探索。

2.5 存在问题

综合以上研究现状分析，现有工作在以下方面存在不足：

2.5.1 非对称博弈建模与策略设计的系统性不足

现有追逃博弈研究对”非对称性”的建模往往过于简化。大多数工作仅考虑速度差异这一单一维度，忽略了加速度、转弯半径、感知范围、能量续航等多维度性能不对称性。更重要的是，缺乏系统性地研究”如何利用多维度优势（数量、机动、地形）弥补单一维度劣势（速度）”的策略设计方法论^[5]。在追击方性能劣势的场景下，策略学习容易陷入简单追击的局部最优，难以涌现出”诱敌入瓮”、”多点封锁”等高级战术。对抗训练虽可提升策略鲁棒性，但对手策略的多样性仍然受限，导致学习到的策略在面对不同类型对手时泛化能力不足^{[31][34]}。

2.5.2 水下场景的针对性研究匮乏

尽管多智能体强化学习与追逃博弈研究已取得显著进展，但专门针对三维水下环境的工作仍相对稀缺。Tang 等人的综述指出^[28]，水下机器人是 DRL 应用中研究相对不足的领域，大多数工作仅涉及基础运动控制与导航，缺乏面向复

杂任务（如追逃围捕）的深入研究。水下环境的特殊性——声学通信受限、感知不确定性大、三维运动与复杂地形——使得陆地/空中场景的方法难以直接移植。Akyildiz 等人^[21]强调的水下通信挑战对 CTDE 范式的“集中训练”假设构成威胁，在水下场景中即使是训练阶段全局信息的获取也可能面临困难，需要发展更加适应通信受限条件的协同学习方法。

2.5.3 动力学建模非线性与学习控制的融合不足

现有学习型控制方法虽可规避显式建模，但存在以下不足^{[42][28]}：**仿真-现实迁移问题**：Tang 等人指出，零样本 Sim-to-Real 迁移的成功高度依赖于仿真环境的保真度以及有效的域随机化策略^[28]。水下流体动力学效应的精确仿真成本高昂，简化仿真与真实系统之间存在显著差距，导致仿真训练的策略在真实环境中性能下降。**安全性保障缺失**：端到端学习策略缺乏形式化的安全保证，在未探索状态区域可能产生危险动作，这在水下机器人等安全关键应用中风险尤为突出。Zhang 等人^[51]的安全强化学习方法为此提供了有益探索，但在复杂水下环境中的适用性仍需验证。**样本效率与稳定性权衡**：在线策略方法（如 PPO）稳健但样本效率低；离线策略方法样本效率高但面临分布偏移导致的稳定性问题^[28]。在水下机器人应用中，真实环境交互成本高昂，如何在有限样本下学习高性能策略是关键挑战。

2.5.4 地形感知与博弈策略的耦合深度不足

当前地形感知技术与多智能体博弈策略之间的融合程度有限，具体表现为：**地形表示与 RL 状态空间的脱节**：环境建图系统输出的通常是高维地图数据（栅格、点云、TSDF 等），难以直接作为强化学习的状态输入。如何设计紧凑且信息丰富的地形战术特征表示，使其既能保留战术相关信息又不增加过多状态维度，是一个开放问题^{[59][53]}。**地形信息在奖励设计中的缺位**：现有追逃博弈研究的奖励函数多基于距离、时间等简单指标，未能体现地形的战术价值。利用地形实现隐蔽接近、构建围堵陷阱、封锁逃逸通道等行为在现有奖励框架下难以被有效

激励，限制了策略的战术丰富度。**缺乏系统性评估框架：**尚无“工作系统量化”地形感知对协同追逃性能提升”的贡献程度，难以判断地形融合的投入产出比，也无法指导地形特征设计的优化方向。

2.5.5 综合性研究空白

综合以上分析，目前尚缺乏一个系统性的研究框架，能够同时处理：（1）三维复杂水下地形建模与感知；（2）追逃双方的多维度性能非对称性；（3）多智能体协同博弈策略学习；（4）分层架构下的低层运动控制与高层战术决策耦合；（5）地形感知与博弈策略的深度融合。本研究旨在填补这一综合性空白，构建融合地形感知的水下多机器人非对称追逃博弈分层强化学习框架。

3 研究展望

3.1 融合地形感知的非对称追逃多智能体强化学习框架

针对现有研究的不足，本领域未来研究将呈现以下发展趋势：

（1）地形感知与博弈策略的深度耦合：未来研究应建立“地形-感知-决策”一体化框架。在地形表示层面，需发展适用于水下环境的高效特征表示方法，将声呐感知数据转化为紧凑的战术语义特征，编码地形的可通行性、遮蔽性、围堵潜力等属性^{[59][23]}。在策略学习层面，需设计地形感知驱动的奖励函数与约束机制，引导智能体学习利用地形优势弥补性能劣势的协同战术。Tang 等人的综述^[28]表明，特权学习是处理高维环境信息的有效范式：可首先训练具有完整地形信息的教师策略，再蒸馏为基于实际传感器的学生策略，从而在保持策略性能的同时适应真实感知条件。

（2）非对称性能的系统性建模与优势补偿策略：应建立更加全面的非对称性能模型，涵盖速度、加速度、转弯半径、感知范围、通信能力等多维度差异。在此基础上，研究“性能劣势方如何通过数量优势、地形优势、战术优势实现博弈胜利”的策略设计方法论。可借鉴游戏 AI 领域的自博弈与课程学习思想，设计

从简单到复杂的训练课程，逐步提升策略复杂度与鲁棒性^{[31][34]}。

(3) 对抗鲁棒性与泛化能力提升：引入对抗训练、域随机化、元学习等方法增强策略的鲁棒性与泛化能力^{[34][63]}。通过在训练中注入对手策略扰动、环境参数变化、感知噪声等不确定性因素，使策略具备应对未知对手与复杂环境的能力。特别地，可设计多样化的逃逸者策略库用于追击者的训练，提升追击策略的泛化性。

3.2 分层多智能体深度强化学习架构设计

(1) 低层运动控制策略学习：基于分层强化学习思想^{[47][48]}，构建适用于水下机器人的低层运动控制框架。低层策略采用 DDPG、SAC 等连续控制算法学习基础运动技能（姿态保持、轨迹跟踪、避障机动等），通过与高保真仿真环境的大量交互隐式学习非线性动力学关系，规避复杂水下动力学的显式建模困难^{[42][50]}。低层策略的训练可采用域随机化方法增强鲁棒性，使其能够适应真实环境的不确定性。

(2) 高层战术决策与协同规划：高层策略专注于多追击者的协同博弈决策，包括围捕编队形态选择、包抄位置分配、诱敌路径规划、封锁时机判断等^[49]。高层策略的状态空间应融合智能体相对位置、地形战术特征、对手行为预测等信息。通过 CTDE 范式的多智能体强化学习算法（如 MADDPG、QMIX 及其变体）学习协同策略^{[29][30]}。高层决策的时间尺度应与低层执行相匹配，通过合理的时间抽象实现高效的层次协同。

(3) 层次间接口设计与安全保障：高低层之间的接口可定义为子目标位置、期望速度方向、或运动原语选择等形式^[47]。接口设计需权衡抽象程度与控制精度：过于抽象可能丧失对低层行为的精细控制，过于具体则增加高层决策的负担。同时引入安全约束机制，参考 Zhang 等人^[51]的事件触发安全强化学习思想，确保低层执行过程中的碰撞规避与能耗约束，在追求任务性能的同时保障系统运行安全。

3.3 高保真水下仿真与系统验证路径

（1）仿真平台构建：未来研究需构建高保真的水下追逃博弈仿真环境，整合以下要素：真实海底地形数据（基于多波束声呐测绘）、复杂流场模拟（洋流、湍流）、水下机器人六自由度动力学模型、声学传感模型（包括噪声、多径、阴影效应）以及水声通信模型（带宽、时延、丢包）^[64]。仿真平台应支持域随机化，允许动力学参数、环境参数的随机扰动，以提升策略的迁移鲁棒性^[63]。

（2）多层级验证路径：遵循渐进式验证策略：纯数字仿真 → 半实物仿真（Hardware-in-the-Loop）→ 水池受控实验 → 开放水域外场试验。每个阶段评估策略的性能、鲁棒性与安全性，识别仿真与真实环境的差距并针对性改进。这种渐进式路径可有效降低研发风险与成本，同时积累真实环境数据用于仿真模型的校准与改进^[28]。

（3）性能评估指标体系：建立系统的性能评估指标，包括：捕获成功率、平均捕获时间、能耗效率、地形利用度（如利用遮蔽次数、成功封锁通道比例）、以及策略鲁棒性（面对不同对手策略、不同地形配置的性能稳定性）。通过消融实验量化地形感知、分层架构、非对称性能建模等各组件对协同追逃性能提升的贡献程度，指导系统的进一步优化。

3.4 技术挑战与未来方向

展望未来，本领域仍面临以下开放挑战：

（1）大规模异构编队：将研究思路扩展至大规模多 AUV 编队的协同攻防，以及空-水跨域协同（无人机与水下机器人联合作战）场景^{[65][36]}。异构编队需要处理不同平台间的能力差异、通信协议差异以及任务分工问题，是多机器人系统研究的重要方向。

（2）部分可观测与通信受限：发展适用于水下通信受限条件的分布式 MARL 算法，减少对全局信息的依赖^[21]。可探索基于通信学习（Learning to Communicate）的方法，让智能体学习何时通信、传递何种信息，在有限通信带宽下实现高效协同。

(3) 在线学习与持续适应：研究策略的在线微调与持续学习机制，使机器人能够在真实部署中适应环境变化与对手策略演化。这要求策略具备快速适应能力，同时避免灾难性遗忘，是强化学习领域的重要开放问题。

(4) 可解释性与人机协同：提升学习策略的可解释性，使人类操作员能够理解策略的决策逻辑，支持人机协同决策与必要时的人工干预。可解释性对于安全关键的军事与搜救应用尤为重要。

综上所述，融合地形感知的水下机器人非对称追逃博弈多智能体深度强化学习策略研究，是一个交叉性强、应用价值高的前沿课题。本研究将深度强化学习、分层控制、地形感知与多智能体协同等多个领域的最新进展有机整合，有望为复杂海洋环境下智能无人系统的自主决策与协同控制提供新的理论方法与技术途径。

4 参考文献

- [1] Chung T H, Hollinger G A, Isler V. Search and pursuit-evasion in mobile robotics: A survey [J]. *Autonomous Robots*, 2011, 31(4): 299-316. DOI: 10.1007/s10514-011-9241-4.
- [2] Isaacs R. *Differential Games: A Mathematical Theory with Applications to Warfare and Pursuit, Control and Optimization*[M]. Dover Publications, 1999.
- [3] Weintraub I E, Pachter M, Garcia E. An introduction to pursuit-evasion differential games [J]. *American Control Conference (ACC)*, 2020: 1675-1692. DOI: 10.23919/ACC45564.2020.9147205.
- [4] 王宏健, 陈卫东, 席裕庚. 多机器人追捕问题的研究[J]. *机器人*, 2003, 25(5): 464-468.
- [5] Huang H, Zhang W, Ding J, et al. Guaranteed decentralized pursuit-evasion in the plane with multiple pursuers[C]//2011 50th IEEE Conference on Decision and Control and European Control Conference. IEEE, 2011: 4835-4840. DOI: 10.1109/CDC.2011.6161237.
- [6] Zhou Z, Takei R, Huang H, et al. A general, open-loop formulation for reach-avoid games [C]//2012 IEEE 51st IEEE Conference on Decision and Control (CDC). IEEE, 2012: 6501-6506. DOI: 10.1109/CDC.2012.6426643.
- [7] Vidal R, Shakernia O, Kim H J, et al. Probabilistic pursuit-evasion games: Theory, implementation, and experimental evaluation[J]. *IEEE Transactions on Robotics and Automation*, 2002, 18(5): 662-669. DOI: 10.1109/TRA.2002.804040.
- [8] Chen M, Zhou Z, Tomlin C J. Multiplayer reach-avoid games via pairwise outcomes[J]. *IEEE Transactions on Automatic Control*, 2017, 62(3): 1451-1457. DOI: 10.1109/TAC.2016.2577619.
- [9] Sun W, Tsiotras P, Lolla T, et al. Multiple-pursuer/one-evader pursuit-evasion game in dynamic flowfields[J]. *Journal of Guidance, Control, and Dynamics*, 2017, 40(7): 1627-1637. DOI: 10.2514/1.G002125.
- [10] Li D, Cruz J B, Chen G, et al. A hierarchical approach to multi-player pursuit-evasion differential games[J]. *IEEE Conference on Decision and Control*, 2005: 5674-5679. DOI: 10.1109/CDC.2005.1583066.
- [11] Vieira M A, Govindan R, Sukhatme G S. Scalable and practical pursuit-evasion with networked robots[J]. *Intelligent Service Robotics*, 2009, 2(4): 247-263. DOI: 10.1007/s11370-009-0047-9.
- [12] Nguyen T T, Nguyen N D, Nahavandi S. Deep Reinforcement Learning for Multiagent Systems: A Review of Challenges, Solutions, and Applications[J]. *IEEE Transactions on Cybernetics*, 2020, 50(9): 3826-3839. DOI: 10.1109/TCYB.2020.2977374.
- [13] Sahoo A, Dwivedy S K, Robi P S. Advancements in the field of autonomous underwater vehicle[J]. *Ocean Engineering*, 2019, 181: 145-160. DOI: 10.1016/j.oceaneng.2019.04.011.
- [14] Yu J, Zhang A, Jin W, et al. Development and experiments of the Sea-Wing underwater glider [J]. *China Ocean Engineering*, 2011, 25(4): 721-736. DOI: 10.1007/s13344-011-0058-x.
- [15] 徐德民, 严浙平, 付明玉. 水下无人航行器发展趋势及关键技术[J]. *火力与指挥控制*, 2011, 36(1): 1-5.
- [16] Cao Y, Yu W, Ren W, et al. An overview of recent progress in the study of distributed multi-agent coordination[J]. *IEEE Transactions on Industrial Informatics*, 2013, 9(1): 427-438. DOI: 10.1109/TII.2012.2219061.
- [17] Zeng Z, Lian L, Sammut K, et al. A survey on path planning for persistent autonomy of autonomous underwater vehicles[J]. *Ocean Engineering*, 2015, 110: 303-313. DOI: 10.1016/j.oceaneng.2015.10.007.
- [18] Hadi B, Khosravi A, Sarhadi P. A review of the path planning and formation control for multiple autonomous underwater vehicles[J]. *Journal of Intelligent and Robotic Systems*, 2021, 101(67). DOI: 10.1007/s10846-021-01330-4.
- [19] 朱大奇, 刘凡, 颜明重. 多水下机器人协调控制研究进展[J]. *智能系统学报*, 2012, 7(5): 377-385.

- [20] 严卫生, 张福斌, 李晔. 自主水下航行器路径规划技术研究综述[J]. 鱼雷技术, 2010, 18(3): 1-7.
- [21] Akyildiz I F, Pompili D, Melodia T. Underwater acoustic sensor networks: Research challenges[J]. *Ad Hoc Networks*, 2005, 3(3): 257-279. DOI: 10.1016/j.adhoc.2005.01.004.
- [22] Yan Z, Joubert N, Rabaste O. A survey of AUV navigation and localization techniques[J]. *Journal of Marine Science and Engineering*, 2021, 9(2): 123. DOI: 10.3390/jmse9020123.
- [23] Wöfl A C, Snaith H, Aber S, et al. Seafloor mapping—the challenge of a truly global ocean bathymetry[J]. *Frontiers in Marine Science*, 2019, 6: 283. DOI: 10.3389/fmars.2019.00283.
- [24] Bhattacharya S, Hutchinson S. On the existence of nash equilibrium for a two-player pursuit-evasion game with visibility constraints[J]. *International Journal of Robotics Research*, 2010, 29(7): 831-839. DOI: 10.1177/0278364909357413.
- [25] Kober J, Bagnell J A, Peters J. Reinforcement learning in robotics: A survey[J]. *International Journal of Robotics Research*, 2013, 32(11): 1238-1274. DOI: 10.1177/0278364913495721.
- [26] Mnih V, Kavukcuoglu K, Silver D, et al. Human-level control through deep reinforcement learning[J]. *Nature*, 2015, 518(7540): 529-533. DOI: 10.1038/nature14236.
- [27] 张铭钧, 姚峰, 刘维新. 水下机器人智能控制技术研究进展[J]. 哈尔滨工程大学学报, 2020, 41(11): 1573-1582.
- [28] Tang C, Abbatematteo B, Hu J, et al. Deep Reinforcement Learning for Robotics: A Survey of Real-World Successes[J]. *Annual Review of Control, Robotics, and Autonomous Systems*, 2025, 8: 153-188. DOI: 10.1146/annurev-control-030323-022510.
- [29] Lowe R, Wu Y, Tamar A, et al. Multi-agent actor-critic for mixed cooperative-competitive environments[C]//*Advances in Neural Information Processing Systems*: vol. 30. 2017: 6379-6390.
- [30] Rashid T, Samvelyan M, Schroeder C, et al. QMIX: Monotonic value function factorisation for deep multi-agent reinforcement learning[C]//*International Conference on Machine Learning*. 2018: 4295-4304.
- [31] Yu L, Song J, Ermon S. Multi-Agent Adversarial Inverse Reinforcement Learning[C]//*2020 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2020: 1-8. DOI: 10.1109/IJCNN48605.2020.9207248.
- [32] 刘明雍, 杨盼盼, 陈宇. 基于深度强化学习的多 AUV 协同围捕策略研究[J]. 西北工业大学学报, 2020, 38(4): 840-848.
- [33] Pierson A, Rus D. Distributed multi-robot pursuit-evasion with limited field of view[C]//*2017 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2017: 5470-5476. DOI: 10.1109/ICRA.2017.7989732.
- [34] Pinto L, Davidson J, Sukthankar R, et al. Robust adversarial reinforcement learning[C]//*International Conference on Machine Learning*. 2017: 2817-2826.
- [35] Cao X, Zhu D, Yang S X. Multi-AUV cooperative target search and tracking in unknown underwater environment[J]. *Ocean Engineering*, 2018, 150: 296-305. DOI: 10.1016/j.oceaneng.2018.01.001.
- [36] Wu Y, Low K H, Lv C. Cooperative path planning for heterogeneous unmanned vehicles in a search-and-track mission aiming at an underwater target[J]. *IEEE Transactions on Vehicular Technology*, 2020, 69(6): 6782-6787. DOI: 10.1109/TVT.2020.2991983.
- [37] Cai W, Zhang M, Zheng Y R. Cooperative multi-robot encirclement and capture using learning-based approach[J]. *Ocean Engineering*, 2021, 238: 109709. DOI: 10.1016/j.oceaneng.2021.109709.
- [38] Masmitja I, Navarro J, Gomariz S, et al. Dynamic robotic tracking of underwater targets using reinforcement learning[J]. *Science Robotics*, 2023, 8(80): eade7811. DOI: 10.1126/scirobotics.ade7811.
- [39] Fossen T I. *Handbook of Marine Craft Hydrodynamics and Motion Control*[M]. John Wiley & Sons, 2011. DOI: 10.1002/9781119994138.
- [40] Phillips A, Furlong M, Turnock S R. The use of computational fluid dynamics to determine

- the dynamic stability of an autonomous underwater vehicle[C]//10th International Symposium on Unmanned Untethered Submersible Technology. 2007.
- [41] Shen C, Shi Y, Buckham B. Trajectory tracking control of an autonomous underwater vehicle using Lyapunov-based model predictive control[J]. IEEE Transactions on Industrial Electronics, 2018, 65(7): 5796-5805. DOI: 10.1109/TIE.2017.2779442.
 - [42] Carlucho I, De Paula M, Wang S, et al. AUV Position Tracking Control Using End-to-End Deep Reinforcement Learning[J]. IEEE Transactions on Robotics, 2023, 39(3): 2376-2393. DOI: 10.1109/TRO.2023.3257541.
 - [43] Chu Z, Zhu D, Yang S X. Observer-based adaptive neural network trajectory tracking control for remotely operated vehicle[J]. IEEE Transactions on Neural Networks and Learning Systems, 2017, 28(7): 1633-1645. DOI: 10.1109/TNNLS.2016.2544786.
 - [44] Sun B, Zhu D, Tian C, et al. Complete coverage autonomous underwater vehicles path planning based on gladius bio-inspired neural network algorithm for discrete points[J]. IEEE Transactions on Cognitive and Developmental Systems, 2019, 11(1): 73-84. DOI: 10.1109/TCDS.2018.2810235.
 - [45] Wang Y, Tong J, Song T, et al. Reinforcement learning for AUV autonomous navigation and target tracking[J]. Ocean Engineering, 2024, 293: 116587. DOI: 10.1016/j.oceaneng.2024.116587.
 - [46] Yan J, Yang X, Luo X, et al. Energy-efficient data collection for AUV-assisted underwater sensor networks using hierarchical reinforcement learning[J]. IEEE Internet of Things Journal, 2023, 10(3): 2686-2696. DOI: 10.1109/JIOT.2022.3214770.
 - [47] Pateria S, Subagdja B, Tan A H, et al. Hierarchical reinforcement learning: A comprehensive survey[J]. ACM Computing Surveys, 2021, 54(5): 1-35. DOI: 10.1145/3453160.
 - [48] Nachum O, Gu S S, Lee H, et al. Data-efficient hierarchical reinforcement learning[C]//Advances in Neural Information Processing Systems: vol. 31. 2018: 3303-3313.
 - [49] Jeon S, Lee K, Kim H J. Learning Hierarchical Multi-Robot Coordination[C]//2022 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2022: 5978-5984. DOI: 10.1109/ICRA46639.2022.9812151.
 - [50] Hachicha M, et al. Hierarchical Deep Reinforcement Learning for Amphibious Robot Control [J]. arXiv preprint arXiv:2410.18969, 2024. DOI: 10.48550/arXiv.2410.18969.
 - [51] Zhang K, et al. Event-Triggered Hierarchical Safe Reinforcement Learning for Multi-Robot Systems[J]. IEEE Robotics and Automation Letters, 2025. DOI: 10.1109/LRA.2025.3540380.
 - [52] Meng W, He Y, Deng L, et al. Terrain-aided navigation for underwater vehicles using Gaussian process regression[J]. IEEE Transactions on Geoscience and Remote Sensing, 2022, 60: 1-13. DOI: 10.1109/TGRS.2021.3106821.
 - [53] Hornung A, Wurm K M, Bennewitz M, et al. OctoMap: An efficient probabilistic 3D mapping framework based on octrees[J]. Autonomous Robots, 2013, 34(3): 189-206. DOI: 10.1007/s10514-012-9321-0.
 - [54] Oleynikova H, Taylor Z, Fehr M, et al. Voxblox: Incremental 3D Euclidean signed distance fields for on-board MAV planning[J]. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2017: 1366-1373. DOI: 10.1109/IROS.2017.8202315.
 - [55] Cadena C, Carlone L, Carrillo H, et al. Past, present, and future of simultaneous localization and mapping: Toward the robust-perception age[J]. IEEE Transactions on Robotics, 2016, 32(6): 1309-1332. DOI: 10.1109/TRO.2016.2624754.
 - [56] Palomeras N, Carreras M, Andrade-Cetto J. Active SLAM for autonomous underwater exploration[J]. Remote Sensing, 2019, 11(23): 2827. DOI: 10.3390/rs11232827.
 - [57] Papadakis P. Terrain traversability analysis methods for unmanned ground vehicles: A survey [J]. Engineering Applications of Artificial Intelligence, 2013, 26(4): 1373-1385. DOI: 10.1016/j.engappai.2013.01.006.
 - [58] Fankhauser P, Bloesch M, Hutter M. Probabilistic terrain mapping for mobile robots with un-

- certain localization[J]. IEEE Robotics and Automation Letters, 2018, 3(4): 3019-3026. DOI: 10.1109/LRA.2018.2849506.
- [59] Frey J, et al. Terrain-Aware Navigation with Deep Reinforcement Learning[J]. IEEE Transactions on Neural Networks and Learning Systems, 2025. DOI: 10.1109/TNNLS.2025.3566548.
- [60] Medagoda L, Kinsey J C, Eilders M. Autonomous underwater vehicle localization in a spatiotemporally varying water current field[J]. IEEE Journal of Oceanic Engineering, 2016, 41(4): 855-866. DOI: 10.1109/JOE.2016.2527499.
- [61] Yang G, Wu Y, Li Z. Multi-AUV cooperative hunting based on improved artificial potential field method[J]. Journal of Marine Science and Technology, 2021, 26: 701-715. DOI: 10.1007/s00773-020-00766-7.
- [62] Huang H, Zhou H, Yang X, et al. Multi-AUV cooperative target search with terrain-based potential field guidance[J]. Ocean Engineering, 2023, 267: 113267. DOI: 10.1016/j.oceaneng.2022.113267.
- [63] Tobin J, Fong R, Ray A, et al. Domain randomization for transferring deep neural networks from simulation to the real world[C]//2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2017: 23-30. DOI: 10.1109/IROS.2017.8202133.
- [64] Koenig N, Howard A. Design and use paradigms for Gazebo, an open-source multi-robot simulator[C]//2004 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS): vol. 3. IEEE, 2004: 2149-2154. DOI: 10.1109/IROS.2004.1389727.
- [65] Queralta J P, Taipalmaa J, Pullinen B C, et al. Collaborative multi-robot search and rescue: Planning, coordination, perception, and active vision[J]. IEEE Access, 2020, 8: 191617-191643. DOI: 10.1109/ACCESS.2020.3030190.

二、开题报告

1 问题提出的背景

1.1 背景介绍

随着海洋强国战略的深入推进，水下机器人已成为海洋探测、资源勘探、水下搜救、海洋国防等领域不可或缺的技术装备^[1]。自主水下航行器（AUV）从早期的单体遥控作业演进为具备自主决策能力的智能平台，在深海科考与海洋观测中发挥着日益重要的作用^[2]。然而，随着海洋任务复杂度的提升，单一水下机器人的能力已难以满足大范围协同搜索、动态目标跟踪、协同围捕等任务需求，多水下机器人协同系统应运而生^[3-4]。

追逃博弈（Pursuit-Evasion Game）是博弈论与控制理论交叉领域的经典问题，其核心在于追击方如何设计最优运动策略以捕获逃逸方^[5]。该问题在军事领域具有广泛应用，包括导弹拦截、无人机编队对抗、海上缉私与反潜作战等场景^[6]。在实际应用中，追逃双方的性能往往呈现显著的非对称性——追击方可能在绝对速度上处于劣势但在加速性能或数量上占优^[7]。如何在运动学劣势条件下通过智慧与战术实现成功捕获，是该领域亟需解决的核心科学问题。

深度强化学习的快速发展为解决高维复杂决策问题提供了新途径^[8]。研究表明，基于多智能体深度强化学习的追逃策略能够涌现出包抄、驱赶、诱捕等复杂协同行为，无需人工设计繁琐的启发式规则^[9-10]。Zhang 等人提出的 Game of Drones 方法通过引入目标预测网络（TP Net），利用 LSTM 预测目标未来轨迹，实现了“先预测后决策”的策略框架，有效应对了追击者的速度劣势问题^[11]。然而，当前研究主要集中在陆地与空中场景，专门针对三维水下环境的工作仍相对匮乏^[12]。

1.1.1 项目提出的原因

本项目的提出源于现有研究在以下方面存在的显著不足：

第一，水下环境的特殊性未得到充分考虑。水下环境具有区别于陆地与空中的显著特性：通信主要依赖声学信道，带宽受限、时延大且易受干扰^[13]；GPS信号不可用导致定位困难^[14]；复杂的流体动力学效应使得机器人动力学模型呈现强非线性和参数不确定性^[15]。这些特性使得陆地/空中场景的方法难以直接移植。

第二，非对称博弈的系统性研究不足。现有追逃博弈研究对非对称性的建模往往过于简化，仅考虑速度差异这一单一维度^[7]。缺乏系统性研究“如何利用多维度优势（数量、机动、地形）弥补单一维度劣势（速度）”的策略设计方法论。在速度劣势条件下，现有方法主要依赖两种思路：一是通过数量优势实现协同围捕^[16]，二是通过目标轨迹预测实现提前拦截^[11]。然而，将地形作为战术资源以弥补运动学劣势的研究仍相对缺乏。

第三，地形利用与博弈策略的融合深度不足。水下复杂地形（沟壑、山脊、礁石群等）既构成导航挑战，也蕴含潜在战术价值。地形对追逃博弈的影响是多层次的：最基础的是避障与可通行性分析^[17]；进一步是复杂地形中的导航规划^[18]；更深层次是利用地形限制目标的观察角和逃逸方向^[19]；最高层次是结合地形进行目标运动预测^[11]。然而，当前研究将地形仅视为“需要规避的障碍”，未能实现向“可利用的战术资源”的认知转变。

第四，动力学建模与学习控制的融合面临挑战。水下机器人动力学建模涉及附加质量效应、非线性阻尼、科氏力耦合等复杂因素^[15]，传统基于模型的控制方法对模型精度要求高，而端到端学习方法在仿真-现实迁移与安全性保障方面仍存在不足^[20]。

1.2 本研究的意义和目的

本研究旨在构建融合地形感知的水下多机器人非对称追逃博弈分层深度强化学习框架，具有重要的理论意义与应用价值。

在理论层面，本研究将深度强化学习、分层控制、地形感知与多智能体协同等多个领域的最新进展有机整合，建立“地形-感知-决策”一体化的博弈策略学习框架。通过系统研究非对称性能条件下的优势补偿机制，揭示地形特征对追逃博弈均衡结构的影响规律^[19]，丰富多智能体博弈与强化学习的理论体系。特别地，本研究将探索如何利用地形限制目标的感知范围和逃逸方向，以及如何结合地形信息进行目标轨迹预测^[11]，为非对称追逃博弈提供新的理论视角。

在应用层面，研究成果可直接服务于海洋国防、海上执法、水下搜救等领域的实际需求^[21]。多 AUV 协同围捕技术可应用于入侵者拦截、目标跟踪与捕获等任务，为复杂海洋环境下智能无人系统的自主决策与协同控制提供技术支撑。

本研究的核心目标包括：（1）建立水下机器人六自由度动力学模型，分析非线性特性并为学习型控制提供理论依据；（2）设计基于 PPO-LSTM 的低层运动控制策略，实现稳定鲁棒的位置追踪与姿态维稳；（3）构建分层多智能体强化学习框架，实现从一追一到多追一的协同策略学习，其中融入地形避障、地形导航、视角利用与目标预测等多层次地形利用能力；（4）使追击者能够利用地形优势弥补运动学劣势，实现有效围捕。

2 项目的主要研究内容和技术路线

本研究面向水下复杂地形环境中的非对称追逃博弈问题，致力于构建一套融合地形感知的多智能体分层深度强化学习框架。研究遵循“由简到繁、由单到多、由抽象到具体”的渐进式思路，借鉴自然界中掠食者利用地形掩护、观察猎物、伺机捕获的狩猎行为，将整体研究内容分解为四个相互关联、逐步深入的子课题。

2.1 主要研究内容

本研究的核心科学问题是：**如何在运动学劣势条件下，通过智能决策与地形利用实现有效围捕**。围绕这一问题，研究内容按照分层控制架构的逻辑递进展开，各阶段之间存在明确的依赖关系：阶段一为阶段二提供理论基础，阶段二为

后续所有高层策略提供可靠的执行器，阶段三在简化场景中验证地形利用的可行性，阶段四将研究扩展到完整的多智能体协同场景。

地形对追逃博弈的影响是渐进式、多层次的^[22]：（1）**基础层——避障**：地形作为需要规避的障碍物；（2）**导航层——路径规划**：在复杂环境中规划可行路径；（3）**战术层——观察角利用**：利用地形遮挡限制目标感知范围^[19]；（4）**预测层——轨迹预测**：结合地形约束预测目标运动^[11]。本研究将这四个层次的地形利用能力有机融入策略学习框架。

2.1.1 阶段一：水下机器人动力学建模与分析

研究目标：建立水下机器人六自由度动力学模型，分析非线性特性，为学习型控制方法的选择提供理论依据。

研究动机与必要性：水下机器人动力学具有区别于陆地/空中机器人的显著特殊性^[15]。附加质量效应、非线性水动力阻尼、科氏力耦合以及洋流扰动等因素使得精确建模极为困难^[23]。传统基于模型的控制方法对模型精度要求高，而CFD 计算成本过高、经验公式适用范围有限、系统辨识对工况敏感^[24]。这些建模困难是本研究选择深度强化学习方法的核心理论依据——通过学习型控制规避显式建模的需求。

主要研究内容：（1）建立包含刚体惯性、附加质量、科氏力、水动力阻尼、恢复力矩的完整动力学方程；（2）分析各非线性项的物理机制及其对控制性能的影响；（3）明确传统建模方法的局限性，论证强化学习方法的合理性。

阶段成果：形成水下机器人动力学建模的理论框架，为后续低层控制策略设计提供依据。

2.1.2 阶段二：低层运动控制策略学习

研究目标：基于深度强化学习设计低层运动控制策略，实现稳定鲁棒的位置追踪与姿态维稳，为高层决策提供可靠的执行器。

研究动机与必要性：低层控制是分层框架的基石^[25]。高层战术决策（如”向

目标位置移动”)必须通过低层控制转化为具体的推进器指令。若低层控制不稳定或精度不足,高层策略的任何决策都无法正确执行。此外,水下环境的部分可观测性(传感器噪声、通信延迟、洋流未知)对控制策略提出了额外挑战^[26]。

主要内容: (1) 设计融合位置追踪、姿态维稳、安全避障、能量效率的多目标控制框架; (2) 采用 PPO 算法保证训练稳定性,引入 LSTM 网络处理部分可观测性^[27-28]; (3) 通过域随机化技术增强策略对模型不确定性的鲁棒性^[29]; (4) 使低层策略具备基础的地形避障能力。

不可替代性: 低层控制策略的质量直接决定整个分层框架的上限。一个稳定、鲁棒、可复用的低层策略是后续所有高层研究的前提条件。

阶段成果: 训练得到可跨任务复用的低层运动控制策略,具备在复杂地形中稳定导航的能力。

2.1.3 阶段三: 融合地形感知的一追一逃博弈策略

研究目标: 在一追一场景中研究非对称追逃博弈策略,探索地形利用的基础能力(导航、视角利用、攻击时机判断)。

研究动机与必要性: 一追一是追逃博弈的最简形式,便于验证分层框架的有效性和调试奖励函数设计^[5]。更重要的是,本阶段需要解决一个关键科学问题: **在速度劣势条件下,追击者如何利用地形弥补运动学不足。**自然界中的捕食者(如狮子)正是通过智慧与战术——利用地形隐蔽接近、选择最佳攻击时机——来克服速度劣势^[30]。

现有追逃博弈研究将地形仅视为“需要规避的障碍”,未能实现向“可利用的战术资源”的认知转变。Bhattacharya 和 Hutchinson 的理论研究表明,视线约束的存在可根本性地改变追逃博弈的均衡结构^[19],这意味着地形利用具有深刻的理论价值。

主要内容:

- **非对称性建模:** 设定追击者速度劣势但加速度占优,引入警戒区域机制模拟猎物警觉行为

- **地形特征提取：**设计可通行性、遮蔽性、死角指数、瓶颈指数等战术语义特征
- **分阶段策略框架：**隐蔽接近阶段利用地形遮挡视线，快速捕获阶段利用地形封锁逃逸路线
- **课程学习：**从开放空间到复杂地形、从简单逃逸者到智能逃逸者的渐进式训练^[31]

不可替代性：本阶段首次系统性地研究“地形作为战术资源”在追逃博弈中的应用，建立地形特征与战术行为之间的映射关系，为多追一协同地形利用奠定基础。

阶段成果：（1）一追一追逃策略，具备隐蔽接近与地形驱赶能力；（2）地形特征提取与编码模块；（3）作为多追一研究的性能基线。

2.1.4 阶段四：融合协同地形利用与目标预测的多追一策略

研究目标：将研究扩展到多追一协同场景，引入协同地形利用和目标轨迹预测两项关键技术，实现速度劣势条件下的有效围捕。

研究动机与必要性：在非对称设定下，单个追击者在开放空间中难以保证捕获逃逸者——根据 Apollonius 圆分析，速度劣势的追击者无法覆盖整个空间^[7]。多追击者协同提供了两种弥补速度劣势的途径：

途径一：协同地形利用。单个追击者只能选择追击或等待，而多个追击者可以分工协作——部分负责驱赶，部分负责封锁关键地形位置。这种协同使地形利用的战术价值倍增^[9]。

途径二：目标轨迹预测。Zhang 等人在 Game of Drones 中提出的“先预测后决策”思想为速度劣势问题提供了另一种解决方案^[11]：通过预测目标未来轨迹，追击者可以提前规划拦截路径而非简单跟随。本研究将目标预测与地形约束相结合，设计地形感知的目标预测网络（TA-TP Net），使预测结果符合地形可通行性约束。

主要研究内容:

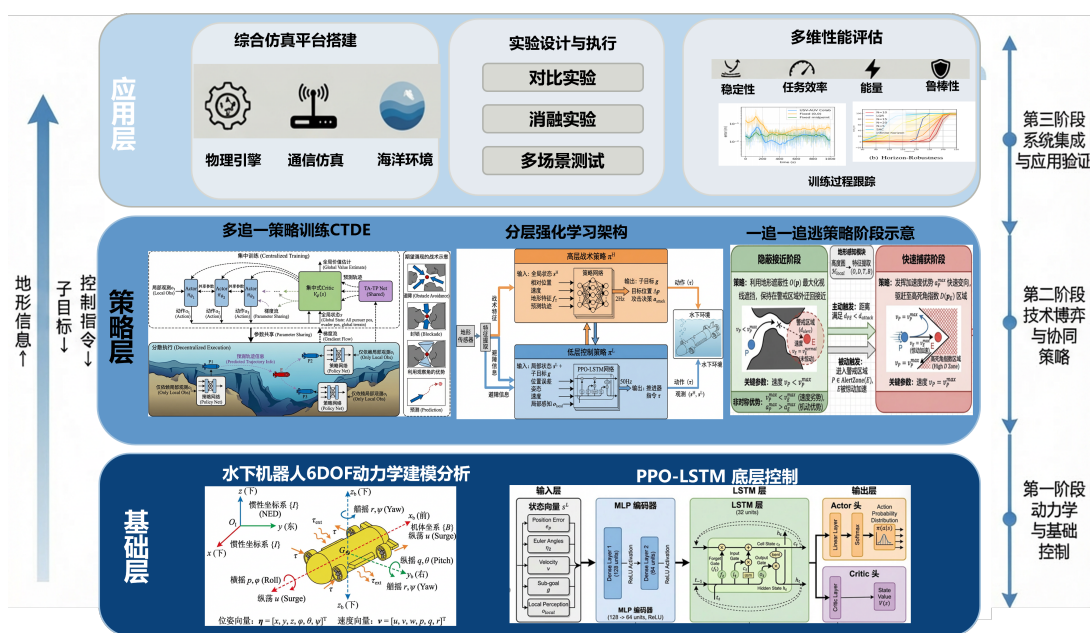
- **马尔可夫博弈建模**: 将多追一场景建模为合作博弈, 设计团队奖励函数^[8]
- **协同地形利用**: 瓶颈封锁策略、协同驱赶策略、地形封锁网构建
- **TA-TP Net 设计**: 融合地形约束的目标轨迹预测网络, 采用监督学习独立训练^[32]
- **MAPPO 算法**: 采用集中训练分散执行范式, 实现多追击者协同决策^[33]
- **战术涌现**: 通过奖励塑形促进包抄、驱赶、封锁、拦截等协同战术的涌现^[34]

不可替代性: 本阶段的创新在于将”协同地形利用”与”目标轨迹预测”两种弥补速度劣势的策略有机融合。现有研究要么单独考虑多智能体协同, 要么单独考虑目标预测, 鲜有工作将两者与地形感知结合。这种融合有望产生”1+1>2”的协同效果。

阶段成果: (1) 完整的多追一协同围捕策略; (2) TA-TP Net 目标预测模块; (3) 涌现的协同战术行为分析。

2.2 技术路线

本研究的技术路线遵循”理论建模—框架设计—底层控制—基础博弈—协同扩展—系统集成”的渐进式逻辑, 各阶段之间存在明确的依赖关系和数据流向。技术路线的设计遵循三个基本原则: (1) **自底向上**: 先解决底层控制问题, 再解决高层决策问题, 底层是高层的执行器, 其稳定性直接决定高层策略能否有效实施; (2) **由简到繁**: 从一追一开始, 逐步扩展到多追一, 地形利用能力从基础避障逐步深化到战术协同, 每一步都建立在前一步成功的基础上; (3) **模块化设计**: 各模块相对独立, 可分别开发、测试和优化, 便于问题定位和调试。



8

图 2.1 技术路线总体框架图。自底向上分为三层：底层（动力学建模 → PPO-LSTM 控制策略）、中层（一追一追逃博弈 → 多追一协同围捕）、顶层（系统集成验证）。箭头表示依赖关系和数据流向。

2.2.1 水下机器人六自由度动力学建模

水下机器人精确动力学建模是实现高性能运动控制的理论基础。本节建立完整的六自由度动力学模型，分析其非线性特性，为后续采用学习型控制方法提供理论依据。

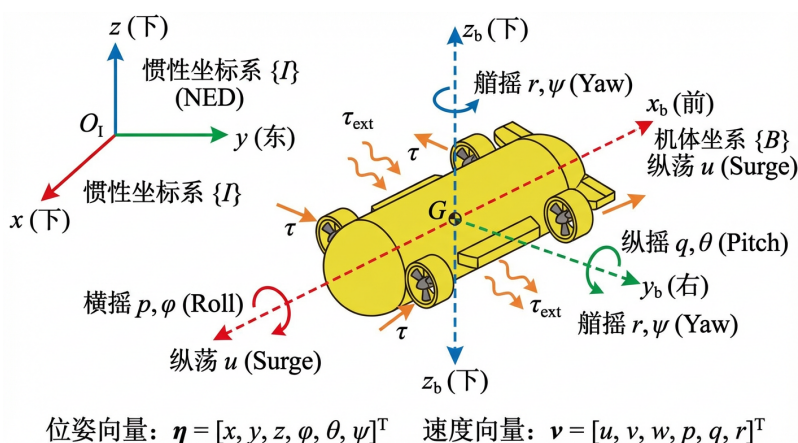


图 2.2 水下机器人六自由度动力学建模示意图

(1) 坐标系统与状态变量定义

定义惯性坐标系 $\{I\}$ 和机体坐标系 $\{B\}$ 。惯性坐标系采用北-东-地 (NED) 约定： x 轴指向北， y 轴指向东， z 轴指向地心方向。机体坐标系原点位于机器人重心， x_b 轴指向前进方向， y_b 轴指向右舷， z_b 轴指向下方。

水下机器人的完整状态由位姿向量 $\boldsymbol{\eta}$ 和速度向量 $\boldsymbol{\nu}$ 描述^[15]：

$$\boldsymbol{\eta} = \begin{bmatrix} \boldsymbol{\eta}_1 \\ \boldsymbol{\eta}_2 \end{bmatrix} = \begin{bmatrix} x & y & z & \phi & \theta & \psi \end{bmatrix}^\top \in \mathbb{R}^6 \quad (2-1)$$

其中 $\boldsymbol{\eta}_1 = [x, y, z]^\top$ 为惯性系下的位置向量， $\boldsymbol{\eta}_2 = [\phi, \theta, \psi]^\top$ 为欧拉角向量，分别表示横摇角 (roll)、纵摇角 (pitch) 和艏摇角 (yaw)。

$$\boldsymbol{\nu} = \begin{bmatrix} \boldsymbol{\nu}_1 \\ \boldsymbol{\nu}_2 \end{bmatrix} = \begin{bmatrix} u & v & w & p & q & r \end{bmatrix}^\top \in \mathbb{R}^6 \quad (2-2)$$

其中 $\boldsymbol{\nu}_1 = [u, v, w]^\top$ 分别为纵荡 (surge)、横荡 (sway)、垂荡 (heave) 速度， $\boldsymbol{\nu}_2 = [p, q, r]^\top$ 分别为横摇、纵摇、艏摇角速度。

(2) 运动学方程

机体坐标系与惯性坐标系之间的运动学关系为：

$$\dot{\boldsymbol{\eta}} = \mathbf{J}(\boldsymbol{\eta})\boldsymbol{\nu} \quad (2-3)$$

其中雅可比变换矩阵 $\mathbf{J}(\boldsymbol{\eta}) \in \mathbb{R}^{6 \times 6}$ 为分块对角矩阵：

$$\mathbf{J}(\boldsymbol{\eta}) = \begin{bmatrix} \mathbf{R}(\boldsymbol{\eta}_2) & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{T}(\boldsymbol{\eta}_2) \end{bmatrix} \quad (2-4)$$

姿态旋转矩阵 $\mathbf{R}(\boldsymbol{\eta}_2) \in SO(3)$ 采用 z - y - x 欧拉角顺序：

$$\mathbf{R}(\boldsymbol{\eta}_2) = \begin{bmatrix} c\psi c\theta & -s\psi c\theta + c\psi s\theta s\phi & s\psi s\theta + c\psi c\theta s\phi \\ s\psi c\theta & c\psi c\theta + s\psi s\theta s\phi & -c\psi s\theta + s\psi c\theta s\phi \\ -s\theta & c\theta s\phi & c\theta c\phi \end{bmatrix} \quad (2-5)$$

其中 $c(\cdot) = \cos(\cdot)$ ， $s(\cdot) = \sin(\cdot)$ 。角速度变换矩阵 $\mathbf{T}(\boldsymbol{\eta}_2)$ 将机体系角速度转换为欧拉角变化率，当 $\theta = \pm 90$ 时存在奇异性，实际应用中通过限制俯仰角范围避免此问题。

（3）六自由度动力学方程

根据 Fossen 标准海洋航行器建模框架^[15]，动力学方程为：

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau} + \boldsymbol{\tau}_{ext} \quad (2-6)$$

其中 $\mathbf{M}$ 为系统惯性矩阵， $\mathbf{C}(\boldsymbol{\nu})$ 为科氏力与向心力矩阵， $\mathbf{D}(\boldsymbol{\nu})$ 为水动力阻尼矩阵， $\mathbf{g}(\boldsymbol{\eta})$ 为恢复力矩向量， $\boldsymbol{\tau}$ 为控制力与力矩， $\boldsymbol{\tau}_{ext}$ 为外部环境干扰。

（4）系统惯性矩阵

系统惯性矩阵由刚体惯性与附加质量组成： $\mathbf{M} = \mathbf{M}_{RB} + \mathbf{M}_A$ 。附加质量矩阵 $\mathbf{M}_A$ 反映机器人加速时带动周围流体运动的等效惯性效应，这是水下机器人区别于陆地/空中机器人的本质特征^[23]。对于具有对称性的 AUV，附加质量矩阵简化为对角形式：

$$\mathbf{M}_A = -\text{diag}(X_{\dot{u}}, Y_{\dot{v}}, Z_{\dot{w}}, K_{\dot{p}}, M_{\dot{q}}, N_{\dot{r}}) \quad (2-7)$$

附加质量的建模困难包括：频率依赖性（高频振荡时数值变化显著）、边界效应（靠近海底时附加质量增加）、非稳态效应（快速机动时流场无法瞬时响应）。

（5）水动力阻尼矩阵

水动力阻尼建模为线性阻尼与非线性阻尼的叠加^[24]：

$$\mathbf{D}(\boldsymbol{\nu}) = \mathbf{D}_L + \mathbf{D}_{NL}(\boldsymbol{\nu}) \quad (2-8)$$

非线性阻尼主要源于形状阻力，与速度平方成正比：

$$\mathbf{D}_{NL}(\boldsymbol{\nu}) = -\text{diag}(X_{u|u}|u|, Y_{v|v}|v|, Z_{w|w}|w|, K_{p|p}|p|, M_{q|q}|q|, N_{r|r}|r|) \quad (2-9)$$

非线性阻尼项具有 $|v|v$ 的形式，在速度过零点处连续但不可微，且阻尼系数强烈依赖于攻角和雷诺数。

（6）非线性特性总结

综合上述分析，水下机器人动力学模型的非线性特性总结如表2.1所示。这些复杂的非线性特性使得传统基于精确模型的控制方法面临根本性困难：CFD 方法计算成本过高；经验公式适用范围有限；系统辨识对工况敏感^[24]。这些建模困难是本研究选择强化学习方法的核心动机。

表 2.1 水下机器人动力学模型非线性特性总结

非线性来源	数学表现	物理机制	建模困难
附加质量效应	M_A 非常值	流体随机器人加速	频率/边界依赖
科氏力耦合	$C(\nu)\nu \propto \ \nu\ ^2$	旋转参考系惯性力	多自由度强耦合
非线性阻尼	$D_{NL} \propto v v$	涡激/形状阻力	攻角/雷诺数依赖
环境干扰	$\nu_c(t, p)$ 时空变化	洋流/湍流	难以感知预测

2.2.2 分层强化学习框架设计

基于上述动力学分析，本节阐述选择分层强化学习作为核心方法的理由，并介绍本研究的分层框架设计。

(1) 强化学习方法的必要性

强化学习提供了一种无模型的控制范式^[35]，其核心优势在于：策略网络通过与环境的大量交互隐式学习状态到动作的映射，无需显式建立动力学模型；直接优化任务目标而非中间指标，避免建模误差向控制性能的传递；通过域随机化等技术，策略可学会对模型不确定性和环境扰动具有鲁棒性^[29]。

(2) 分层架构的动机

单一的端到端策略在本研究的追逃博弈问题中面临严峻挑战^[25]：(1) **时间尺度差异**——底层运动控制需要高频决策（如 50Hz），而高层战术决策在较长时间尺度进行（如 2Hz），单一策略难以同时学习两种时间尺度的行为；(2) **探索空间爆炸**——若直接学习从高层目标到底层推力的映射，探索空间急剧膨胀；(3) **奖励信号稀疏**——追逃的最终奖励是稀疏的，底层动作与最终奖励之间因果链很长；(4) **技能复用性**——底层运动控制技能在不同高层任务中是通用的。

(3) 本研究的分层框架

本研究采用目标条件分层强化学习框架^[36]。高层策略 π^H 在时间步 $t =$

$0, T_H, 2T_H, \dots$ 输出子目标:

$$\mathbf{g}_t = \pi^H(\mathbf{s}_t^H) \quad (2-10)$$

低层策略 π^L 在每个时间步以子目标为条件执行动作:

$$\mathbf{a}_t = \pi^L(\mathbf{s}_t^L, \mathbf{g}_t) \quad (2-11)$$

两层策略的职责划分如下：**低层控制策略**负责追踪子目标位置/速度、维持姿态稳定、避免碰撞（地形利用的第一层次），决策频率与仿真步长一致；**高层战术策略**负责选择子目标位置、判断攻击时机、协调多追击者配合、利用地形进行战术决策（地形利用的高级层次），决策频率较低（每 T_H 步决策一次）。

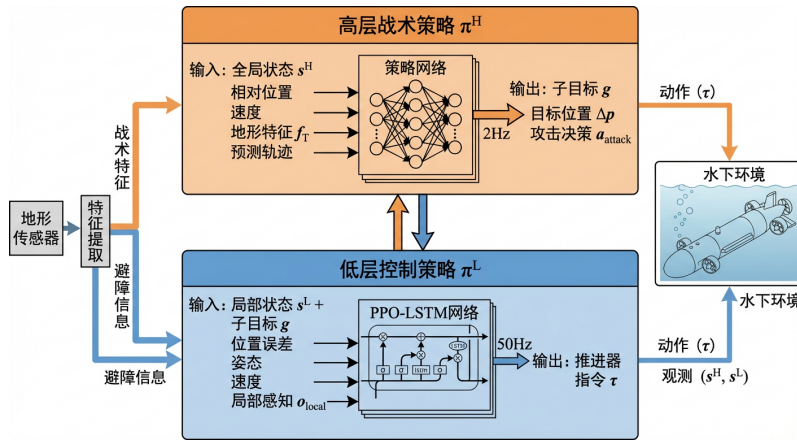


图 2.3 分层强化学习框架架构图。高层策略（战术层）以低频率输出子目标（目标位置/航向），低层策略（控制层）以高频率执行推进器控制。两层通过子目标接口解耦，分别处理战术规划和运动控制问题。地形信息分层传递：低层接收局部避障信息，高层接收战术语义特征。

（4）分阶段训练策略

本研究采用分阶段训练：**阶段一**，在简化的子目标追踪任务中预训练低层策略；**阶段二**，冻结低层参数，将其视为可靠执行器，训练高层策略；**阶段三**（可选），解冻低层进行联合微调。这种策略降低了训练复杂度，便于调试分析，且低层策略可跨任务复用。

2.2.3 基于 PPO-LSTM 的低层运动控制策略学习

低层控制策略是分层框架的基石，目标是学习稳定、鲁棒的运动控制技能，同时具备基础的地形避障能力。

(1) 低层控制问题的 MDP 建模

将低层控制建模为 MDP $\mathcal{M}^L = \langle S^L, \mathcal{A}^L, P^L, R^L, \gamma^L \rangle$ 。

状态空间 S^L 设计充分考虑六自由度维稳和局部环境感知需求：

$$\mathbf{s}^L = \begin{bmatrix} \mathbf{e}_p^\top & \boldsymbol{\eta}_2^\top & \mathbf{e}_\psi & \boldsymbol{\nu}^\top & \mathbf{g}^\top & \mathbf{o}_{local}^\top \end{bmatrix}^\top \in \mathbb{R}^{d_s} \quad (2-12)$$

其中 $\mathbf{e}_p = \mathbf{p} - \mathbf{g}_{pos}$ 为位置误差， $\boldsymbol{\eta}_2$ 为欧拉角状态（横摇 ϕ 和纵摇 θ 需保持在 $\pm 15^\circ$ 内）， $\boldsymbol{\nu}$ 为速度向量， $\mathbf{g}$ 为高层子目标， $\mathbf{o}_{local}$ 为局部环境感知信息（包含前方障碍物距离、局部地形坡度等，用于实现基础避障）。

动作空间 $\mathcal{A}^L$ 采用离散化设计，每个动作 a_k 对应一组推进器推力档位组合：

$$\mathcal{A}^L = \{a_1, a_2, \dots, a_K\}, \quad a_k \leftrightarrow (\tau_1^{(k)}, \tau_2^{(k)}, \dots, \tau_n^{(k)}) \quad (2-13)$$

选择离散动作空间的理由包括：学习稳定性更好、与执行器物理约束兼容、便于与安全层结合^[28]。

(2) 多目标奖励函数设计

奖励函数设计为多目标复合形式：

$$R^L = R_{track} + R_{stable} + R_{safe} + R_{efficiency} \quad (2-14)$$

位置追踪奖励：

$$R_{track} = -\omega_1 \|\mathbf{e}_p\|^2 - \omega_2 |\mathbf{e}_\psi|^2 + \omega_3 \cdot (\|\mathbf{e}_p^t\| - \|\mathbf{e}_p^{t+1}\|) \quad (2-15)$$

姿态稳定奖励：

$$R_{stable} = -\omega_4 (|\phi|^2 + |\theta|^2) - \omega_5 \|[p, q, r]\|^2 - \omega_6 \cdot \mathbb{I}[|\phi| > \phi_{max} \text{ or } |\theta| > \theta_{max}] \quad (2-16)$$

该奖励项确保低层控制器严格维持横摇角和纵摇角在安全范围内。

安全奖励（包含地形避障）：

$$R_{safe} = \begin{cases} -R_{collision} & \text{if collision with terrain/obstacle} \\ -\omega_7 \cdot (d_{safe} - d_{obs}) & \text{if } d_{obs} < d_{safe} \\ 0 & \text{otherwise} \end{cases} \quad (2-17)$$

其中 d_{obs} 为到最近障碍物/地形的距离。该奖励项使低层策略学会基础的地形避障能力。

能效奖励惩罚过大控制量以节省能量：

$$R_{efficiency} = -\omega_8 \sum_{i=1}^n |\tau_i| \quad (2-18)$$

(3) PPO 算法

选择近端策略优化（PPO）作为核心算法^[28]，其通过裁剪概率比限制策略更新幅度。定义概率比 $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{old}}(a_t|s_t)$ ，裁剪目标函数为：

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (2-19)$$

其中 $\hat{A}_t$ 为优势函数估计， ϵ 为裁剪参数。该机制确保每次更新不会让策略偏离太远，保证训练稳定性。

(4) LSTM 网络

在策略网络中引入 LSTM 层处理部分可观测性^[26]。水下环境中传感器噪声、通信延迟、洋流未知等因素导致状态部分可观测，LSTM 通过维护隐状态 $\mathbf{h}_t$ 从历史观测序列推断真实状态。LSTM 的门控机制包括遗忘门、输入门、输出门：

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f) \quad (2-20)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i) \odot \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c) \quad (2-21)$$

$$\mathbf{h}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o) \odot \tanh(\mathbf{c}_t) \quad (2-22)$$

网络架构采用 MLP 编码器（2 层，128-64 单元）提取特征，LSTM 层（32 单元）处理时序依赖，Actor 头输出动作概率分布，Critic 头输出状态价值。

(5) 域随机化

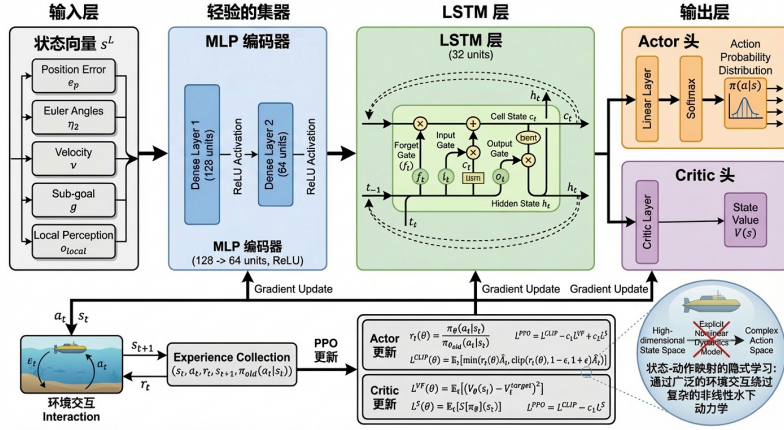


图 2.4 PPO-LSTM 低层控制网络架构图。输入状态经 MLP 编码器（128-64 单元）提取特征，送入 LSTM 层（32 单元）处理时序依赖。输出分为 Actor 头（Softmax 输出动作概率分布）和 Critic 头（线性层输出状态价值）。LSTM 隐状态在 episode 内持续传递。

为提升策略鲁棒性，训练中引入域随机化^[29]，随机化参数包括：附加质量系数 $M_A^{(i)} \sim \mathcal{N}(\bar{M}_A^{(i)}, 0.2\bar{M}_A^{(i)})$ 、阻尼系数 $D^{(i)} \sim \mathcal{N}(\bar{D}^{(i)}, 0.3\bar{D}^{(i)})$ 、推进器效率 $\eta_\tau \sim \mathcal{U}[0.85, 1.0]$ 、洋流速度 $V_c \sim \mathcal{U}[-V_{c,max}, V_{c,max}]$ 、地形高度扰动等。

2.2.4 融合地形感知的一追一非对称追逃博弈策略学习

在低层控制策略稳定收敛后，本阶段研究一追一追逃博弈的高层策略，并在此场景中逐步引入地形利用能力。

（1）非对称性设定与警戒区域机制

本研究的问题设定受自然界捕食行为启发^[30]。设定追击者与逃逸者的非对称运动学参数：

$$v_P^{max} < v_E^{max}, \quad a_P^{max} > a_E^{max} \quad (2-23)$$

即追击者最大速度劣于逃逸者，但加速度（机动能力）占优。

引入警戒区域机制：逃逸者 E 拥有半径为 d_{alert} 的感知区域，当追击者 P 在区域外时， E 以正常速度 v_E^{normal} 运动；当 P 进入警戒区域时， E 被惊动，加速至 v_E^{max} 逃离。该机制增加问题难度，迫使追击者学会利用地形遮挡视线进行隐蔽接近^[19]。

（2）地形特征提取与编码

为支持地形利用策略的学习，需要从环境中提取战术相关的地形特征。采用分层的地形特征提取方案：

底层感知：从传感器（仿真中直接获取）获得局部高度图 $\mathcal{H}_{local} = \{h_{ij}\}$ ，覆盖以当前位置为中心的局部区域。

几何特征提取：

$$\text{坡度: } \|\nabla h\| = \sqrt{(\partial h/\partial x)^2 + (\partial h/\partial y)^2} \quad (2-24)$$

$$\text{粗糙度: } \sigma_h = \sqrt{\frac{1}{|\mathcal{N}|} \sum_{(x',y') \in \mathcal{N}} (h(x',y') - \bar{h})^2} \quad (2-25)$$

战术语义特征提取：

可通行性评分^[17]：

$$T(x, y, z) = \mathbb{I}[z - h(x, y) > \delta_{safe}] \cdot \mathbb{I}[\|\nabla h\| < \theta_{max}] \cdot \mathbb{I}[\sigma_h < \sigma_{max}] \quad (2-26)$$

遮蔽性评分（量化视线遮挡能力）^[19]：

$$O(x, y, z) = \frac{1}{2\pi} \int_0^{2\pi} \mathbb{I}[\text{LOS}(x, y, z, \theta) \text{ blocked within } r_{view}] d\theta \quad (2-27)$$

死角指数（可逃逸方向的受限程度）：

$$D(x, y) = 1 - \frac{1}{2\pi} \int_0^{2\pi} \mathbb{I}[T(x + r \cos \theta, y + r \sin \theta) = 1] d\theta \quad (2-28)$$

瓶颈指数（通道宽度）：

$$B(x, y) = \min_{\theta \in [0, 2\pi)} W(x, y, \theta) \quad (2-29)$$

其中 $W(x, y, \theta)$ 为从 (x, y) 沿方向 θ 的可通行宽度。

特征编码网络：

$$\mathbf{f}_T = \text{CNN}(\mathcal{H}_{local}) \oplus \mathbf{f}_{semantic} \quad (2-30)$$

其中 CNN 提取局部高度图的空间特征， $\mathbf{f}_{semantic} = [T_{avg}, O_{max}, D_{local}, B_{min}]$ 为语义特征向量， $\oplus$ 表示拼接。

（3）分阶段追逃策略框架

将追逃过程分解为两个阶段：

阶段一：隐蔽接近——目标是在不惊动逃逸者的前提下尽可能接近；需要利用地形的遮蔽性特征 $O(x, y, z)$ ，选择在视线盲区内的移动路径；策略特点为低速、隐蔽、迂回接近、观察逃逸者运动模式；终止条件为距离小于攻击距离 d_{attack} 或被发现。

阶段二：快速捕获——触发条件为追击者主动发起攻击或被发现；需要利用地形的死角指数 $D(x, y)$ ，将逃逸者驱赶向不利地形；策略特点为高速、利用加速度优势快速变向响应；终止条件为成功捕获（距离小于 $r_{capture}$ ）或超时。

阶段切换由策略自主决定，综合考虑距离和逃逸者所处地形的有利程度。

(4) 高层策略的 MDP 建模

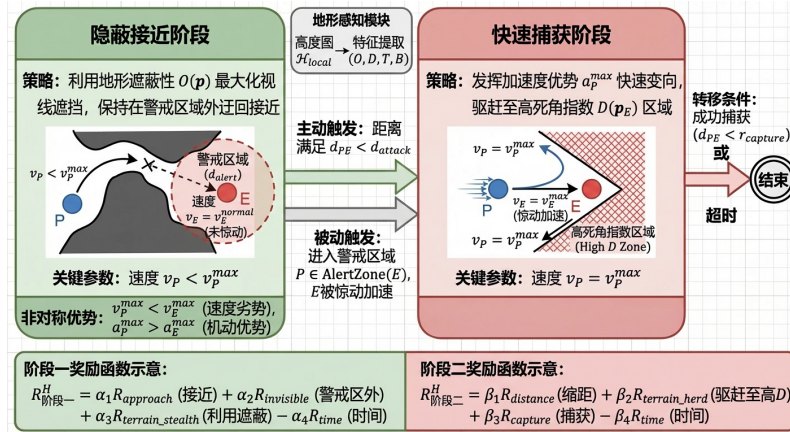


图 2.5 一追一逃过程的两阶段示意图

高层状态空间融入地形感知信息：

$$\mathbf{s}^H = \left[\mathbf{p}_{rel}^\top \quad \mathbf{v}_P^\top \quad \mathbf{v}_E^\top \quad d_{PE} \quad \theta_{PE} \quad \mathbf{f}_{vis}^\top \quad \mathbf{f}_T^\top \quad phase \right]^\top \quad (2-31)$$

其中 $\mathbf{p}_{rel}$ 为相对位置， d_{PE} 为追逃距离， $\mathbf{f}_{vis}$ 为视线遮挡特征（追击者是否在逃逸者视野内）， $\mathbf{f}_T$ 为地形战术特征， $phase$ 为当前阶段。

高层动作空间设计：

$$\mathbf{a}^H = (\Delta \mathbf{p}_{target}, a_{attack}) \quad (2-32)$$

其中 $\Delta \mathbf{p}_{target} \in \mathbb{R}^3$ 为传递给低层的目标位置增量， $a_{attack} \in \{0, 1\}$ 为是否发起攻击的决策。

（5）融合地形感知的奖励函数设计

高层奖励函数针对两阶段设计，并融入地形利用激励：

$$R^H = \begin{cases} \alpha_1 R_{approach} + \alpha_2 R_{invisible} + \alpha_3 R_{terrain_stealth} - \alpha_4 R_{time} & \text{阶段一} \\ \beta_1 R_{distance} + \beta_2 R_{terrain_herd} + \beta_3 R_{capture} - \beta_4 R_{time} & \text{阶段二} \end{cases} \quad (2-33)$$

隐蔽接近阶段的接近奖励：

$$R_{approach} = (d_{PE}^t - d_{PE}^{t+1}) + \lambda \cdot \mathbb{I}[d_{PE}^{t+1} < d_{threshold}] \quad (2-34)$$

隐蔽奖励激励在不惊动目标的前提下接近：

$$R_{invisible} = \begin{cases} +r_{inv} & \text{if } P \notin \text{AlertZone}(E) \text{ and } d_{PE}^{t+1} < d_{PE}^t \\ -r_{alert} & \text{if } P \in \text{AlertZone}(E) \text{ and } E \text{ accelerates} \end{cases} \quad (2-35)$$

地形隐蔽奖励激励利用遮蔽性高的位置：

$$R_{terrain_stealth} = \omega_S \cdot O(\mathbf{p}_P) \cdot (d_{PE}^t - d_{PE}^{t+1}) \cdot \mathbb{I}[E \text{ not alerted}] \quad (2-36)$$

地形驱赶奖励激励将逃逸者驱赶向不利地形：

$$R_{terrain_herd} = \omega_T \cdot (D(\mathbf{p}_E^{t+1}) - D(\mathbf{p}_E^t)) \quad (2-37)$$

捕获奖励：

$$R_{capture} = \begin{cases} +R_{success} & \text{if } d_{PE} \leq r_{capture} \\ -R_{timeout} & \text{if } t > T_{max} \end{cases} \quad (2-38)$$

（6）逃逸者策略库与课程学习

设计多种规则驱动的逃逸策略，逐步增加智能程度^[31]：静止目标、随机游走、纯贪婪逃离（ $\mathbf{v}_E = v_E^{max} \cdot (\mathbf{p}_E - \mathbf{p}_P) / \|\mathbf{p}_E - \mathbf{p}_P\|$ ）、边界感知逃离、地形感知逃离（主动利用地形掩护）。

训练采用课程学习：Level 1 静止目标（开放空间），Level 2 慢速随机游走（开放空间），Level 3 随机游走加警戒（简单障碍物），Level 4 纯贪婪逃离加警戒（复杂地形），Level 5 地形感知逃离（复杂地形）。通过渐进式课程，策略逐步掌握从基础追踪到地形利用的完整能力。

2.2.5 融合地形利用与目标预测的多追一协同追逃策略学习

在一追一策略成熟后，研究扩展到多追一协同场景，并引入目标轨迹预测机制以进一步弥补速度劣势。

(1) 多追击者协同的必要性

在非对称设定下，单个追击者在开放空间中难以保证捕获逃逸者^[7]。根据 Apollonius 圆分析，当 $v_P < v_E$ 时，逃逸者只需保持在追击者主导区域外即可永远不被捕获。多追击者协同的价值在于：空间封锁限制逃逸方向、区域收缩压缩安全空间、协同地形利用形成封锁网、战术配合实现包抄驱赶诱捕^[9]。

(2) 地形感知的目标预测网络 (TA-TP Net)

在速度劣势条件下，“先预测后决策”是一种有效的补偿策略^[11]。本研究借鉴 Zhang 等人提出的目标预测网络思想，设计地形感知的目标预测网络 (Terrain-Aware Target Prediction Network, TA-TP Net)。

网络架构：采用基于 LSTM 的序列到序列结构^[32]：

$$\hat{\mathbf{p}}_E^{t+1:t+k} = \text{TA-TP Net}(\mathbf{p}_E^{t-n+1:t}, \{\mathbf{p}_{P_i}^{t-n+1:t}\}, \mathbf{v}_E^{t-n+1:t}, \mathcal{H}_{local}^E) \quad (2-39)$$

其中输入包括：逃逸者历史位置序列、所有追击者历史位置序列、逃逸者历史速度、以及逃逸者周围的局部地形高度图 $\mathcal{H}_{local}^E$ 。输出为未来 k 步的预测轨迹 $\hat{\mathbf{p}}_E^{t+1:t+k}$ 。

地形约束的融入：与原始 TP Net 不同，本研究的 TA-TP Net 显式考虑地形对目标运动的约束：

- **可通行性掩码：**预测结果通过可通行性函数 $T(x, y, z)$ 过滤，确保预测轨迹不穿越不可通行区域
- **地形特征输入：**局部地形高度图作为额外输入，使网络学会目标在地形约束下的运动模式
- **概率化输出：**可扩展为输出多条可能轨迹及其概率，反映逃逸者决策的不确定性

训练方式：TA-TP Net 采用监督学习独立训练^[11]。在强化学习探索过程中，维护一个目标轨迹回放缓冲区 $\mathcal{T}$ ，存储 $(,)$ 对。训练损失为：

$$\mathcal{L}_{TP} = \frac{1}{k} \sum_{i=1}^k \|\hat{\mathbf{p}}_E^{t+i} - \mathbf{p}_E^{t+i}\|^2 \quad (2-40)$$

(3) 马尔可夫博弈建模

将多追一场景建模为马尔可夫博弈^[8]：

$$\mathcal{G} = \langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}_{i \in \mathcal{N}}, P, \{R_i\}_{i \in \mathcal{N}}, \gamma \rangle \quad (2-41)$$

其中 $\mathcal{N} = \{P_1, P_2, \dots, P_N, E\}$ 为参与者集合。追击者团队共享奖励（合作博弈）。

对于每个追击者 i ，其局部观测融入地形特征和目标预测：

$$\mathbf{o}_i = \left[\mathbf{p}_{P_i} - \mathbf{p}_E \quad \mathbf{v}_{P_i} \quad \mathbf{v}_E \quad \{\mathbf{p}_{P_j} - \mathbf{p}_{P_i}\}_{j \neq i} \quad \mathbf{f}_{T,i} \quad \hat{\mathbf{p}}_E^{t+1:t+k} \right]^\top \quad (2-42)$$

包含：相对于逃逸者的位置、自身速度、逃逸者速度、队友相对位置、局部地形特征、以及目标预测轨迹。

(4) 融合地形与预测的协同奖励函数

团队奖励设计综合考虑协同接近、地形利用和预测拦截：

$$R_{team} = R_{min_dist} + R_{encircle} + R_{terrain_block} + R_{predict_intercept} + R_{capture} + R_{collision} \quad (2-43)$$

最近距离奖励（激励整体接近目标）：

$$R_{min_dist} = \omega_1 \cdot \left(\min_i d_{i,E}^t - \min_i d_{i,E}^{t+1} \right) \quad (2-44)$$

围捕奖励（激励形成包围圈）^[34]：

$$R_{encircle} = \omega_2 \cdot \text{AngularCoverage}(\{P_i\}, E) = \omega_2 \cdot \frac{1}{2\pi} \sum_{i=1}^N \Delta\theta_i \quad (2-45)$$

其中 $\Delta\theta_i$ 为相邻追击者从逃逸者视角的张角，覆盖度接近 1 表示完全包围。

地形封锁奖励（激励占据关键地形位置）：

$$R_{terrain_block} = \omega_3 \cdot \sum_i \mathbb{I}[\mathbf{p}_{P_i} \in \mathcal{B}_{critical}] \cdot \mathbb{I}[d(\mathbf{p}_{P_i}, \mathbf{p}_E) < d_{relevant}] \quad (2-46)$$

其中 $\mathcal{B}_{critical}$ 为瓶颈指数 $B(x, y)$ 低于阈值的关键封锁位置集合。

预测拦截奖励（激励基于预测的提前布局）：

$$R_{predict_intercept} = \omega_4 \cdot \sum_i \mathbb{I}[d(\mathbf{p}_{P_i}^{t+1}, \hat{\mathbf{p}}_E^{t+k}) < d(\mathbf{p}_{P_i}^t, \mathbf{p}_E^t)] \quad (2-47)$$

当追击者向预测的未来位置移动（而非简单追向当前位置）时获得奖励，鼓励“提前拦截”行为。

分散奖励（防止聚集）：

$$R_{spread} = -\omega_5 \cdot \mathbb{I}[\min_{i \neq j} \|\mathbf{p}_{P_i} - \mathbf{p}_{P_j}\| < d_{min}] \quad (2-48)$$

（5）MAPPO 算法与 CTDE 范式

采用集中训练、分散执行（CTDE）范式下的多智能体 PPO（MAPPO）算法^[33]。关键设计包括：

- **共享策略网络**：所有追击者使用同一策略网络 π_θ ，输入为局部观测 $\mathbf{o}_i$ （包含地形特征和目标预测）
- **集中式价值网络**：价值函数 $V_\phi(\mathbf{s})$ 以全局状态为输入，包含所有追击者和逃逸者的完整信息以及全局地形特征
- **TA-TP Net 共享**：所有追击者共享同一个目标预测网络，输出统一的预测轨迹供各追击者决策使用
- **独立优势估计**：各追击者使用自己的优势函数估计 $\hat{A}_i$ 进行策略更新

通过适当的奖励设计，期望涌现出以下协同战术^[9]：包抄（多方向接近形成包围）、驱赶（迫使目标向不利地形移动）、封锁（占据瓶颈切断逃逸路线）、拦截（基于预测提前移动到拦截点）。

（6）分层信息流设计

地形信息和预测信息在分层框架中的传递遵循关注点分离原则：

- **低层接收**：局部避障信息 $\mathbf{o}_{local}$ （前方障碍物距离、可通行性、局部坡度），专注于“如何安全到达目标位置”

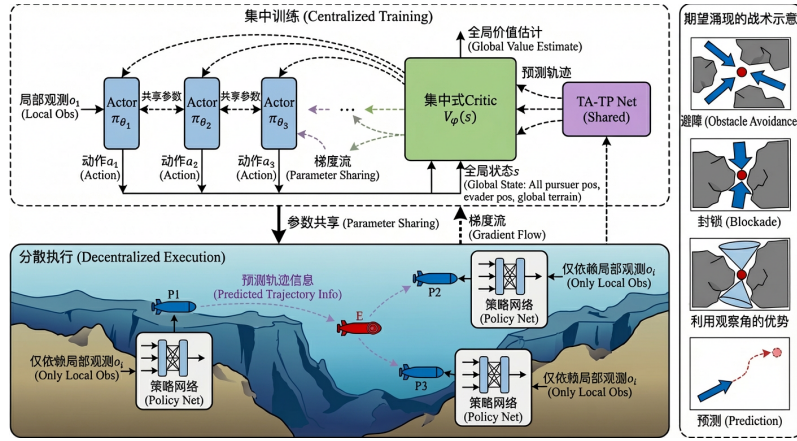


图 2.6 多追一协同追逃策略的 CTDE 网络架构图

- **高层接收：**地形战术特征 f_T （死角分布、瓶颈位置、遮蔽性）和目标预测 $\hat{p}_E^{t+1:t+k}$ ，专注于”选择什么目标位置能最好地利用地形和预测优势”

这种设计实现了信息的层次化利用：底层处理短时域的安全导航问题，高层处理长时域的战术规划问题，两者通过子目标接口有机协同。

2.3 可行性分析

本研究的可行性从理论基础、技术条件、研究基础三个维度进行分析。

2.3.1 理论可行性

分层强化学习的成熟性：分层强化学习已在复杂控制任务中得到广泛验证^[25,36]。Option 框架、Feudal Networks、HIRO 等方法为时间抽象和目标条件策略提供了坚实的理论基础。本研究采用的高低层分离架构在机器人导航、操作等领域已有成功应用先例。

多智能体强化学习的理论支撑：MAPPO 等 CTDE 算法在 StarCraft 多智能体挑战（SMAC）等基准测试中展现出优异的协同学习能力^[33]。共享参数机制和集中式价值函数的设计为多追击者协同提供了可靠的算法框架。追逃博弈作为经典的多智能体问题，其马尔可夫博弈建模方法已有完善的理论体系^[5]。

目标预测方法的有效性：基于 LSTM 的轨迹预测方法在多个领域（行人轨

迹预测、车辆轨迹预测)已得到验证^[32]。Zhang 等人的工作证明了将目标预测与强化学习策略结合的有效性^[11],为本研究的 TA-TP Net 设计提供了直接参考。

地形表示与处理的理论基础: 地形特征提取方法在地形分析领域已有成熟理论^[17]。卷积神经网络处理空间结构数据的能力已在图像、点云等领域得到充分验证,将其应用于高度图编码具有理论合理性。视线约束对追逃博弈均衡结构的影响已有理论分析^[19]。

2.3.2 技术可行性

仿真平台的成熟性: Unity 引擎结合 ML-Agents 工具包为强化学习研究提供了高效的仿真训练环境,支持复杂物理仿真、并行化训练和灵活的环境配置^[37]。

算法实现的可行性: PPO、LSTM、CNN 等核心算法组件在主流深度学习框架中均有成熟实现。ML-Agents 原生支持 PPO 和 LSTM 网络,MAPPO 可在此基础上扩展实现。目标预测网络采用标准的序列到序列架构,实现难度适中。

渐进式研究设计降低风险: 技术路线采用由简到繁的渐进设计,每个阶段都建立在前一阶段成功的基础上。若某阶段遇到困难,可通过简化问题设定、调整算法参数或采用替代方案解决。

2.3.3 研究基础

相关领域研究积累: 研究团队在强化学习、多智能体系统、机器人控制等领域具有研究基础,熟悉 PPO、MARL 等核心算法的原理和实现。前期文献调研为本研究提供了清晰的问题定义和方法选择依据。

开发工具链的掌握: 研究者具备 Unity 开发和 ML-Agents 使用经验,能够快速搭建仿真环境和训练流程。Python、PyTorch 等工具的熟练使用为算法实现和实验分析提供保障。

综上所述,本研究在理论、技术和研究基础方面均具备可行性,研究目标的实现有充分保障。

3 研究计划进度安排及预期目标

3.1 进度安排

本研究计划从 2024 年 12 月底启动，至 2025 年 5 月中旬完成，总周期约 20 周。按照“理论建模 → 底层控制 → 基础博弈 → 协同扩展 → 地形融合 → 论文撰写”的顺序展开，具体进度安排如表 2.2 所示。

各阶段之间存在递进依赖关系：阶段 1 完成理论基础和实验平台搭建，为后续所有算法开发提供仿真环境；阶段 2 训练的低层控制策略是阶段 3-5 高层博弈策略的执行基础；阶段 3 的一追一策略为阶段 4 多追一协同提供行为原语和经验参考；阶段 4 的协同框架为阶段 5 地形融合提供决策架构；阶段 6 完成论文撰写和答辩准备。为应对可能的技术风险，各阶段采用迭代式开发策略，确保各模块可独立测试验证，即使后期阶段遇到困难，前期成果也具有独立价值。

3.2 预期目标

3.2.1 理论与方法目标

- 提出融合地形感知的分层多智能体强化学习框架，实现“底层稳定控制—高层博弈决策—地形战术融合”的有机结合
- 设计适用于非对称追逃博弈的奖励体系，有效引导隐蔽接近、时机选择、协同围捕等智能行为的涌现
- 建立地形特征与追逃博弈战术价值的映射关系，实现从“地形避障”到“地形利用”的范式转变

3.2.2 系统实现目标

- 实现基于深度强化学习的水下机器人低层运动控制策略，满足位置追踪与姿态稳定的基本要求

- 实现基于多智能体强化学习的协同围捕策略，使多追击者能够有效捕获具有速度优势的智能逃逸者
- 开发地形特征提取与编码模块，支持可通行性、遮蔽性、瓶颈区域等战术特征的表示
- 构建完整的水下追逃仿真平台，支持多种地形场景和追逃配置的实验验证

3.2.3 验证与成果目标

- 通过对比实验验证地形感知模块对协同围捕性能的提升效果
- 通过消融实验分析分层架构、协同奖励设计、地形感知等各组成部分的贡献
- 完成毕业论文一篇，形成可复用的仿真平台与训练框架代码库

表 2.2 研究进度安排

序号	时间节点	主要任务
1	2025.01.15 前	完成文献调研与理论学习；建立水下机器人六自由度动力学模型并分析非线性特性；基于 Unity ML-Agents 搭建水下追逃仿真平台，实现机器人动力学仿真、场景渲染、强化学习训练接口
2	2025.02.05 前	设计 PPO-LSTM 网络架构与离散动作空间；设计多目标奖励函数（位置追踪、姿态稳定、避障安全、能量效率）；完成低层控制策略训练与域随机化实现；验证追踪精度与姿态稳定性指标
3	2025.03.05 前	设定一追一非对称场景参数，实现多种逃逸者策略；设计分阶段追击框架（隐蔽接近 + 快速捕获）与对应奖励函数；采用课程学习训练追击策略；验证策略性能并与基线方法对比
4	2025.04.05 前	扩展仿真环境支持多追击者（3-5 个）；实现 MAPPO 算法与参数共享机制；设计团队奖励函数（最近距离、围捕覆盖、分散度等）；训练协同策略并分析涌现的协同战术；进行消融实验
5	2025.05.01 前	设计多种典型水下地形场景；实现地形特征提取模块（几何特征、语义特征、CNN 编码）；设计地形相关奖励并训练地形感知增强策略；整合全系统并进行端到端测试与性能评估
6	2025.05.15 前	整理实验数据与结果；撰写毕业论文各章节；完成论文格式校对与修订；准备答辩 PPT 与演示材料

4 参考文献

- [1] Sahoo A, Dwivedy S K, Robi P S. Advancements in the field of autonomous underwater vehicle[J]. *Ocean Engineering*, 2019, 181: 145-160. DOI: 10.1016/j.oceaneng.2019.04.011.
- [2] 俞建成, 张艾群, 金文明, 等. 水下滑翔机技术发展现状与展望[J]. *中国造船*, 2018, 59(3): 206-216.
- [3] Cao Y, Yu W, Ren W, et al. An overview of recent progress in the study of distributed multi-agent coordination[J]. *IEEE Transactions on Industrial Informatics*, 2013, 9(1): 427-438. DOI: 10.1109/TII.2012.2219061.
- [4] Hadi B, Khosravi A, Sarhadi P. A review of the path planning and formation control for multiple autonomous underwater vehicles[J]. *Journal of Intelligent and Robotic Systems*, 2021, 101(67). DOI: 10.1007/s10846-021-01330-4.
- [5] Chung T H, Hollinger G A, Isler V. Search and pursuit-evasion in mobile robotics: A survey [J]. *Autonomous Robots*, 2011, 31(4): 299-316. DOI: 10.1007/s10514-011-9241-4.
- [6] Weintraub I E, Pachter M, Garcia E. An introduction to pursuit-evasion differential games [J]. *American Control Conference (ACC)*, 2020: 1675-1692. DOI: 10.23919/ACC45564.2020.9147205.
- [7] Huang H, Zhang W, Ding J, et al. Guaranteed decentralized pursuit-evasion in the plane with multiple pursuers[C]//2011 50th IEEE Conference on Decision and Control and European Control Conference. IEEE, 2011: 4835-4840. DOI: 10.1109/CDC.2011.6161237.
- [8] Nguyen T T, Nguyen N D, Nahavandi S. Deep Reinforcement Learning for Multiagent Systems: A Review of Challenges, Solutions, and Applications[J]. *IEEE Transactions on Cybernetics*, 2020, 50(9): 3826-3839. DOI: 10.1109/TCYB.2020.2977374.
- [9] De Souza C, Newbury R, Cosgun A, et al. Decentralized multi-agent pursuit using deep reinforcement learning[J]. *IEEE Robotics and Automation Letters*, 2021, 6(3): 4552-4559. DOI: 10.1109/LRA.2021.3068952.
- [10] 王越, 董璐, 孙长银. 基于强化学习的多智能体追逃博弈协同控制[J]. *计算机学报*, 2020, 43(8): 1455-1469.
- [11] Zhang R, Zong Q, Zhang X, et al. Game of drones: Multi-UAV pursuit-evasion game with online motion planning by deep reinforcement learning[J]. *IEEE Transactions on Neural Networks and Learning Systems*, 2022, 34(10): 7900-7909. DOI: 10.1109/TNNLS.2022.3146976.
- [12] Tang C, Abbatematteo B, Hu J, et al. Deep Reinforcement Learning for Robotics: A Survey of Real-World Successes[J]. *Annual Review of Control, Robotics, and Autonomous Systems*, 2025, 8: 153-188. DOI: 10.1146/annurev-control-030323-022510.
- [13] Akyildiz I F, Pompili D, Melodia T. Underwater acoustic sensor networks: Research challenges[J]. *Ad Hoc Networks*, 2005, 3(3): 257-279. DOI: 10.1016/j.adhoc.2005.01.004.
- [14] 严卫生, 高剑, 李鹏. 水下自主航行器组合导航技术研究进展[J]. *中国惯性技术学报*, 2017, 25(3): 315-322.
- [15] Fossen T I. *Handbook of Marine Craft Hydrodynamics and Motion Control*[M]. John Wiley & Sons, 2011. DOI: 10.1002/9781119994138.
- [16] Chen M, Zhou Z, Tomlin C J. Multiplayer reach-avoid games via pairwise outcomes[J]. *IEEE Transactions on Automatic Control*, 2017, 62(3): 1451-1457. DOI: 10.1109/TAC.2016.2577619.
- [17] Papadakis P. Terrain traversability analysis methods for unmanned ground vehicles: A survey [J]. *Engineering Applications of Artificial Intelligence*, 2013, 26(4): 1373-1385. DOI: 10.1016/j.engappai.2013.01.006.
- [18] Zeng Z, Lian L, Sammut K, et al. A survey on path planning for persistent autonomy of autonomous underwater vehicles[J]. *Ocean Engineering*, 2015, 110: 303-313. DOI: 10.1016/j.oceaneng.2015.10.007.
- [19] Bhattacharya S, Hutchinson S. On the existence of nash equilibrium for a two-player pursuit-

- evasion game with visibility constraints[J]. International Journal of Robotics Research, 2010, 29(7): 831-839. DOI: 10.1177/0278364909357413.
- [20] Carlucho I, De Paula M, Wang S, et al. AUV Position Tracking Control Using End-to-End Deep Reinforcement Learning[J]. IEEE Transactions on Robotics, 2023, 39(3): 2376-2393. DOI: 10.1109/TRO.2023.3257541.
- [21] 徐玉如, 庞永杰, 甘永. 智能水下机器人技术展望[J]. 智能系统学报, 2006, 1(1): 9-16.
- [22] Hornung A, Wurm K M, Bennewitz M, et al. OctoMap: An efficient probabilistic 3D mapping framework based on octrees[J]. Autonomous Robots, 2013, 34(3): 189-206. DOI: 10.1007/s10514-012-9321-0.
- [23] Phillips A, Furlong M, Turnock S R. The use of computational fluid dynamics to assess the hull resistance of concept autonomous underwater vehicles[J]. Ocean Engineering, 2007, 34(5-6): 727-746. DOI: 10.1016/j.oceaneng.2006.04.013.
- [24] Shen C, Shi Y, Buckham B. Trajectory tracking control of an autonomous underwater vehicle using Lyapunov-based model predictive control[J]. IEEE Transactions on Industrial Electronics, 2018, 65(7): 5796-5805. DOI: 10.1109/TIE.2017.2779442.
- [25] Pateria S, Subagdja B, Tan A H, et al. Hierarchical reinforcement learning: A comprehensive survey[J]. ACM Computing Surveys, 2021, 54(5): 1-35. DOI: 10.1145/3453160.
- [26] Hochreiter S, Schmidhuber J. Long short-term memory[J]. Neural Computation, 1997, 9(8): 1735-1780. DOI: 10.1162/neco.1997.9.8.1735.
- [27] Hachicha M, et al. Hierarchical Deep Reinforcement Learning for Amphibious Robot Control [C]//arXiv preprint arXiv:2410.18969. 2024. DOI: 10.48550/arXiv.2410.18969.
- [28] Schulman J, Wolski F, Dhariwal P, et al. Proximal policy optimization algorithms[C]//arXiv preprint arXiv:1707.06347. 2017.
- [29] Tobin J, Fong R, Ray A, et al. Domain randomization for transferring deep neural networks from simulation to the real world[C]//2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2017: 23-30. DOI: 10.1109/IROS.2017.8202133.
- [30] Wilson A M, Lowe J C, Roskilly K, et al. Locomotion dynamics of hunting in wild cheetahs [J]. Nature, 2013, 498(7453): 185-189. DOI: 10.1038/nature12295.
- [31] Narvekar S, Peng B, Leonetti M, et al. Curriculum learning for reinforcement learning domains: A framework and survey[J]. Journal of Machine Learning Research, 2020, 21(1): 1-50.
- [32] Alahi A, Goel K, Ramanathan V, et al. Social LSTM: Human trajectory prediction in crowded spaces[C]//Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2016: 961-971.
- [33] Yu C, Velu A, Vinitsky E, et al. The surprising effectiveness of PPO in cooperative, multi-agent games[C]//Advances in Neural Information Processing Systems (NeurIPS). 2021: 24611-24624.
- [34] Ma J, Lu H, Xiao J, et al. Multi-robot target encirclement control with collision avoidance via deep reinforcement learning[J]. Journal of Intelligent and Robotic Systems, 2020, 99(2): 371-386. DOI: 10.1007/s10846-019-01106-x.
- [35] Kober J, Bagnell J A, Peters J. Reinforcement learning in robotics: A survey[J]. International Journal of Robotics Research, 2013, 32(11): 1238-1274. DOI: 10.1177/0278364913495721.
- [36] Nachum O, Gu S S, Lee H, et al. Data-efficient hierarchical reinforcement learning[C]//Advances in Neural Information Processing Systems: vol. 31. 2018: 3303-3313.
- [37] Juliani A, Berges V P, Teng E, et al. Unity: A general platform for intelligent agents[J]. arXiv preprint arXiv:1809.02627, 2018.

三、外文翻译

译文题目：多障碍环境下围捕控制的策略引导强化学习方法

原文题目： **A Policy-Guided Reinforcement Learning
Method for Encirclement Control in Multiobstacle
Environment**

原稿出处： M.Han,L.Zhang,J.Wang,andW.Pan,IEEE Robotics and Automation
Letters,2020,Vol.5,No.4,pp.6217–6224.

摘要

多智能体围捕与多障碍避障 (EMOCA) 问题一直具有挑战性, 因为在同时围捕移动目标和规避障碍物之间难以取得平衡。为了解决 EMOCA 问题, 我们提出了一种新型策略引导强化学习 (RL) 方法, 即多调节器辅助围捕强化学习 (MRA-RLEC)。该方法利用预启动学习 (Jump-start Learning) 和课程学习 (CL) 机制来提高训练效率。MRA-RLEC 将复杂的围捕任务分解为一系列难度逐渐增加的子任务。在此过程中, 利用多个调节器来调整各种训练维度, 包括围捕条件、障碍物规避以及从引导策略到自主学习策略的过渡。此外, 本文提出了一种全局围捕奖励分解 (GERD) 方法以缓解奖励稀疏问题, 并设计了一种双向通信协议以减少通信负荷。通过大量实验展示了该方法的鲁棒性和优越性, 并在基于机器人操作系统 2 (ROS2) 的 Gazebo 仿真平台中, 利用自主设计的全向移动机器人模型验证了 MRA-RLEC 的实际应用性。

关键词: 课程学习 (CL), 围捕控制, 多智能体强化学习 (MARL), 多障碍避障。

1 引言

多智能体系统 (Multi-Agent System) 因其在解决复杂挑战方面优于单个智能体的能力而成为研究焦点。它们具有广泛的应用, 包括搜索与救援^[1]、编队控制^[2]和追逃博弈 (Pursuit-Evasion Game)^[3]。其中, 围捕控制 (Encirclement Control) 因其在民用和军事领域的适用性而受到广泛关注。围捕控制旨在以预定义的队形包围逃避目标, 军事应用侧重于使用无人机 (UAV) 或无人水面艇 (USV) 进行目标捕获^[4-5], 以及监视^[6]和护送高价值目标^[7]。在民用领域, 它有助于野生动物监测^[8]和保护地面车辆^[9]。随着围捕环境日益复杂, 在密林或礁石水域等复杂背景下, 防止与障碍物碰撞对于智能体的安全至关重要^[10]。由于任务复杂性, 多智能体围捕与多障碍物避碰问题提出了巨大挑战。

目前的围捕控制方法大致分为传统方法和基于强化学习的方法。传统方法包括基于控制理论的方法和基于优化的方法。在基于控制理论的方法^[11-13]中, 特

定的围捕点被分配给追捕者（Pursuer），并设计控制器使追捕者与其目标点之间的距离趋于零，从而实现围捕。然而，该方法依赖于智能体运动学且可能耗时。对于基于优化的方法，围捕过程被建模为最优控制问题，其中模型预测控制（Model Predictive Control, MPC）被用于最小化控制能量和围捕误差^[4,14-15]。尽管其有效，但模型预测控制的计算需求巨大，可能阻碍实时实现。

随着人工智能的进步，深度强化学习（Deep Reinforcement Learning, DRL）已被广泛应用于复杂的多智能体任务^[16]。在围捕控制中，许多研究提出了基于 DRL 的框架来解决这一挑战^[17-19]。Ma 等人^[17]提出了一种基于多智能体深度确定性策略梯度（Multi-Agent Deep Deterministic Policy Gradient, MADDPG）的策略，采用使用局部信息的分布式控制架构。然而，它通过在围捕移动目标时移除障碍物并在有障碍环境中假设静态目标来简化场景。Zhang 等人^[18-19]提出了图注意力网络（Graph Attention Network）来建模实体之间的空间关系，能够适应不同数量的机器人和障碍物，但其实验仅限于少量障碍物。其他工作，如文献^[20]和^[21]，侧重于有界区域内的多智能体包围问题，其中不考虑障碍物，且终止条件可以借助环境边界轻松实现。

为解决多障碍物环境中的避碰问题，现有方法包括人工势场（Artificial Potential Field, APF）、Voronoi 分区、离线规划和感知-避障技术^[2]。其中，APF 因其固有的对称性和简洁的表述而被广泛用于围捕控制^[22-23]。基于 Voronoi 分区的方法也解决追逃和避碰问题^[24-25]。具体而言，Wang 等人^[25]通过构建以逃脱者为中心的有界 Voronoi 单元（Evader-Centered Bounded Voronoi Cell, ECBVC）提出了一种围捕算法，在具有杂乱障碍物的无界环境中协作追捕对抗性逃脱者。然而，Voronoi 分区和基于 APF 的方法都可能遇到局部极小值、控制振荡和依赖全面环境信息等问题，使其在密集障碍区域效果较差。离线规划方法需要预计算轨迹且计算代价高昂，特别是在目标不断移动的动态围捕中^[26]。感知-避障技术在围捕中较少使用，因为其前瞻性有限且状态预测不可靠^[27]。

在基于 DRL 的围捕方法中，大多数算法仅在障碍物较少时表现良好，原因有二。一方面，DRL 的避碰机制是在发生碰撞时分配负奖励，而大量障碍物的存在可能导致长期的负反馈，进而导致训练发散。另一方面，终止阶段碰撞产生

的负奖励可能抵消成功围捕的正奖励，使终止状态的学习变得复杂。因此，研究通常在不同阶段分别处理障碍物避让和围捕。从上述文献可以看出，尽管相当多的研究取得了令人满意的围捕结果，但在围捕移动目标和避开多个障碍物之间取得平衡仍然是一个挑战。

幸运的是，结合传统方法和 DRL 方法的创新框架已经出现，以解决多智能体任务中的环境复杂性。Sui 等人率先将模仿学习（Imitation Learning）与 DRL 相结合用于编队控制^[2]，在模仿阶段教授基本的编队控制，并在更复杂的场景中用强化学习进行优化。后续研究提出了一种结合基于模型范式的 DRL 方法^[28]，分别以并行和串行结构进行融合。这些方法表明了解决移动目标围捕和障碍物避让复杂性的潜力。

受上述讨论的启发，我们提出了一种新颖的方案，将基于模型的方法和基于 DRL 的方法与跳跃启动强化学习（Jump-Start Reinforcement Learning, JSRL）^[29]的概念相结合。我们设计了一种面向引导策略（Guide Policy）的技术，配合多个课程学习调节器来解决 EMOCA 的挑战。具体而言，引入了一个基于模型的引导策略来辅助训练，这显著缓解了奖励稀疏问题，并应用于各种多智能体强化学习（Multi-Agent Reinforcement Learning, MARL）算法。此外，我们提出了一个能够适应更复杂环境的两阶段框架。我们还改进了奖励函数和通信协议的设计。本文的主要贡献如下：

1. 提出了一种新颖的用于围捕控制的多调节器辅助强化学习（MRA-RLEC）方案，以解决无界多障碍物环境中移动目标的围捕问题。通过多个调节器和引导策略提高了训练效率和收敛性。
2. 提出了一种全局围捕奖励分解（GERD）方法来缓解奖励稀疏问题，鼓励实现终止围捕条件。此外，设计了一种双向通信协议以减少通信负载。
3. 进行了对比实验以证明该方法的可扩展性和鲁棒性。此外，基于自主设计的全向车辆平台，在 Gazebo 中使用 ROS2 对模型进行测试，以验证实际应用的可行性。

本文其余部分组织如下：第二节给出围捕场景的问题描述；第三节介绍预备知识；第四节详细阐述 MRA-RLEC 方法的设计过程，包括引导策略、奖励函数和网络结构；第五节在不同设置下进行仿真和实验；最后，第六节总结全文。

2 问题描述

2.1 围捕控制场景

围捕控制问题在一个包含 M 个静态障碍物的二维无界空间 $\mathcal{W} \subseteq \mathbb{R}^2$ 中考虑，其中 N 个追捕者需要在避免碰撞的同时围捕一个移动目标。追捕者和障碍物的索引集分别表示为 $\mathcal{I} = \{1, 2, \dots, N\}$ 和 $\mathcal{O} = \{1, 2, \dots, M\}$ ，单个追捕者和目标分别称为 P 和 T 。对于 $i \in \mathcal{I}$ ，用 $\mathcal{N}(i)$ 表示 P_i 的邻居。假设 $i, j \in \mathcal{I}$ ， P_i 和 P_j 相对于目标 T 的夹角定义为 $\alpha_{i,j}$ ， P_i 和 P_j 之间的距离定义为 $d_{i,j}$ 。

定义 1（邻居）：当 $\alpha_{i,j}$ 之间没有其他追捕者时， P_i 和 P_j 被定义为邻居。特别地，当向量叉积 $\overrightarrow{TP_i} \times \overrightarrow{TP_j} > 0$ 时， P_j 被视为 P_i 的右邻居，左邻居的定义相反。我们定义垂直于 $\mathcal{W}$ 平面并向上指的方向为正方向。

所有智能体和障碍物都以圆形建模，其中 r^i 和 δ^i 分别表示每个实体的半径和安全缓冲距离。在时刻 t ， P_i 的位置和速度分别表示为 $p^i = [p_x^i, p_y^i]$ 和 $v^i = [v_x^i, v_y^i]$ ， $i \in \mathcal{I}$ ，而目标的位置和速度分别表示为 p^T 和 v^T 。本文中我们用上标 T 表示与目标相关的变量。将障碍物 i 表示为 O_i ， $i \in \mathcal{O}$ ，且 $O_i = [x_o^i, y_o^i, r_o^i, \delta_o^i]$ 。每个智能体的动力学用二阶积分器形式描述：

$$\begin{cases} \dot{p}^i = v^i \\ \dot{v}^i = u^i \end{cases} \quad (1)$$

其中 $u^i = [u_x^i, u_y^i]$ 表示 P_i 的加速度， u^i 的每个分量都有限制 $u_{\max}$ 。目标也是如此。此外，追捕者和目标的最大速度存在约束，意味着其速度分量分别不能超过 $v_{\max}$ 和 $v_{\max}^T$ 。为保证成功围捕，本文设置 $u_{\max} = u_{\max}^T$ ， $v_{\max} > v_{\max}^T$ 。

初始时，追捕者排列成一条直线，目标随机位于其前方的圆形区域内。追捕者和目标的初始速度均为零。对于围捕条件，所有追捕者应均匀分布在以目标为圆

心、捕获距离 d_c 为半径的圆周上。 P_i 与其邻居的期望围捕角度设为 $\alpha_{\text{exp}} = 2\pi/N$ 。为便于学习，允许最终围捕角度和距离有轻微偏差 Δ_α 和 Δ_d 。 P_i 与 T 之间的距离表示为 $d_{i,T}$ 。

定义 2 (围捕条件): 对于任意追捕者 $i \in \mathcal{I}$ ，如果以下条件同时成立，则认为围捕完成：

$$|\alpha_{i,j} - \alpha_{\text{exp}}| < \Delta_\alpha, \quad j \in \mathcal{N}(i) \quad (2)$$

$$|d_{i,T} - d_c| < \Delta_d \quad (3)$$

围捕完成后，所有智能体减速至静止。此外，任何追捕者都不能与其他追捕者或障碍物碰撞，训练需要安全缓冲距离。当满足条件 $\|p^i - p^j\| < r^i + r^j + k_\delta \times (\delta^i + \delta^j)$ 时（其中 k_δ 是系数），追捕者将受到相应惩罚。障碍物之间的平均距离设置为接近捕获距离 d_c ，使围捕场景更加复杂。

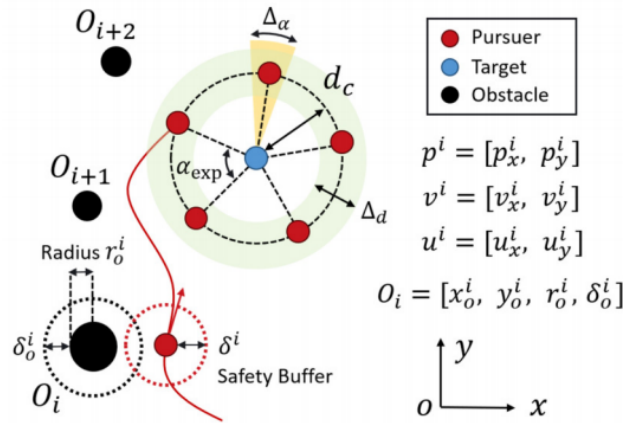


图 2.1 围捕控制场景示意图。浅绿色和浅黄色条带分别表示距离偏差和角度偏差。

2.2 基于 ROS2 的仿真平台

我们使用高保真自主导航仿真器 Gazebo 来构建半物理实验环境。部署了一个自主设计的全向车辆模型平台进行围捕实验。此外，我们采用开源的 ROS2 来促进我们的系统与 Gazebo 仿真器之间的通信^[30]。值得注意的是，通过运动学调整，我们的算法可扩展到其他平台如无人机和无人水面艇。

全向车辆的逆运动学用于根据沿各轴的期望速度计算每个车轮的速度。在 Gazebo 仿真中，车轮被视为旋转连杆，其速度由仿真器内置的动力学引擎控制。

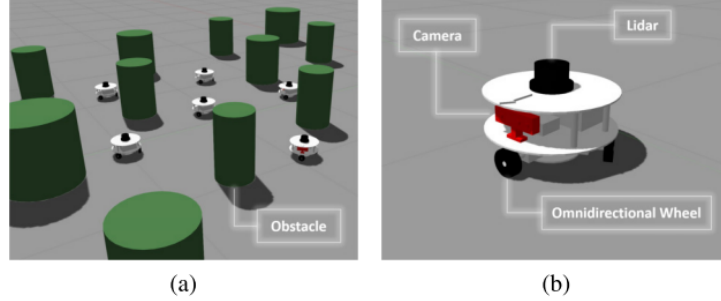


图 2.2 基于 ROS2 的物理引擎仿真器 Gazebo。(a) 围捕环境。(b) 自主设计的全向车模型。

表 3.1 模型车辆参数

参数	参数解释	追捕者	逃脱者
m	质量（千克）	0.8	0.8
r	车辆半径（米）	0.15	0.15
$v_{\max}$	最大线速度（米/秒）	0.5	0.25
$u_{\max}$	最大加速度（米/秒 ² ）	0.5	0.5
I_{xx}	X 轴转动惯量（千克·米 ² ）	0.006	0.006
I_{yy}	Y 轴转动惯量（千克·米 ² ）	0.006	0.006

3 预备知识

3.1 多智能体强化学习

多智能体强化学习是解决多智能体序贯决策任务的有效方法，可通过马尔可夫决策过程（Markov Decision Process, MDP）建模。该过程用元组 $\langle \mathcal{S}, \mathcal{A}, \mathcal{T}, R, \gamma \rangle$ 表示，其中 $\mathcal{S}$ 是状态集， $s \in \mathcal{S}$ 表示环境的全局状态。 $\mathcal{A} = [\mathcal{A}_1, \dots, \mathcal{A}_n]$ 是联合动作集， $\mathbf{a} \in \mathcal{A}$ ，其中 $\mathbf{a} = [a_1, \dots, a_n]$ 。分量 $a_i \in \mathcal{A}_i$ 是智能体 i ($i \in \{1, \dots, n\}$) 采取的动作。 $\mathcal{T} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \mapsto [0, 1]$ 表示状态转移函数， $R : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}$ 是奖

励。 $\gamma \in [0, 1]$ 是决定未来奖励重要性的折扣因子 (Discount Factor)。智能体的目标是采用策略 π 以最大化折扣累积奖励的期望:

$$J(\pi) = E_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t R^t(s^t, a_1^t, \dots, a_n^t) \right] \quad (4)$$

其中 R^t 是所有智能体在时刻 t 获得的奖励。

多智能体近端策略优化 (Multi-Agent Proximal Policy Optimization, MAPPO) 是 PPO 在多智能体场景中的扩展^[31], 是一种基于策略梯度的 MARL 方法, 采用集中式训练分布式执行 (Centralized Training with Decentralized Execution, CTDE) 框架^[32]。策略 π 表示为参数化的深度神经网络 π^{θ} , 其中 θ 表示参数。评估网络由 ϕ 参数化。参数通过以梯度 $\nabla_{\theta} J(\pi^{\theta})$ 为优化方向最大化目标函数 $J(\pi^{\theta})$ 来更新:

$$\nabla_{\theta} J(\pi^{\theta}) = \nabla_{\theta} \log \pi^{\theta}(a^t | s^t) \sum_{t'=t}^{\infty} \gamma^{t'-t} R^{t'}(s^{t'}, a^{t'}) \quad (5)$$

更新参数时, 引入重要性采样比 (Importance Sampling Ratio) $l_t(\theta)$ 以提高样本效率:

$$l_t(\theta) = \frac{\pi^{\theta}(a^t | s^t)}{\pi^{\theta_{\text{old}}}(a^t | s^t)} \quad (6)$$

其中 $l_t(\theta)$ 表示似然比, θ_{old} 是更新前的策略参数。策略网络的损失函数定义如下:

$$L^{\pi}(\theta) = E \left[\min \left(l_t(\theta) \hat{A}_t^{\theta}(s^t, a^t), \text{clip}(l_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t^{\theta}(s^t, a^t) \right) \right] \quad (7)$$

其中 $\hat{A}_t^{\theta}$ 是广义优势估计 (Generalized Advantage Estimation, GAE), clip 函数 $\text{clip}(l_t(\theta), 1 - \epsilon, 1 + \epsilon)$ 将 $l_t(\theta)$ 约束在 $[1 - \epsilon, 1 + \epsilon]$ 范围内以避免策略过度更新。价值网络的损失表示为 $L^V(\phi)$ 。

与其他 MARL 算法相比, MAPPO 在收敛性和采样效率方面具有优越的性能。本文选择带有循环层的 MAPPO 变体 rMAPPO 作为基础训练算法。

3.2 课程学习

课程学习受人类学习过程启发, 在难度逐渐增加的任务上逐步训练模型, 从简单到复杂的数据。CL 范式在 DRL 领域有广泛的应用^[33], 因为它增强了样本

效率、组织了模型探索并解决了稀疏奖励问题。此外，它有助于加速学习过程的收敛^[34]。

CL 在 DRL 中的一个具体应用是在数据收集过程中控制数据难度，课程通常可以从以下几个方面设计：初始状态、奖励函数、目标、环境和对手^[35]。在初始训练阶段，设计一个简化任务的简单课程以便于获取奖励。随着训练的推进，任务逐渐恢复到原始难度，从而增强模型对更具挑战性场景的适应性。

3.3 跳跃启动强化学习

在 JSRL 引入之前，基础控制器的概念已被用于促进机械臂控制的训练^[36]。在此概念上改进，JSRL 提供了一个令人信服的解决方案来解决 RL 中遇到的稀疏奖励挑战。它用预先存在的策略初始化 RL 策略，并与任何 RL 方法兼容^[29]。JSRL 使用两个策略来解决特定任务：引导策略 π^g 和学习策略 π^θ 。引导策略由课程控制，并随着学习策略性能的提高而逐渐退出。经验和理论分析都表明，由于在样本效率方面的优势，JSRL 显著优于现有的模仿学习算法。

JSRL 的机制描述如下。设 H 表示 RL 任务的时域， β_{thre} 表示模型性能阈值。首先，选择一个引导步序列来展开引导策略 $(H_1, H_2, \dots, H_n)$ ，其中 $H_i \in \{H\}$ 表示当前迭代中的引导步。设 h 为引导步序列 $(H_1, H_2, \dots, H_n)$ 的迭代器。在训练过程中，展开 $\pi^g h$ 步， π^θ 展开剩余的 $H - h$ 步，即 $\pi_{1:h} = \pi^g$ ， $\pi_{h+1:H} = \pi^\theta$ 。每次参数更新后，通过函数 $V^*(\pi)$ 评估模型性能。当 $V^*(\pi) > \beta_{\text{thre}}$ 时，迭代器 h 前进到引导步序列中的下一个值。这个训练过程循环继续，直到遍历完整个引导步序列。

4 面向策略的围捕控制学习

本节详细介绍 MRA-RLEC 和 MARL 框架的设计过程。训练分为多个线程，在每个线程内收集轨迹数据并调整为批次后再优化参数。

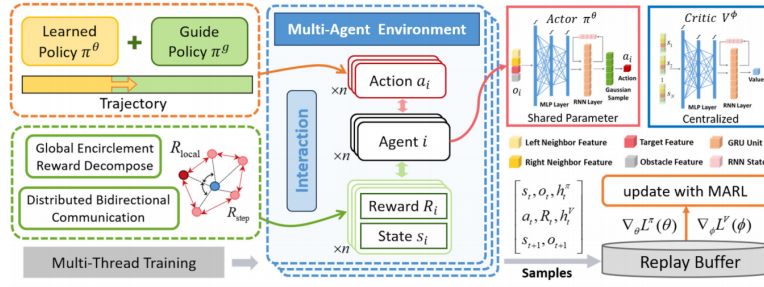


图 4.1 MRA-RLEC 整体架构。

4.1 JSRL 的引导策略

人工势场是一种经典的路径规划算法，将目标视为吸引源，将障碍物视为排斥源，引导智能体沿合力方向移动。然而，如第一节所述，APF 方法有显著的缺点。尽管如此，APF 仍能驱动追捕者向目标移动，这在一定程度上减少了探索空间。因此，我们采用 APF 作为 JSRL 的引导策略。为便于形成期望的围捕队形并避免局部极小值，我们只考虑三种不同的力：追捕者之间的排斥力 ($f_r^{i,j}$)、追捕者与障碍物之间的排斥力 ($f_o^{i,k}$) 和追捕者与目标之间的吸引力 ($f_a^{i,T}$)，其中 $i, j \in \mathcal{I}$, $k \in \mathcal{O}$ 。在期望的围捕队形中，相邻追捕者之间的距离为 $L = 2d_c \sin(\alpha_{\text{exp}}/2)$ 。力的生成如下：

$$f_r^{i,j} = \begin{cases} \frac{k_r(L-d_{i,j})}{d_{i,j}}(p^i - p^j), & d_{i,j} \leq L \\ 0, & d_{i,j} > L \end{cases} \quad (8)$$

$$f_o^{i,k} = \begin{cases} \frac{k_o(L_s-d_{i,k})}{d_{i,k}}(p^i - p^k), & d_{i,k} \leq L_s \\ 0, & d_{i,k} > L_s \end{cases} \quad (9)$$

$$f_a^{i,T} = \frac{k_a(d_{i,T} - d_c)}{d_{i,T}}(p^T - p^i) \quad (10)$$

为缓解振动，添加了包含 v^i 的阻尼项。追捕者 i 受到的总力为：

$$u^i = \sum_{j=1, j \neq i}^N f_r^{i,j} + \sum_{k=1}^M f_o^{i,k} + f_a^{i,T} - k_b v^i \quad (11)$$

其中 k_r 、 k_o 、 k_a 和 k_b 是正的控制增益， L_s 是与避碰相关的阈值参数。

随着训练的推进，引导策略的初始步数逐渐减少，最终让位于学习策略。在 JSRL 中，引导步的减少基于性能阈值 β_{thre} ，要求学习策略在减少发生前满足特定标准。然而，评估策略性能给训练带来不确定性，难以预测策略何时达到目标水平，可能影响后续训练任务。为解决这个问题，我们引入一个非线性调节器来管理引导策略的退出，确保 JSRL 有一致的、预定的训练持续时间。

设计的调节器旨在遵循 S 形曲线，原因如下。当 JSRL 开始时，追捕者需要在引导策略下有更多的探索空间，因此缓慢减少引导步对于鼓励更广泛的探索是必要的。随着训练的推进，追捕者已识别出高奖励的终止状态，允许更快地减少引导步。然而，在 JSRL 接近结束时，引导策略的缓慢退出是必要的，因为初始状态对未来轨迹有持久影响。如果学习策略尚未完全掌握初始状态附近的最优行为，此阶段的突然减少可能损害学习轨迹。因此，引导步的减少遵循 S 形曲线，确保从引导策略到学习策略执行的平滑过渡。

为构建 JSRL 调节器的 S 形曲线，我们选择两个奇函数作为基础： $\tanh(x)$ 和 x^3 。由于 $\tanh(x)$ 本身具有 S 形轮廓， x^3 项用于调整端点处的函数值。为调节 S 曲线的陡峭程度，在变量 x 中引入缩放因子 k_s 。接下来，通过函数变换将水平和垂直坐标都映射到 0-1 范围。最终的调节器函数表示引导策略步数减少的速率，表述如下：

$$f(x) = \frac{1}{2}(-\tanh(k_s X) - \varepsilon X^3 + 1) \quad (12)$$

其中 $X = 2x - 1$ ， $\varepsilon = 1 + \tanh(-k_s)$ ， $f(\cdot)$ 表示 JSRL 调节器函数。

设 E 表示总训练回合数， $c_g \in (0, 1)$ 是比率系数， $E_g = c_g E$ 定义为 JSRL 的训练回合数， E_t 表示已完成的当前训练回合数。比率 $E_t/E_g \in [0, 1]$ 作为 JSRL 阶段进度的指标。设 M 表示一个回合的总步数， ρ_0 表示引导策略执行步数的初始比例，则当前引导步数 M^g 由 $M^g = M \times \rho_0 \times f(E_t/E_g)$ 计算。学习策略执行的剩余步数 M^θ 由 $M^\theta = M - M^g$ 给出。当 E_t/E_g 达到 1 时，JSRL 过程完成，训练继续，学习策略接管剩余训练回合的所有步数。

4.2 通过多课程调节器训练

由于 EMOCA 是一个复杂任务，仅依靠 JSRL 实现围捕具有挑战性。鉴于此，我们实施两阶段训练策略，第一阶段训练 Base 模型，在 Fine Tune 阶段训练更复杂的场景。我们根据不同目标将 Base 训练过程分为多个子任务。每个子任务在一定过程中通过 CL 训练，从简单到困难。之后，我们在 Fine Tune 阶段考虑增加的目标速度和各种逃跑策略。

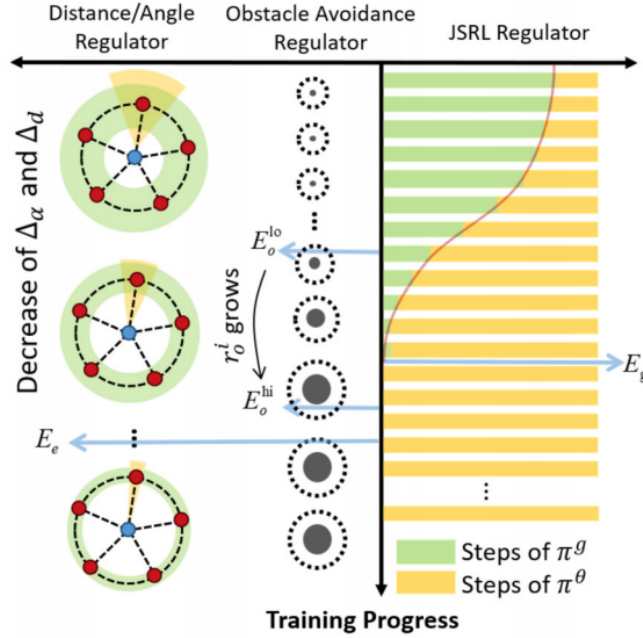


图 4.2 辅助训练的多调节器示意图。从左到右依次为角度和距离偏差、障碍物半径以及引导策略步数的演化过程。

Base 训练中有四个 CL 调节器：JSRL 调节器、角度调节器、距离调节器和障碍物避让调节器。JSRL 调节器如前所述帮助减少探索空间。角度调节器和距离调节器通过将角度偏差和距离偏差从大调整到小来控制终止围捕条件的难度。这种调整允许追捕者在训练过程早期获得奖励，即使他们还没有达到精确的围捕条件，从而缓解稀疏奖励问题。这两个调节器的训练回合数为 $E_e = c_e E$ ，两

个调节器都以线性形式定义：

$$\Delta_{\alpha} = \begin{cases} \Delta_{\alpha}^{\text{sta}} - (\Delta_{\alpha}^{\text{sta}} - \Delta_{\alpha}^{\text{exp}})E_t/E_e, & E_t \leq E_e \\ \Delta_{\alpha}^{\text{exp}}, & E_t > E_e \end{cases} \quad (13)$$

$$\Delta_d = \begin{cases} \Delta_d^{\text{sta}} - (\Delta_d^{\text{sta}} - \Delta_d^{\text{exp}})E_t/E_e, & E_t \leq E_e \\ \Delta_d^{\text{exp}}, & E_t > E_e \end{cases} \quad (14)$$

其中上标”sta”表示起始值，”exp”表示期望值。与 c_g 类似，参数 c_e 是定义受调节器影响的回合比例的比率。后面提到的参数 c^{lo} 和 c^{hi} 同理。

对于障碍物避让调节器，它在特定训练阶段调节障碍物的大小。考虑到障碍物显著影响围捕过程，我们引入一种机制，在训练期间障碍物不会立即以全尺寸呈现。相反，它们逐步出现，从点开始逐渐扩展到达原始尺寸。此过程发生在训练阶段 E_o^{lo} 和 E_o^{hi} 之间，其中 $E_o^{\text{lo}} = c^{\text{lo}}E$ 和 $E_o^{\text{hi}} = c^{\text{hi}}E$ 。对于 $i \in \mathcal{O}$ ，在此期间障碍物 i 的半径定义如下：

$$r_o^i = \begin{cases} 0, & E_t < E_o^{\text{lo}} \\ r_o^{\text{exp}} \times \frac{E_t - E_o^{\text{lo}}}{E_o^{\text{hi}} - E_o^{\text{lo}}}, & E_o^{\text{lo}} \leq E_t \leq E_o^{\text{hi}} \\ r_o^{\text{exp}}, & E_t > E_o^{\text{hi}} \end{cases} \quad (15)$$

其中 r_o^{exp} 是障碍物 i 的原始半径。

在 Fine Tune 阶段，目标以不同策略逃跑，其速度通过线性课程调节器逐渐增加。此外，碰撞惩罚增加，学习率设置更小。完整的 MRA-RLEC 方法在算法 1 中给出。

4.3 MARL 框架设计

4.3.1 奖励函数

对于 EMOCA 问题，仅考虑最终状态来描述系统奖励是困难的，因为最终状态难以触发，通常导致稀疏奖励问题。为此，我们提出 GERD 方法允许对部

算法 3.1 MRA-RLEC

Require: 总训练回合数 E , 最大回合步数 M , 学习率 l_r , PPO 梯度更新次数 K

Ensure: Actor 网络 θ 和 Critic 网络 ϕ

- 1: 初始化 Actor 参数 θ 和 Critic 参数 ϕ , 定义 E_t 为当前训练回合数
 - 2: **for** $E_t = 1$ to E **do**
 - 3: 计算 E_t/E 并选择相关调节器。通过式 (13) 和 (14) 计算 Δ_α 和 Δ_d 。通过式 (15) 计算 r_o^i 并设置围捕场景
 - 4: 计算 E_t/E_g 并通过 JSRL 调节器函数 (12) 获得引导步 M^g
 - 5: **for** step = 1 to M **do**
 - 6: 执行联合动作 $a_{1:N}^t = \pi(o_1, o_2, \dots, o_N)$ 。如果 step < M^g , 则设 $\pi = \pi^g$, 否则设 $\pi = \pi^\theta$
 - 7: 运行 π^g 和 π^θ 的混合策略共 M 个时间步, 为每个智能体收集 $\{o, a, r\}$ 并获得轨迹 τ 。将 τ 存入回放缓冲区
 - 8: **end for**
 - 9: 从回放缓冲区的 τ 估计优势函数 A_t
 - 10: **for** $k = 1$ to K **do**
 - 11: 通过式 (7) 计算策略损失 $L^\pi(\theta)$
 - 12: 为每个智能体更新网络参数: $\theta' \leftarrow \theta - l_r \nabla_\theta L^\pi(\theta)$, $\phi' \leftarrow \phi - l_r \nabla_\phi L^V(\phi)$
 - 13: **end for**
 - 14: **end for**
-

分围捕给予奖励，奖励设计如下：在围捕完成之前，每个追捕者获得一个步奖励 R_{step} 作为个体奖励。为激励局部合作，只要追捕者 i 满足围捕条件，它将获得局部包围奖励 R_{local} ，此时我们认为追捕者 i 完成。完成后，追捕者继续调整其位置以维持部分围捕并获得持续奖励。这个过程持续到所有追捕者都完成，最终导致完全围捕。当所有追捕者成功围捕时，每个追捕者将获得全局捕获奖励 R_{global} 。追捕者 i 的奖励定义为：

$$R_i = \begin{cases} R_{\text{global}}, & \text{所有追捕者完成} \\ R_{\text{local}}, & \text{追捕者 } i \text{ 完成} \\ R_{\text{step}}, & \text{其他情况} \end{cases} \quad (16)$$

在围捕过程中，追捕者团队需要同时实施包围和追逐动作。考虑到围捕任务的对称性质，追捕者与目标之间所有相对位置向量之和提供了围捕状态的自然表示。理想情况下，在实现完美围捕时，该聚合向量的幅度应收敛到零。将此定义为编队奖励 R_f ，它同时鼓励包围和追逐动作。为归一化此奖励， R_f 定义为：

$$R_f = \exp \left(-k_1 \cdot \left\| \sum_{i=1}^N (p^T - p^i) \right\| \right) - 1 \quad (17)$$

为确保追捕者达到指定的捕获距离 d_c ，我们引入距离奖励 R_d 定义如下：

$$R_d = \exp \left(-k_2 \cdot \sum_{i=1}^N (d_{i,T} - d_c)^2 \right) - 1 \quad (18)$$

其中 k_1 和 k_2 是可调系数。最后，我们为碰撞建立固定惩罚，表示为 R_l 。步奖励 R_{step} 表述为编队奖励 R_f 、距离奖励 R_d 和碰撞惩罚 R_l 的加权和：

$$R_{\text{step}} = w_1 R_f + w_2 R_d + w_3 R_l \quad (19)$$

其中 w_1 、 w_2 和 w_3 是决定每个分量在总奖励中相对重要性的权重。具体地，我们设置碰撞惩罚 $R_l = -3$ ，局部合作奖励 $R_{\text{local}} = 5$ ，成功完成围捕的奖励 $R_{\text{global}} = 10$ 。选择这些值是为了创建平衡的激励结构，引导学习过程朝着有效的围捕策略发展同时避免碰撞。

4.3.2 状态空间和通信协议

在大多数多智能体协作任务中，智能体理想情况下应与所有其他智能体通信以获取完整信息。然而，在围捕等动态多智能体任务中，与所有智能体通信消耗资源并对通信带宽提出更高要求。考虑到终止围捕状态是一个标准圆，我们提出分布式双向通信协议，使每个追捕者只与其左邻居和右邻居交换信息。

定义 S_t^i 为时刻 t 追捕者 i 的状态向量，其邻居为追捕者 $j, k \in \mathcal{N}(i)$ 。 S_t^i 由四个不同分量组成： $S_t^{i,j}$ 、 $S_t^{i,k}$ 、 $S_t^{i,T}$ 和 $S_t^{i,o}$ 。具体地， $S_t^{i,j}$ 和 $S_t^{i,k}$ 表示追捕者 i 相对于其两个邻居的相对状态。例如，相对于其左邻居 P_j 的状态为 $S_t^{i,j} = [\alpha_{i,j}, d_{i,j}, v_{i,j}, Q_{i,j}, Q_{j,i}]^\top$ ，其中 $v_{i,j}$ 是相对速度的幅度， $Q_{i,j}$ 是从 P_i 的速度到指向 P_j 的位置向量的相对方位角。右邻居状态 $S_t^{i,k}$ 具有相同的分量。项 $S_t^{i,T}$ 表示观测到的目标状态，定义为 $S_t^{i,T} = [d_{i,T}, \dot{d}_{i,T}, Q_{i,T}, \dot{Q}_{i,T}, \omega, \dot{\omega}]^\top$ ，其中 $Q_{i,T}$ 是追捕者速度与相对位置向量之间的方位角， ω 表示从追捕者 i 视角观测的目标旋转速度。最近障碍物 o 的观测由 $S_t^{i,o} = [x_{i,o}, y_{i,o}, d_{i,o}, d_{j,o}, d_{k,o}, r_o, v_x^i, v_y^i]^\top$ 捕获，其中 $x_{i,o}$ 和 $y_{i,o}$ 分别反映追捕者 i 与障碍物 o 沿 x 轴和 y 轴的位置差异。最后，导出级联状态 S_t^i ：

$$S_t^i = \left[(S_t^{i,j})^\top, (S_t^{i,k})^\top, (S_t^{i,T})^\top, (S_t^{i,o})^\top \right]^\top \quad (20)$$

S_t^i 是 Actor 网络的直接输入。

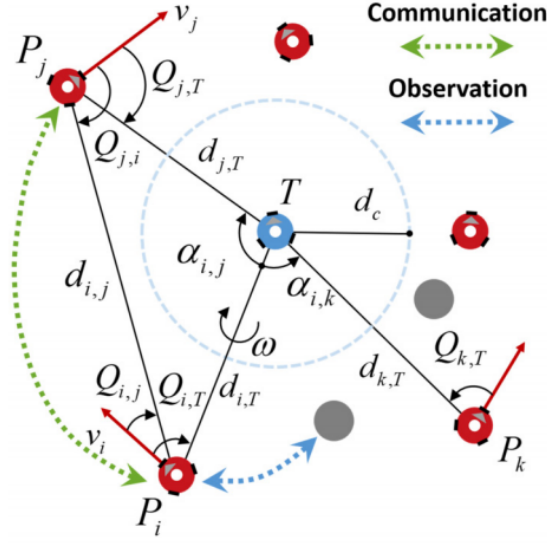


图 4.3 围捕特征示意图。

4.3.3 网络结构

鉴于 MAPPO 采用 Actor-Critic 架构，每个追捕者配备一个 Actor 网络将观测状态映射到其动作，整个系统有一个集中式 Critic 网络评估整体状态。Actor 网络和 Critic 网络都基于全连接和循环神经网络（Recurrent Neural Network, RNN）结构，其中 RNN 中的隐藏状态 $h^{\pi, V}$ 恢复历史信息并促进决策。全连接结构使用多层感知器（Multilayer Perceptron, MLP）构建，门控循环单元（Gated Recurrent Unit, GRU）作为 RNN 层来捕获时间依赖性。此外，选择 $\tanh(\cdot)$ 作为激活函数，所有 Actor 网络具有相同的结构并共享模型参数。

5 仿真与结果

在本节中，我们将 MRA-RLEC 应用于不同的 MARL 算法以验证其鲁棒性和有效性，并使用不同数量的智能体训练模型以展示所提方法的可扩展性。不失一般性，我们选择默认智能体数量为 5，即 $N = 5$ 。此外，我们将 MRA-RLEC 与其他围捕算法进行比较以验证其有效性。我们还进行消融测试，并通过在 Gazebo 中使用自主设计的全向车辆进行实验来证明我们方法的实用性。

5.1 仿真设置

在 Base 训练过程中，我们将调节器参数 c_g 、 c_e 、 c^{lo} 和 c^{hi} 分别设置为 0.6、0.75、0.45 和 0.75。折扣因子 γ 设置为 0.985，裁剪系数 ϵ 设置为 0.15。步奖励的权重设置为 $\omega_1 = 0.3$ ， $\omega_2 = 0.3$ ， $\omega_3 = 0.4$ 。在追捕动力学方面，追捕者的最大速度 $v_{\max}$ 设置为 0.5 m/s，而 $v_{\max}^T$ 在 Base 中设置为 0.15 m/s，在 Fine Tune 中设置为 0.25 m/s。双方的加速度限制 $u_{\max}$ 设置为 0.5 m/s²。

对于围捕条件参数，我们设置 $d_c = 1$ m， $\Delta_\alpha = 0.3$ rad， $\Delta_d = 0.1$ m。我们还设置引导策略中的参数为 $k_r = 4.5$ ， $k_o = 4.0$ ， $k_a = 2.0$ ， $k_b = 1.5$ ，碰撞距离阈值为 $L_s = 0.3$ m。此外，每个回合的最大步数设置为 200，每个仿真时间间隔 dt 代表 0.1 s。Base 阶段设计进行总共 7.68×10^6 训练步，而 Fine Tune 阶段包含 5.12×10^6 步。

表 3.2 网络和训练相关参数

名称	参数值
Actor 网络 MLP 层数	2
Actor 网络隐藏单元数	128, 64
Actor 网络 RNN 隐藏单元数	32
Critic 网络 MLP 层数	3
Critic 网络隐藏单元数	256, 128, 64
Critic 网络 RNN 隐藏单元数	32
RNN 数据块长度	10
更新批量大小	100
Base 阶段学习率	0.0002
Fine Tune 阶段学习率	0.00015
PPO 更新轮数	15

5.2 MRA-RLEC 的可扩展性和鲁棒性

为验证 MRA-RLEC 的可扩展性，我们在第二节介绍的环境中实现算法，追捕者数量从 3 到 6 变化。在 Base 训练期间，我们设置 JSRL 参数 $k_s = 2.0$ 。在 Fine Tune 阶段，我们增加目标最大速度并使用线性 CL 调节器逐渐增加速度。选择势场策略（Potential Field Policy, PFP）^[37]作为目标的逃跑策略，实验在 rMAPPO 方案下进行。

从实验结果可以看出，在 Base 中奖励曲线通常在上升之前会经历下降。这种现象可归因于引导策略的退出，这可能导致训练奖励下降。而在 Fine Tune 中，没有引导策略，奖励稳步增加。奖励的收敛表明 MRA-RLEC 可扩展到不同数量的追捕者。

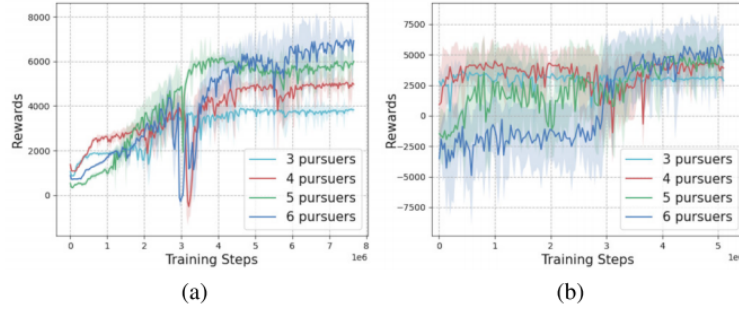


图 5.1 不同追击者数量的训练奖励。(a) Base 阶段奖励。(b) Fine Tune 阶段奖励。

模型进一步验证，围捕过程的轨迹可视化说明了成功的围捕队形形成，追捕者以目标为中心形成正多边形，满足 $|\alpha_{i,j} - \alpha_{\text{exp}}| < \Delta_\alpha$, $j \in \mathcal{N}(i)$ 和 $|d_{i,T} - d_c| < \Delta_d$ 。此外，追捕者轨迹中任何明显的重叠是由于时间差异而非物理碰撞造成的。该图还确认轨迹与障碍物之间没有交叉，表明围捕过程中智能体与障碍物之间没有发生碰撞。

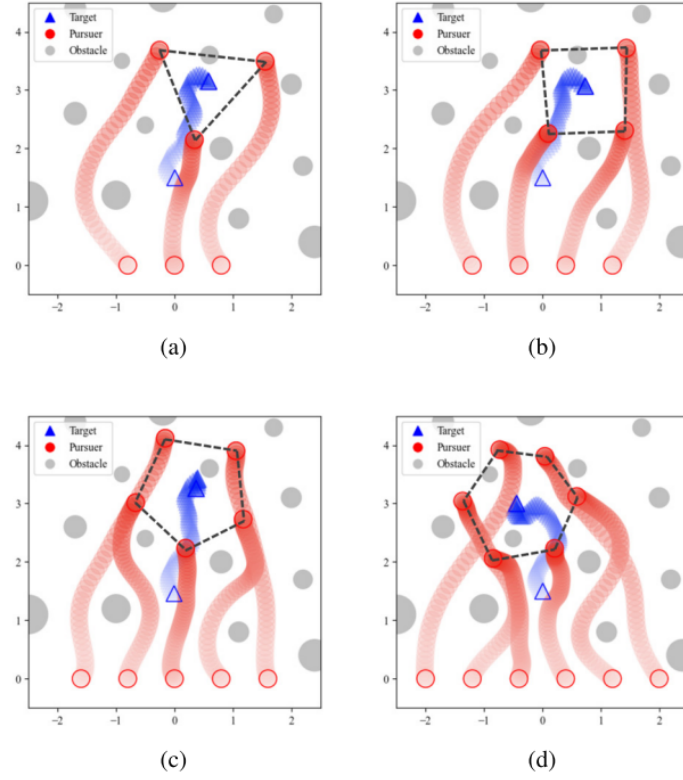


图 5.2 使用 MRA-RLEC 和 rMAPPO 训练的不同追击者数量的围捕轨迹。坐标轴单位为米 (m)。(a) 3 个追击者围捕。(b) 4 个追击者围捕。(c) 5 个追击者围捕。(d) 6 个追击者围捕。

为验证 MRA-RLEC 的鲁棒性，我们将其应用于各种 MARL 算法来训练五追捕者围捕场景。在这些算法中，独立近端策略优化 (Independent Proximal Policy Optimization, IPPO) 方法提供了一种方案，其中每个智能体独立估计其局部价值函数^[38]。与 rMAPPO 相比，MAPPO 没有 RNN 层，在多智能体 Transformer (MATransformer) 中网络使用 Transformer 结构而非卷积神经网络^[39]。此外，异构智能体 PPO (Heterogeneous-Agent PPO, HAPPO)^[40]、MADDPG^[32]和多智能体双延迟 DDPG (Multi-Agent Twin Delayed DDPG, MATD3)^[41]都是 MARL 领域的突出算法。在我们的框架下所有方法的训练奖励都正确收敛。

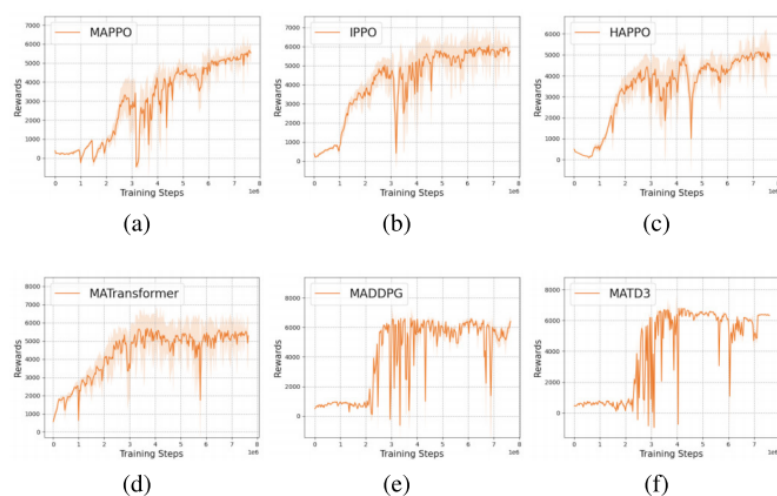


图 5.3 使用 MRA-RLEC 训练的不同 MARL 算法的训练奖励。(a) MAPPO。(b) IPPO。(c) HAPPO。(d) MATransformer。(e) MADDPG。(f) MATD3。

在这些方法中，离线策略算法（Off-Policy Algorithm）（MADDPG 和 MATD3）区分目标策略和行为策略，更可能探索全局最优策略。这可能导致更多的奖励振荡，因为它们探索更多策略。因此，离策略算法具有更高的最大奖励和平均奖励标准差。相比之下，基于 PPO 的算法是在策略方法（On-Policy Method），可能陷入局部最优。此外，RNN 层的存在，如 MAPPO 和 rMAPPO 之间的对比所示，显著影响平均奖励。

表 3.3 基于不同 MARL 算法使用 MRA-RLEC 的奖励指标

训练方法	平均奖励 (↑)	达到平均奖励的 训练步数 (↓)	最大奖励 (↑)	平均轮次 标准差 (↓)	平均奖励 标准差 (↓)
rMAPPO	4345.177	2630400	6390.158	599.2639	1908.99
MAPPO	2870.418	2598400	5941.362	567.4647	1942.78
IPPO	4141.887	2342400	6372.418	727.4509	1790.40
HAPPO	3366.252	1862400	5518.821	612.6576	1533.79
MATransformer	4290.802	2374400	6312.289	864.2967	1304.26
MADDPG	4017.395	2438400	6836.482	405.1290	2414.49
MATD3	3977.205	2790400	6848.849	252.7119	2658.52

5.3 消融测试

我们首先进行消融测试以证明 JSRL 调节器的有效性。在此实验中，关注五追捕者围捕场景，在训练过程中引入 JSRL 调节器。引导策略步数的初始比例 ρ_0 设置为 0.8。通过改变 JSRL 参数 k_s ，我们生成一系列调节器函数，每个都用于训练模型。

从实验结果可以看出，当比例曲线在初始阶段呈现凸形 ($k_s > 1$) 时，训练奖励实现收敛。然而，比例曲线不能过于凸，因为这可能导致引导策略在中间突然退出，从而影响探索。相反，当比例曲线为凹形 ($k_s < 1$) 时，模型的收敛性能下降。这与分析一致，即模型在初始阶段需要额外时间探索终止条件。此外，我们在训练几个回合后完全撤出引导策略以展示 JSRL 的影响，结果显示在如此复杂的场景中奖励无法收敛。

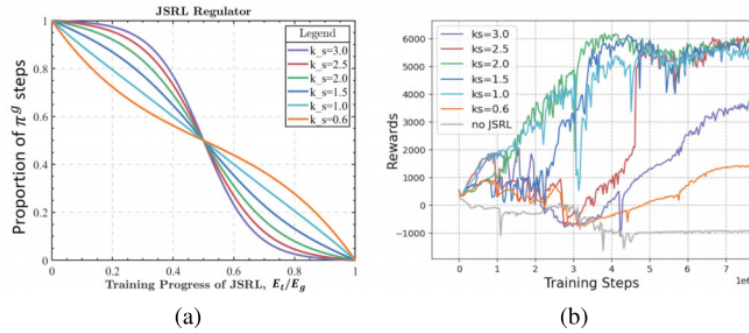


图 5.4 JSRL 调节器消融实验。(a) 不同 k_s 值下引导策略步数的比例曲线。(b) 不同 JSRL 调节器下的训练奖励。

接下来，我们测试 GERD 的有效性。为简化场景，我们移除障碍物并使用 JSRL 调节器辅助训练。第一个实验包含 GERD，为部分围捕给予局部奖励。在对比实验中，只有在形成完全围捕时才给予奖励。GERD 的引入加速了奖励的增加并导致更高的收敛值。相比之下，没有 GERD 训练的模型花费更多时间探索完全围捕，产生较低的初始奖励。这一对比突出了 GERD 在增强训练效率和收敛方面的作用。

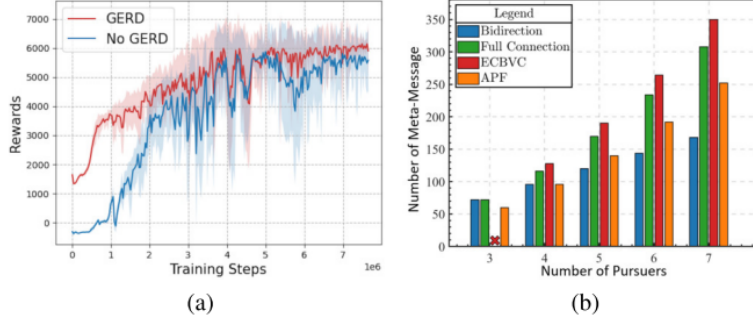


图 5.5 奖励函数和通信设计的对比结果。(a) 有无 GERD 的训练奖励。(b) 不同围捕算法的元消息通信开销。

我们进一步使用元消息 (Meta-Message)^[20] 评估双向协议的通信效率, 这可以衡量算法在单个时间步中的通信负载复杂度。定义 N_p 、 N_T 和 N_o 分别为追捕者、目标和障碍物的观测维度。双向协议下的总通信元消息计算为 $(2N_p + N_T + N_o)N$, 负载复杂度为 $O(N)$ 。而在全连接情况下, 通信负载为 $((N-1)N_p + N_T + N_o)N$, 复杂度为 $O(N^2)$ 。ECBVC 和 APF 方法要求每个追捕者对所有实体有完整了解, 导致元消息随 N 的平方增长, 负载复杂度也是 $O(N^2)$ 。这一分析强调了我们的双向协议在围捕任务中节省通信资源方面的效率。

5.4 与其他围捕算法的比较

在本节中, 我们将 MRA-RLEC 与其他围捕算法进行比较。我们使用 rMAPPO、MADDPG 和 MATD3 作为基本 RL 框架, 并在这些训练基础上集成 MRA-RLEC, 以 APF 作为引导策略。作为比较, 我们选择了三种典型的传统方法: ECBVC^[25]、基于伪力的围捕控制 (Pseudo-Force-based Encirclement Control, PFEC)^[23] 和基于编队的围捕控制 (Formation-Based Encirclement Control, FBEC)^[11]。性能通过以下关键指标评估: 平均步奖励、围捕时间、危险率和成功率。危险率表示两个实体过于接近的情况频率, 这种情况由安全缓冲确定, 特别是当 $\|p^i - p^j\| < r^i + r^j + k_\delta \times (\delta^i + \delta^j)$ 时。

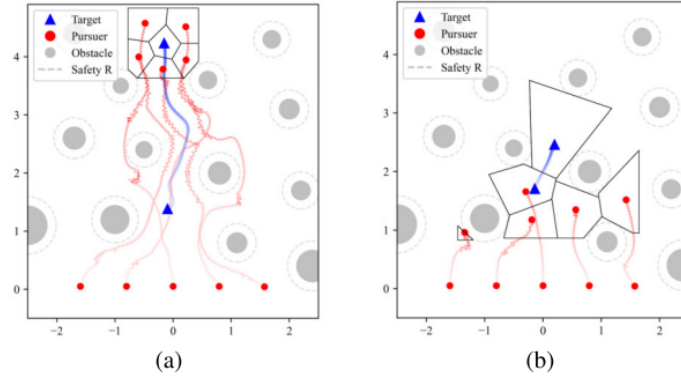


图 5.6 基于 ECBVC 的围捕轨迹。(a) 成功围捕。(b) 未成功围捕。

为进一步评估不同围捕算法的通用性，我们为目标指定了不同的逃跑策略，概述如下：

1 势场策略（PFP）考虑追捕者团队的联合威胁并帮助目标做出响应。相应的控制律 v_1^T 为：

$$v_1^T = \|v_{\max}^T\| \frac{\sum_i \frac{\hat{d}_{i,T}}{\|p^T - p^i\|} (p^T - p^i)}{\left\| \sum_i \frac{\hat{d}_{i,T}}{\|p^T - p^i\|} (p^T - p^i) \right\|} \quad (21)$$

其中 $\hat{d}_{i,T} = d_{i,T} - (r^T + r^i)$ ，涉及实体半径来计算势力。

2 贪婪策略（Greedy Policy, GP）驱动目标远离最近的追捕者 g 。相应的控制律 v_2^T 为：

$$v_2^T = \|v_{\max}^T\| \frac{p^T - p^g}{\|p^T - p^g\|} \quad (22)$$

3 阿波罗尼斯策略（Apollonius Policy, AP）^[42]帮助目标保持在其优势区域内并最大化追捕者的捕获时间。相应的控制律 v_3^T 为：

$$v_3^T = \|v_{\max}^T\| \frac{p^A - p^T}{\|p^A - p^T\|} \quad (23)$$

其中 p^A 是所有目标-追捕者对中最近阿波罗尼斯圆的最远边界点。

4 移动到质心策略（Move-to-Centroid Policy, MCP）^[25]通过向其 Voronoi 质心移动来使目标远离相邻追捕者。相应的控制律 v_4^T 为：

$$v_4^T = \|v_{\max}^T\| \frac{C_{V_T^W} - p^T}{\|C_{V_T^W} - p^T\|} \quad (24)$$

其中 $C_{V_T^w}$ 是目标 Voronoi 单元的中心。

我们进行了蒙特卡洛仿真（Monte Carlo Simulation），每个算法运行 100 次。结合 MRA-RLEC 的 MADDPG 和 MATD3 在围捕时间方面表现最佳，这归功于它们找到全局最优解的能力。对于基于 RL 的方法，追捕者形成围捕队形越快，累积的终止奖励越大，导致围捕时间越短。在危险率和成功率方面，基于 RL 的方法通常比传统方法表现更好。因为在传统方法中，历史信息几乎不被考虑，控制律很大程度上依赖于障碍物的位置，这在复杂的多障碍物环境中可能效果较差。结果是，大量时间花在避免碰撞上，导致围捕性能较差。

另一个比较维度是计算复杂度。使用我们的双向协议和固定的网络结构，每个追捕者只处理来自两个邻居和最具威胁性障碍物的信息。因此，从信息角度来看，前向推理复杂度保持恒定。相比之下，对于需要全连接通信的其他方法，每个追捕者必须整合来自所有其他实体的信息，导致计算复杂度为 $O(NM)$ 。从这个角度来看，我们的方法在效率方面展现出显著优势。

5.5 性能分析

此外，我们进行了实验以验证算法在不同任务复杂度水平下的性能，并分析了在围捕移动目标和避开障碍物之间取得平衡的有效性。任务复杂度由目标最大速度决定，我们选择一系列典型速度进行蒙特卡洛测试，每个参数进行 1000 次试验。

从实验结果可以观察到，当 $v_{\max}^T$ 超过临界阈值时性能下降。这种下降归因于追捕者追上快速逃跑目标的难度增加。因此，模型优先最大化围捕奖励，可能以牺牲障碍物避让为代价，以获得更高的整体回报。这种行为反映了基于 RL 方法中固有的权衡，学习到的策略适应以最大化整体任务性能。因此，算法在任务简单时达到最佳效果，在高度复杂条件下可能表现较差。在极端情况下，特别是当 $v_{\max}^T$ 接近 $v_{\max}$ 时，追捕者无法追上目标，使围捕更加难以实现。

表 3.4 不同方法在各种逃逸策略下的围捕指标

逃逸策略	围捕方法	步奖励	围捕时间 (s)	危险率 (%)	成功率 (%)
PFP	rMAPPO	21.44	9.79 ± 0.43	7	100
	MADDPG	19.57	10.61 ± 0.98	16	99
	MATD3	21.28	9.65 ± 0.66	12	100
	ECBVC	8.73	13.07 ± 1.23	42	28
	PFEC	9.04	13.58 ± 2.21	79	41
	FBEC	10.69	14.33 ± 1.51	60	47
GP	rMAPPO	24.48	8.89 ± 0.11	11	100
	MADDPG	23.65	9.21 ± 0.59	15	100
	MATD3	22.97	8.74 ± 0.32	9	100
	ECBVC	7.29	12.38 ± 1.12	49	26
	PFEC	13.25	12.20 ± 2.58	58	67
	FBEC	11.99	13.51 ± 3.97	70	32
AP	rMAPPO	21.85	9.81 ± 0.17	12	99
	MADDPG	22.29	9.52 ± 0.38	7	100
	MATD3	23.91	8.66 ± 0.49	4	100
	ECBVC	8.83	13.21 ± 1.96	45	43
	PFEC	9.82	13.54 ± 3.08	79	52
	FBEC	8.80	14.57 ± 1.86	62	42
MCP	rMAPPO	25.43	8.13 ± 0.11	8	100
	MADDPG	26.78	7.45 ± 0.20	6	100
	MATD3	26.39	7.93 ± 0.09	5	100
	ECBVC	7.87	11.58 ± 1.15	39	37
	PFEC	10.77	11.96 ± 2.45	58	74
	FBEC	13.79	13.23 ± 1.68	34	72

5.6 ROS2 中不同逃跑策略下的实验

为未来的实际车辆部署，我们在 Gazebo 中进行了额外实验，使用 ROS2 管理车辆控制和通信。使用 MRA-RLEC 结合 rMAPPO 训练的模型从 Actor 网络生成加速度。车轮速度通过运动学进一步计算，控制命令通过 ROS2 系统发布。车辆通过共享消息池发布和订阅状态进行通信。我们使用不同的目标逃跑策略进行了五追捕者围捕测试。

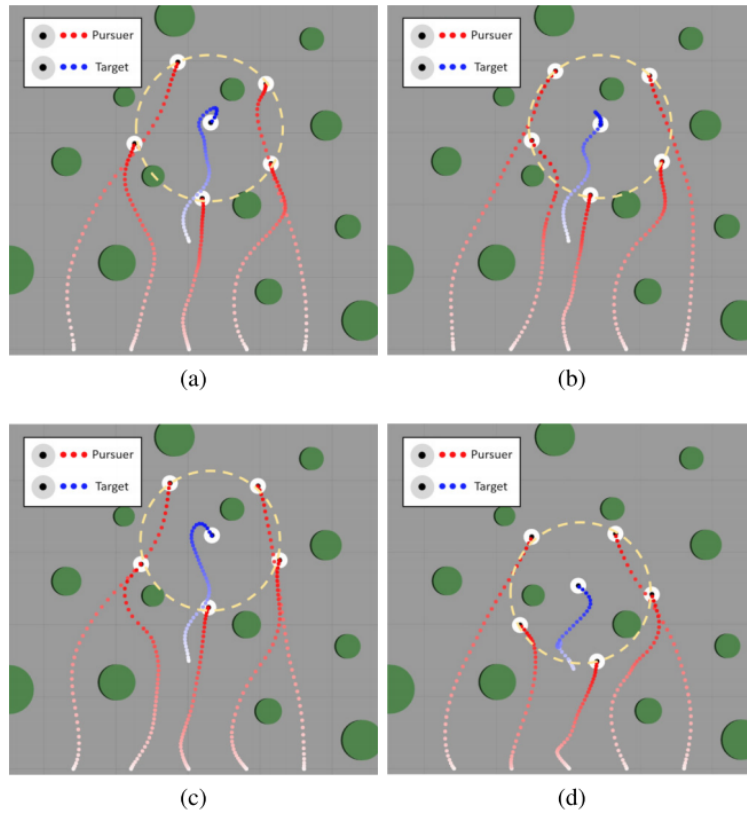


图 5.7 使用 MRA-RLEC 框架和 rMAPPO 训练，在 ROS2 中运行的不同逃逸策略下的围捕轨迹。(a) PFP 逃逸策略。(b) GP 逃逸策略。(c) AP 逃逸策略。(d) MCP 逃逸策略。

此外，可视化最小距离有助于评估避碰有效性。 D_o 定义为任意追捕者到障碍物的最小距离，计算为 $D_o = \min_{i \in \mathcal{I}, j \in \mathcal{O}} (d_{i,j} - r_o^j)$ ，其中 r_o^j 是障碍物 j 的半径。实验结果说明 D_o 始终高于追捕者的半径，表明有效的障碍物避让。追捕者之间的最小距离为 $D_p = \min_{i,j \in \mathcal{I}, i \neq j} (d_{i,j} - r^j)$ ，其中 r^j 是 P_j 的半径，确认围捕过程中追捕者之间没有发生碰撞。

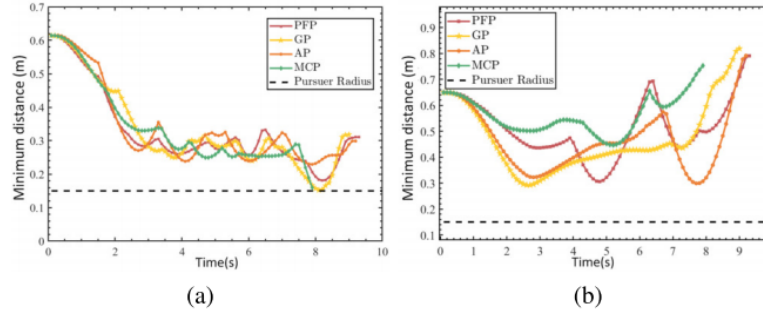


图 5.8 围捕目标时用于评估碰撞的距离。(a) 追击者与障碍物之间的最小距离 $D_o(t)$ 。(b) 所有追击者之间的最小距离 $D_p(t)$ 。

与围捕条件相关的误差分析表明，追捕者的平均距离误差，表示为 $e_d = \frac{1}{N} \sum_{i \in \mathcal{I}} |d_{i,T} - d_c|$ ，逐渐下降到 Δ_d 以下并收敛到零。类似地，期望围捕队形的平均角度误差，表示为 $e_\alpha = \frac{1}{N} \sum_{i=1}^N |\alpha_{i,i+1} - \alpha_{\text{exp}}|$ ，也下降到 Δ_α 以下并正确收敛。

表 3.5 不同围捕复杂度下的性能表现

方法	$v_{\max}^T = 0.10 \text{ m/s}$		$v_{\max}^T = 0.20 \text{ m/s}$		$v_{\max}^T = 0.25 \text{ m/s}$		$v_{\max}^T = 0.27 \text{ m/s}$		$v_{\max}^T = 0.30 \text{ m/s}$	
	S (%)	R	S (%)	R	S (%)	R	S (%)	R	S (%)	R
MADDPG	99.7	36.74	99.2	23.53	99.1	20.24	59.8	9.59	31.4	3.87
MATD3	99.8	35.97	99.5	25.08	99.3	21.17	63.2	11.82	29.8	3.07
RMAPPO	99.8	39.48	99.3	25.76	98.7	20.65	65.7	13.37	34.2	4.16

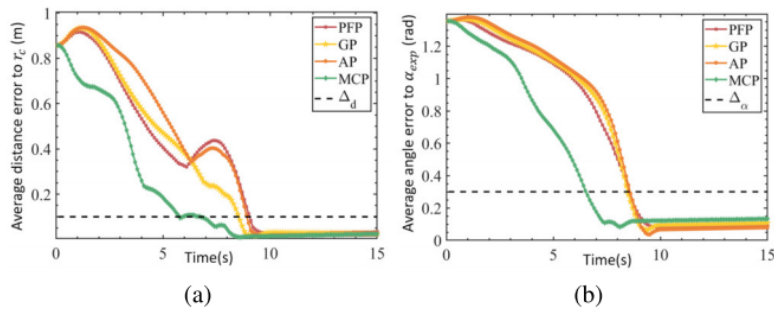


图 5.9 四种逃逸策略下随时间变化的距离误差和角度误差。(a) 距离误差 e_d ，由平均距离减去捕获半径计算得出。(b) 角度误差 e_α ，由平均夹角减去期望围捕角度计算得出。

6 结论与未来工作

本文提出了一种新颖的策略引导强化学习框架 MRA-RLEC，用于解决多障碍物环境中移动目标的围捕问题。在 MRA-RLEC 中，设计了多个调节器来辅助训练，并提出了 GERD 方法来缓解奖励稀疏问题。此外，我们设计了一种节省资源的双向协议，其通信复杂度随追捕者数量线性增长。我们使用不同的追捕者数量和 MARL 结构测试了 MRA-RLEC 的可扩展性和鲁棒性，并将我们的方法与其他围捕算法进行比较，证明了我们的优越性。同时，我们还在物理仿真引擎 Gazebo 中进行了各种逃跑策略下的实验，以展示 MRA-RLEC 在实际应用中的潜力。

虽然我们的实验结果在定义的设置中表现良好，但仍存在一定的局限性。一些存在于真实世界系统中的因素在我们的模型中没有完全考虑，如通信范围，且目标以相对较低的速度移动。在未来的工作中，我们打算通过进行具有更广泛感知范围的高机动性目标的真实世界实验来解决这些挑战。此外，值得注意的是，引导策略和课程调节器的概念可扩展到其他多智能体任务。该框架可能在任何采用 RL 方法的多智能体系统中提高训练效率。

7 外文翻译参考文献

- [1] Ashish M, Vilela J, Nejat G, et al. A multirobot path-planning strategy for autonomous wilderness search and rescue[J]. IEEE Transactions on Cybernetics, 2014, 45(9): 1784-1797. DOI: 10.1109/TCYB.2014.2360370.
- [2] Sui Z, Pu Z, Yi J, et al. Formation control with collision avoidance through deep reinforcement learning using model-guided demonstration[J]. IEEE Transactions on Neural Networks and Learning Systems, 2021, 32(6): 2358-2372. DOI: 10.1109/TNNLS.2020.3004893.
- [3] Zhang R, Zong Q, Zhang X, et al. Game of drones: Multi-UAV pursuit-evasion game with online motion planning by deep reinforcement learning[J]. IEEE Transactions on Neural Networks and Learning Systems, 2022, 34(10): 7900-7909. DOI: 10.1109/TNNLS.2022.3146976.
- [4] Hafez A T, Iskandarani M, Givigi S N, et al. UAVs in formation and dynamic encirclement via model predictive control[C]//IFAC Proceedings Volumes: vol. 47: 3. 2014: 1241-1246. DOI: 10.3182/20140824-6-ZA-1003.01579.
- [5] Qu X, Gan W, Song D, et al. Pursuit-evasion game strategy of USV based on deep reinforcement learning in complex multi-obstacle environment[J]. Ocean Engineering, 2023, 273: 114016. DOI: 10.1016/j.oceaneng.2023.114016.
- [6] Sarwal A, Agrawal D, Chaudhary S. Surveillance in an open environment by co-operative tracking amongst sensor enabled robots[C]//Proceedings of International Conference on Information Acquisition. 2007: 345-349. DOI: 10.1109/ICIA.2007.4295757.
- [7] Antonelli G, Arrichiello F, Chiaverini S. The Entrapment/Escorting mission[J]. IEEE Robotics and Automation Magazine, 2008, 15(1): 22-29. DOI: 10.1109/MRA.2007.914692.
- [8] Khan A, Rinner B, Cavallaro A. Cooperative robots to observe moving targets: Review[J]. IEEE Transactions on Cybernetics, 2018, 48(1): 187-198. DOI: 10.1109/TCYB.2016.2628161.
- [9] Chen F, Ren W, Cao Y. Surrounding control in cooperative agent networks[J]. Systems and Control Letters, 2010, 59(11): 704-712. DOI: 10.1016/j.sysconle.2010.08.005.
- [10] Sun Z, Sun H, Li P, et al. Cooperative strategy for pursuit-evasion problem with collision avoidance[J]. Ocean Engineering, 2022, 266: 112742. DOI: 10.1016/j.oceaneng.2022.112742.
- [11] Huang N, Zhuge J, Sun Z, et al. Dynamic encirclement for anonymous agents with arbitrary formations[J]. IEEE Transactions on Control of Network Systems, 2024, 11(3): 1644-1654. DOI: 10.1109/TCNS.2024.3366714.
- [12] Xu B, Zhang H T, Ding Y, et al. Event-triggered surrounding formation control of multiagent systems for multiple dynamic targets[J]. IEEE Transactions on Control of Network Systems, 2023, 10(2): 752-764. DOI: 10.1109/TCNS.2022.3211148.
- [13] Liu F, Yuan S, Meng W, et al. Multiple noncooperative targets encirclement by relative distance-based positioning and neural antisynchronization control[J]. IEEE Transactions on Industrial Electronics, 2024, 71(2): 1675-1685. DOI: 10.1109/TIE.2023.3257380.
- [14] Hafez A T, Marasco A J, Givigi S N, et al. Solving multi-UAV dynamic encirclement via model predictive control[J]. IEEE Transactions on Control Systems Technology, 2015, 23(6): 2251-2265. DOI: 10.1109/TCST.2015.2411632.
- [15] Liao J, Liu C, Liu H H T. Model predictive control for cooperative hunting in obstacle rich and dynamic environments[C]//2021 IEEE International Conference on Robotics and Automation (ICRA). 2021: 5089-5095. DOI: 10.1109/ICRA48506.2021.9561766.
- [16] Nguyen T T, Nguyen N D, Nahavandi S. Deep reinforcement learning for multiagent systems: A review of challenges, solutions, and applications[J]. IEEE Transactions on Cybernetics, 2020, 50(9): 3826-3839. DOI: 10.1109/TCYB.2020.2977374.
- [17] Ma J, Lu H, Xiao J, et al. Multi-robot target encirclement control with collision avoidance via deep reinforcement learning[J]. Journal of Intelligent and Robotic Systems, 2020, 99(2):

- 371-386. DOI: 10.1007/s10846-019-01106-x.
- [18] Zhang T, Liu Z, Pu Z, et al. Multi-target encirclement with collision avoidance via deep reinforcement learning using relational graphs[C]//2022 IEEE International Conference on Robotics and Automation (ICRA). 2022: 8794-8800. DOI: 10.1109/ICRA46639.2022.9812361.
- [19] Zhang T, Liu Z, Wu S, et al. Multi-robot cooperative target encirclement through learning distributed transferable policy[C]//2020 International Joint Conference on Neural Networks (IJCNN). 2020: 1-8. DOI: 10.1109/IJCNN48605.2020.9207612.
- [20] Wang Y, Dong L, Sun C. Cooperative control for multi-player pursuit-evasion games with reinforcement learning[J]. *Neurocomputing*, 2020, 412: 101-114. DOI: 10.1016/j.neucom.2020.06.031.
- [21] De Souza C, Newbury R, Cosgun A, et al. Decentralized multi-agent pursuit using deep reinforcement learning[J]. *IEEE Robotics and Automation Letters*, 2021, 6(3): 4552-4559. DOI: 10.1109/LRA.2021.3068952.
- [22] Spears W M, Spears D F, Hamann J C, et al. Distributed, physics-based control of swarms of vehicles[J]. *Autonomous Robots*, 2004, 17(2-3): 137-162. DOI: 10.1023/B:AURO.0000033970.96785.f2.
- [23] Liu F, Yuan S, Cao K, et al. Distance-based multiple noncooperative ground target encirclement for complex environments[J]. *IEEE Transactions on Control Systems Technology*, 2025, 33(1): 261-273. DOI: 10.1109/TCST.2024.3451413.
- [24] Tian B, Li P, Lu H, et al. Distributed pursuit of an evader with collision and obstacle avoidance[J]. *IEEE Transactions on Cybernetics*, 2021, 52(12): 13512-13520. DOI: 10.1109/TCYB.2021.3126371.
- [25] Wang X, Xi L, Ding Y, et al. Distributed encirclement and capture of multiple pursuers with collision avoidance[J]. *IEEE Transactions on Industrial Electronics*, 2024, 71(7): 7520-7530. DOI: 10.1109/TIE.2023.3310015.
- [26] Richards A, How J P. Aircraft trajectory planning with collision avoidance using mixed integer linear programming[C]//Proceedings of the 2002 American Control Conference: vol. 3. 2002: 1936-1941. DOI: 10.1109/ACC.2002.1023909.
- [27] Chen Y F, Liu M, Everett M, et al. Decentralized non-communicating multiagent collision avoidance with deep reinforcement learning[C]//2017 IEEE International Conference on Robotics and Automation (ICRA). 2017: 285-292. DOI: 10.1109/ICRA.2017.7989037.
- [28] Pu Z, Zhang T, Ai X, et al. A deep reinforcement learning approach combined with model-based paradigms for multiagent formation control with collision avoidance[J]. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 2023, 53(7): 4189-4204. DOI: 10.1109/TSMC.2023.3243832.
- [29] Uchendu I, et al. Jump-start reinforcement learning[C]//Proceedings of the 40th International Conference on Machine Learning. 2022: 34556-34583.
- [30] Macenski S, Foote T, Gerkey B, et al. Robot operating system 2: Design, architecture, and uses in the wild[J]. *Science Robotics*, 2022, 7(66): 6074. DOI: 10.1126/scirobotics.abm6074.
- [31] Yu C, Velu A, Vinitisky E, et al. The surprising effectiveness of PPO in cooperative, multi-agent games[C]//Advances in Neural Information Processing Systems (NeurIPS). 2021: 24611-24624.
- [32] Lowe R, Wu Y, Tamar A, et al. Multi-agent actor-critic for mixed cooperative-competitive environments[C]//Advances in Neural Information Processing Systems (NeurIPS). 2017: 1-12.
- [33] Wang X, Chen Y, Zhu W. A survey on curriculum learning[J]. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2022, 44(9): 4555-4576. DOI: 10.1109/TPAMI.2021.3069908.
- [34] Narvekar S, Peng B, Leonetti M, et al. Curriculum learning for reinforcement learning domains: A framework and survey[J]. *Journal of Machine Learning Research*, 2020, 21(1):

- 1-50.
- [35] Portelas R, Colas C, Weng L, et al. Automatic curriculum learning for deep RL: A short survey[J]. arXiv preprint arXiv:2003.04664, 2020.
- [36] Wang G, Xin M, Wu W, et al. Learning of long-horizon sparse-reward robotic manipulator tasks with base controllers[J]. IEEE Transactions on Neural Networks and Learning Systems, 2022, 35(3): 4072-4081. DOI: 10.1109/TNNLS.2022.3187612.
- [37] Fang X, Wang C, Xie L, et al. Cooperative pursuit with multi-pursuer and one faster free-moving evader[J]. IEEE Transactions on Cybernetics, 2020, 52(3): 1405-1414. DOI: 10.1109/TCYB.2019.2958548.
- [38] Schroeder de Witt C, et al. Is independent learning all you need in the StarCraft multi-agent challenge?[J]. arXiv preprint arXiv:2011.09533, 2020.
- [39] Wen M, et al. Multi-agent reinforcement learning is a sequence modeling problem[C]// Advances in Neural Information Processing Systems (NeurIPS). 2022: 16509-16521.
- [40] Zhong Y, Kuba J G, Hu S, et al. Heterogeneous-agent reinforcement learning[J]. Journal of Machine Learning Research, 2023, 25: 1-67.
- [41] Ackermann J, Gabler V, Osa T, et al. Reducing overestimation bias in multi-agent domains using double centralized critics[J]. arXiv preprint arXiv:1910.01465, 2019.
- [42] Von Moll A, Casbeer D W, Garcia E, et al. Pursuit-evasion of an evader by multiple pursuers [C]//2018 International Conference on Unmanned Aircraft Systems (ICUAS). 2018: 133-142. DOI: 10.1109/ICUAS.2018.8453470.

四、外文原文

A Policy-Guided Reinforcement Learning Method for Encirclement Control in Multiobstacle Environment

Fandi Gou[✉], Haikuo Du, Chenyu Zhao[✉], and Yunze Cai[✉]

Abstract—The problem of multiagent encirclement with multiobstacle collision avoidance (EMOCA) has been challenging since it is difficult to balance the tradeoff between surrounding a mobile target and avoiding obstacles simultaneously. To address the EMOCA problem, we proposed a novel policy-guided reinforcement learning (RL) method, namely, multiregulator-assisted RL for encirclement control (MRA-RLEC) which leverages the jump-start learning and curriculum learning (CL) mechanism to enhance training efficiency. MRA-RLEC divides the complex encirclement task into a sequence of subtasks, progressively increasing in difficulty. In this process, multiple regulators are utilized to adjust various training aspects, including encirclement condition, obstacle avoidance, and the transition from guide to learned policy execution. Besides, a global encirclement reward decomposition (GERD) method is presented to alleviate reward sparsity, and we design a bidirectional communication protocol to reduce communication. Extensive experiments are carried out to showcase the robustness and superiority of our method, and the practical applicability of MRA-RLEC is demonstrated through experiments conducted in the robot operating system 2 (ROS2)-based simulation platform, Gazebo, using a self-designed omnidirectional vehicle model.

Index Terms—Curriculum learning (CL), encirclement control, multiagent reinforcement learning (MARL), multiobstacle collision avoidance.

I. INTRODUCTION

MULTIAGENT systems have become a focal point of research for their ability to outperform individual agents in tackling sophisticated challenges. They have extensive applications, including search and rescue [1], formation control [2], and pursuit-evasion games [3]. Among these, encirclement control has gained significant attention for its

applicability in both civil and military fields. Encirclement control aims to surround an evading target within a predefined formation, with military applications focusing on target capture using UAVs or USVs [4], [5], as well as surveillance [6] and the escorting of high-value targets [7]. In civilian contexts, it aids in wildlife monitoring [8] and safeguarding ground vehicles [9]. As environments for encirclement become increasingly intricate, preventing collisions with obstacles is crucial for the safety of agents in complicated backgrounds such as dense forests or waters with reefs [10]. Due to the task complexity, the problem of multiagent encirclement with multiobstacle collision avoidance (EMOCA) has posed huge challenges.

Present encirclement control methods are broadly categorized into traditional and reinforcement learning (RL)-based approaches. Traditional approaches include control theory-based and optimization-based methods. In control theory-based methods [11], [12], [13], specific encirclement points are assigned to pursuers, and controllers are designed to drive the distances between pursuers and their target points to zero, thereby achieving encirclement. However, this method is reliant on agent kinematics and can be time-consuming. As for the optimization-based method, the encirclement process is modeled as an optimal control problem where model predictive control is deployed to minimize control effort and encirclement errors [4], [14], [15]. Despite its effectiveness, the computational demands of model predictive control are substantial, which may hinder real-time implementation.

With advancements in artificial intelligence, deep RL (DRL) has been widely applied to complex multiagent tasks [16]. In encirclement control, many studies have put forth DRL-based frameworks to address this challenge [17], [18], [19]. Ma et al. [17] proposed a multiagent deep deterministic policy gradient (MADDPG)-based strategy with a distributed control architecture using local information. However, it simplified scenarios by removing obstacles when encircling a moving target and assuming a static target in obstructed environments. Zhang et al. [18], [19] proposed graph attention networks to model spatial relationships between entities, enabling adaptation to varying numbers of robots and obstacles, but their experiments were limited to a small number of obstacles. Other works, such as [20] and [21], focused on multiagent surrounding problems within a bounded area, where no obstacle is considered and the terminal condition can be easily achieved with the help of environmental boundaries.

Received 13 June 2024; revised 22 December 2024 and 6 March 2025; accepted 27 April 2025. Date of publication 16 May 2025; date of current version 4 September 2025. This work was supported in part by the Open Project of the National Key Laboratory of Air-Based Information Perception and Fusion under Grant 202437 and in part by the Aeronautical Science Foundation of China under Grant 20220001057001 and Grant 20240001057002. (Corresponding author: Yunze Cai.)

Fandi Gou and Haikuo Du are with the Department of Automation, Shanghai Jiao Tong University, Shanghai 200240, China, and also with the Key Laboratory of System Control and Information Processing, Ministry of Education of China, Shanghai 200240, China (e-mail: finley-gou3@sjtu.edu.cn; duhaikuo1013@sjtu.edu.cn).

Chenyu Zhao is with the Department of Electronic Engineering, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: pxwzq6@sjtu.edu.cn).

Yunze Cai are with the Department of Automation, Shanghai Jiao Tong University, Shanghai 200240, China, also with the Key Laboratory of System Control and Information Processing, Ministry of Education of China, Shanghai 200240, China, and also with the National Key Laboratory of Air-based Information Perception and Fusion, Luoyang 471000, China (e-mail: yzcai@sjtu.edu.cn).

Digital Object Identifier 10.1109/TNNLS.2025.3566548

2162-237X © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

Authorized licensed use limited to: Zhejiang University. Downloaded on September 22, 2025 at 12:44:02 UTC from IEEE Xplore. Restrictions apply.

To tackle collision avoidance in multiobstacle environments, prevailing methods include artificial potential field (APF), Voronoi partition, off-line planning, and sense-and-avoid techniques [2]. Among them, APF is widely used in encirclement control due to its inherent symmetry and concise formulation [22], [23]. Voronoi partition-based approach also addresses pursuit-evasion and collision avoidance problems [24], [25]. Specifically, Wang et al. [25] proposed an encirclement algorithm by constructing an evader-centered bounded Voronoi cell (ECBVC), which cooperatively pursues an adversarial evader in an unbounded environment with cluttered obstacles. Nonetheless, both Voronoi partitioning and APF-based methods can encounter issues like local minima, control oscillation, and reliance on comprehensive environmental information, making them less effective in densely obstructed areas. The off-line planning method requires precomputing trajectories and is computationally expensive, especially in dynamic encirclement where targets are constantly moving [26]. Sense-and-avoid techniques are less common in encirclement due to their limited foresight and unreliable state prediction [27].

In DRL-based encirclement methods, most algorithms perform well only when obstacles are few, and the reasons are twofold. For one thing, the collision avoidance mechanism of DRL is to assign negative rewards upon collisions, while the existence of numerous obstacles can lead to prolonged periods of negative feedback and further cause training divergence. For another, the negative rewards from collisions near the terminal phase may counteract the positive rewards from successful encirclement, complicating the learning of the terminal state. Therefore, studies often address obstacle avoidance and encirclement in separate stages. From the above literature, it is evident that even though considerable studies have achieved satisfying encirclement results, it remains a challenge to balance the tradeoff between encircling a moving target and avoiding multiple obstacles.

Fortunately, innovative frameworks combining traditional and DRL methods have emerged to address environmental complexity in multiagent tasks. Sui et al. pioneered the integration of imitation learning with DRL for formation control [2], teaching the basic formation control in the imitation phase and refining them with RL in more complex scenarios. A subsequent study put forward a DRL approach combined with model-based paradigms [28], where they are merged in a parallel and a serial structure, respectively. These methods suggest the potential for tackling the complexities in moving target encirclement and obstacle avoidance.

Inspired by the abovementioned discussions, we propose a novel scheme to combine the model-based and DRL-based methods with the concept of jump-start RL (JSRL) [29]. We devise a guide policy-oriented technique with multiple curriculum learning (CL) regulators to address the challenge of EMOCA. Specifically, a model-based guide policy is introduced to assist training which significantly alleviates the issue of reward sparsity, and is applied to various multiagent RL (MARL) algorithms. Besides, we put forward a two-stage framework capable of adapting to more complex environments. We have also improved the design of the reward

function and the communication protocol. The main contributions of our work are listed as follows.

- 1) A novel multiregulator-assisted RL for encirclement control (MRA-RLEC) scheme is proposed to tackle the encirclement problem of a moving target in an unbounded, multiobstacle environment. The training efficiency and convergence are improved through multiple regulators and the guide policy.
- 2) A global encirclement reward decomposition (GERD) method is presented to alleviate reward sparsity, encouraging the achievement of terminal encirclement condition. Moreover, a bidirectional communication protocol is designed to reduce communication load.
- 3) Comparative experiments are conducted to demonstrate the scalability and robustness of our method. Besides, the model is tested in Gazebo with robot operating system 2 (ROS2), based on a self-designed omnidirectional vehicle platform to verify the probability of real-world application.

The rest of this article is organized as follows. Section II gives the problem formulation of the encirclement scenario. Section III describes the preliminaries. Section IV elaborates the design procedure of the MRA-RLEC method, including guide policy, reward function, and network structure. Section V carries out simulations and experiments under different settings. Finally, Section VI concludes the article.

II. PROBLEM FORMULATION

A. Encirclement Control Scenario

The encirclement control problem is considered within a 2-D unbounded space $\mathcal{W} \subseteq \mathbb{R}^2$ with M static obstacles, where a collective of N pursuers are required to encircle a mobile target while avoiding collision. The set of indices for the pursuers and obstacles are represented by $\mathcal{I} = \{1, 2, \dots, N\}$ and $\mathcal{O} = \{1, 2, \dots, M\}$, respectively, with individual pursuers and the target being referred to as P and T . For $i \in \mathcal{I}$, use $\mathcal{N}(i)$ to represent the neighbors of P_i . Suppose $i, j \in \mathcal{I}$, the included angle of P_i and P_j relative to the target T is defined as $\alpha_{i,j}$, and the distance between P_i and P_j is defined as $d_{i,j}$.

Definition 1 (Neighbors): When there are no other pursuers between $\alpha_{i,j}$, then P_i and P_j are defined as neighbors. Particularly, when the vector cross product $\vec{TP}_i \times \vec{TP}_j > 0$, P_j is deemed as the right neighbor of P_i , and the left neighbor is defined as the opposite. We define the direction perpendicular to the plane of $\mathcal{W}$ and pointing upward as the positive direction.

All the agents and obstacles are modeled in a circular form, where r^i and δ^i denote the radius and the safety buffer distance of each entity. At time t , the position and velocity of P_i are denoted as $p^i = [p_x^i, p_y^i]$ and $v^i = [v_x^i, v_y^i]$, $i \in \mathcal{I}$, while target's position and velocity are denoted as p^T and v^T . We use the superscript T to denote variables associated with the target in this article. Denote the obstacle i as O_i , $i \in \mathcal{O}$, and $O_i = [x_o^i, y_o^i, r_o^i, \delta_o^i]$. The dynamics of each agent are described in a second-order integrator form as

$$\begin{cases} \dot{p}^i = v^i \\ \dot{v}^i = u^i \end{cases} \quad (1)$$

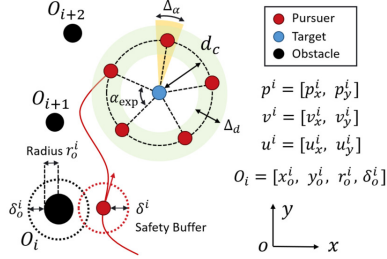


Fig. 1. Illustration of encirclement control scenario. The bands in light green and light yellow indicate distance bias and angle bias, respectively.

where $u^i = [u_x^i, u_y^i]$ denotes the acceleration of P_i , and each component of u^i has a limit $u_{\max}^i$. The same is true for the target. Besides, there are constraints in the maximum velocity for pursuers and the target, which means their velocity component cannot exceed $v_{\max}$ and $v_{\max}^T$, respectively. To guarantee the successful encirclement, we set $u_{\max} = u_{\max}^T, v_{\max} > v_{\max}^T$ in this article.

Initially, pursuers are aligned in a straight line with the target positioned randomly in a circular area ahead of them. Both pursuers and the target have zero initial speeds. For the encirclement condition, all the pursuers are supposed to be uniformly distributed around a circle centered by the target, with capture distance d_c as the radius. The expected encirclement angle for P_i with its neighbors is set to be $\alpha_{exp} = (2\pi/N)$. To facilitate learning, a slight bias in the final encirclement angle Δ_α and distance Δ_d is permitted. The distance between P_i and T is denoted as $d_{i,T}$.

Definition 2 (Encirclement Condition): For any pursuer $i \in \mathcal{I}$, if the following conditions hold at the same time, the encirclement is considered as accomplished:

$$\begin{aligned} |\alpha_{i,j} - \alpha_{exp}| &< \Delta_\alpha, \quad j \in \mathcal{N}(i) \\ |d_{i,T} - d_c| &< \Delta_d. \end{aligned} \quad (2)$$

Upon achieving the encirclement, all agents decelerate to static. A visual illustration of this encirclement condition is depicted in Fig. 1. Moreover, any pursuer cannot collide with other pursuers or obstacles and a safety buffer distance is necessary for the training. When the condition $\|p^i - p^j\| < r^i + r^j + k_\delta \times (\delta^i + \delta^j)$ is satisfied, where k_δ is a coefficient, the pursuers will get relevant punishment. The average distance between obstacles is set close to the capture distance d_c , making the encirclement scenario more complex.

B. ROS2-Based Simulation Platform

We utilize a high-fidelity autonomous navigation simulator, Gazebo, to construct a semi-physical experimental setting. A self-designed omnidirectional vehicle model platform is deployed for encirclement experiments, as shown in Fig. 2. Furthermore, we employ the open-source ROS2 to facilitate communication between our system and the Gazebo simulator [30]. To be noted, our algorithms are extendable to other platforms such as UAVs and USVs through kinematic adjustments.

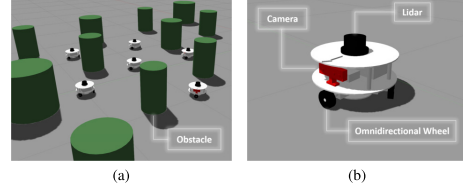


Fig. 2. Physical engine simulator Gazebo with ROS2. (a) Encirclement environment. (b) Self-designed omnidirectional vehicle model.

TABLE I
PARAMETERS OF OMNIDIRECTIONAL CAR

Feature	Description	Pursuer	Evader
m	Mass (kg)	0.8	0.8
r	Vehicle Radius (m)	0.15	0.15
$v_{\max}$	Max velocity (m/s)	0.5	0.25
$u_{\max}$	Max acceleration (m/s ²)	0.5	0.5
I_{xx}	Inertia moment (kg · m ²)	0.006	0.006
I_{yy}	Inertia moment (kg · m ²)	0.006	0.006

The inverse kinematic of the omnidirectional vehicle is calculated to obtain the speed of each wheel according to the desired velocity along each axis. Within the Gazebo simulation, the wheels are treated as rotational links, with their velocities being controlled by the simulator's inbuilt dynamics engine. The parameters of the model vehicle are shown in Table I.

III. PRELIMINARIES

A. Multiagent RL

MARL is an effective approach to solving sequential decision-making tasks for multiple agents, which can be modeled through a Markov decision process. This process is expressed by a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{T}, R, \gamma \rangle$, where $\mathcal{S}$ is the state set and $s \in \mathcal{S}$ represents the global state of the environment. $\mathcal{A} = [\mathcal{A}_1, \dots, \mathcal{A}_n]$ is the set of joint action and $\mathbf{a} \in \mathcal{A}$, where $\mathbf{a} = [a_1, \dots, a_n]$. The component $a_i \in \mathcal{A}_i$ is the action taken by agent i ($i \in \{1, \dots, n\}$). $\mathcal{T}: \mathcal{S} \times \mathcal{A} \times \mathcal{S} \mapsto [0, 1]$ represents the state transition function and $R: \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}$ is the reward. $\gamma \in [0, 1]$ is the discount factor determining the importance of future rewards. The objective of the agents is to adopt a policy π that maximizes the expectation of discounted cumulative rewards

$$J(\pi) = E_\pi \left[\sum_{t=0}^{\infty} \gamma^t R^t(s^t, a_1^t, \dots, a_n^t) \right] \quad (3)$$

where R^t is the reward that all agents obtain at time t .

Multiagent proximal policy optimization (MAPPO) is an extension of PPO in the multiagent scenarios [31], and is a policy gradient-based MARL method that utilizes the centralized training and decentralized execution framework [32]. The policy π is expressed as a parameterized deep neural network π^θ , where θ represents the parameters. The evaluation network is parameterized by ϕ . The parameters are updated by

maximizing the objective function $J(\pi^\theta)$ with gradient $\nabla_\theta J(\pi^\theta)$ as the optimization direction

$$\nabla_\theta J(\pi^\theta) = \nabla_\theta \log(\pi^\theta(a'|s')) \sum_{t'=t}^{\infty} \gamma^{t'-t} R^{t'}(s', a'). \quad (4)$$

When updating parameters, the importance sampling ratio $l_t(\theta)$ is introduced to improve sample efficiency

$$l_t(\theta) = \frac{\pi^\theta(a'|s')}{\pi^{\theta_{\text{old}}}(a'|s')} \quad (5)$$

where $l_t(\theta)$ represents the likelihood ratio and θ_{old} is the policy parameter before the update. The loss function in the policy network is defined as follows:

$$L^\pi(\theta) = E[\min(l_t(\theta)\hat{A}_t^\theta(s', a'), \text{clip}(l_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t^\theta(s', a'))] \quad (6)$$

where $\hat{A}_t^\theta$ is the generalized advantage estimate and the clip function $\text{clip}(l_t(\theta), 1 - \epsilon, 1 + \epsilon)$ constraints the $l_t(\theta)$ within $[1 - \epsilon, 1 + \epsilon]$ to avoid over update of the policy. The loss of value network is denoted by $L^V(\phi)$.

Compared with other MARL algorithms, MAPPO has superior performance in convergence and sampling efficiency. In this article, we select a variant of MAPPO with recurrent layers, rMAPPO, as the basic training algorithm. The details will be presented in Section IV.

B. Curriculum Learning

CL is inspired by the process of human learning, which progressively trains the model on tasks of increasing difficulty, starting from simpler to more complicated data. CL paradigm has a wide application scope in the realm of DRL [33], since it enhances sample efficiency, organizes model exploration, and solves sparse reward problems. Moreover, it contributes to accelerating the convergence of the learning process [34].

One specific application of CL in DRL is to control the data difficulty during the data collection, and a curriculum can be generally designed from the following aspects: initial states, reward functions, goals, environments, and opponents [35]. In the initial training stage, an easy curriculum is designed which simplifies the task to facilitate the acquisition of rewards. As the training advances, the task gradually restores to its original difficulty, thereby enhancing the model's adaptability to more challenging scenarios.

C. Jump-Start RL

Prior to the introduction of JSRL, the concept of base controllers had been utilized to facilitate training in the control of robotic arms [36]. Improving upon this concept, JSRL offers a compelling solution to address the sparse reward challenge encountered in RL. It initializes an RL policy with a preexisting policy and is compatible with any RL approach [29]. JSRL employs two policies to solve a specific task: a guide policy π^g and a learned policy π^θ . The guide policy is controlled by a curriculum, and is progressively withdrawn as the performance of learned policy increases. Both empirical and theoretical analyses have demonstrated that

JSRL significantly outperforms the existing imitation learning algorithms due to its advantage in sample efficiency.

The mechanism of JSRL is described as follows. Let H symbolize the RL task horizon, and denote the model performance threshold as β_{thre} . Initially, select a sequence of guide-steps to roll out the guide policy, $(H_1, H_2, \dots, H_n)$, where $H_i \in \{H\}$ denotes the guide-steps in the present iteration. Denote h as the iterator over the guide-step sequence $(H_1, H_2, \dots, H_n)$. During the training, roll out π^g for h steps and π^θ for the remaining $H - h$ steps, which is $\pi_{1:h} = \pi^g$, $\pi_{h+1:H} = \pi^\theta$. After each parameter update, the model performance is evaluated through a function $V^*(\pi)$. When $V^*(\pi) > \beta_{\text{thre}}$, the iterator h advances to the subsequent value in the guide-step sequence. This process of training continues in a loop until the entire sequence of guide-steps is traversed.

IV. POLICY-ORIENTED LEARNING FOR ENCIRCLEMENT CONTROL

This section gives a detailed design procedure for MRA-RLC and the MARL framework. The overall training architecture is depicted in Fig. 3. The training is divided into multiple threads, within each thread the trajectory data are collected and adjusted into a batch before optimizing parameters.

A. Guide Policy for JSRL

The APF is a classic path planning algorithm, which regards the target as an attractive source and the obstacles as repulsive sources, guiding the agent to move along the resultant force. However, the APF method has significant drawbacks as previously discussed in Section I. Nonetheless, APF is still able to drive the pursuers toward the target, which reduces the exploration space to some extent. In this regard, we employ APF as the guide policy for JSRL. To facilitate the formation of the desired encirclement and avoid local minima, we only account for three distinct forces: the repelling force between pursuers ($f_r^{i,j}$), between pursuers and obstacles ($f_o^{i,k}$), and the attraction force between pursuers and target ($f_a^{i,T}$), where $i, j \in \mathcal{I}$ and $k \in \mathcal{O}$. In the desired encirclement formation, the distance between neighboring pursuers is $L = 2 d_c \sin(\alpha_{\text{exp}}/2)$. The forces are generated as follows:

$$f_r^{i,j} = \begin{cases} k_r(L - d_{i,j}) (p^j - p^i), & d_{i,j} \leq L \\ 0, & d_{i,j} > L \end{cases} \quad (7)$$

$$f_o^{i,k} = \begin{cases} k_o(L_s - d_{i,k}) (p^i - p^k), & d_{i,k} \leq L_s \\ 0, & d_{i,k} > L_s \end{cases} \quad (8)$$

$$f_a^{i,T} = \frac{k_a(d_{i,T} - d_c)}{d_{i,T}} (p^T - p^i). \quad (9)$$

To alleviate vibration, a damping term containing v^i is added. The total force exerted on pursuer i is

$$u^i = \sum_{j=1, j \neq i}^N f_r^{i,j} + \sum_{k=1}^M f_o^{i,k} + f_a^{i,T} - k_b v^i \quad (10)$$

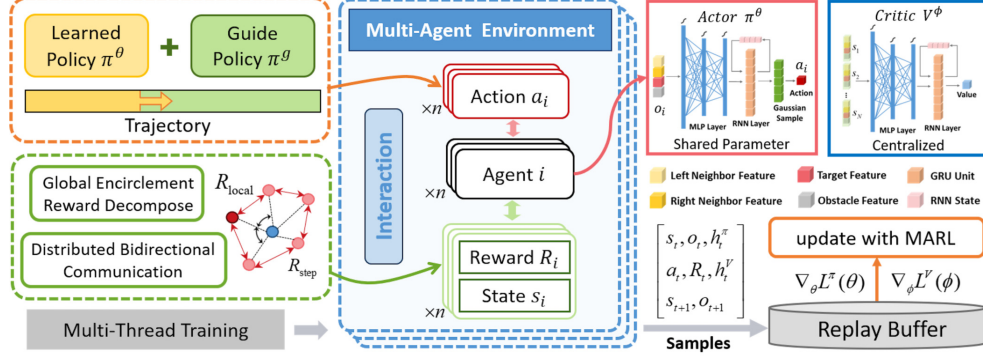


Fig. 3. Overall architecture of MRA-RLC.

where k_r, k_o, k_d , and k_b are positive control gains, and L_s is a collision avoidance related threshold parameter.

As training advances, the initial steps of guide policy are gradually reduced, eventually giving way to the learned policy. In JSRL, the reduction of guide steps is based on a performance threshold β_{thre} , requiring the learned policy to meet specific criteria before reduction occurs. However, evaluating the policy performance brings uncertainty to the training, making it difficult to predict when the policy will meet the target level, potentially impacting subsequent training tasks. To address this, we introduce a nonlinear regulator to manage the guide policy's phase-out, ensuring a consistent, predetermined training duration for JSRL.

The designed regulator is intended to follow an S-shaped curve and here are the reasons. When the JSRL starts, pursuers require more space to explore under the guide policy, hence the slow reduction of guide-steps is necessary to encourage broader exploration. As the training progresses, the pursuers have identified a high reward terminal state, allowing for a more rapid reduction in guide-steps. However, near the end of JSRL, a slow retreat of the guide policy is essential due to the lasting influence of initial states on the future trajectory. If the learned policy has not fully grasped the optimal behavior around the initial states, abrupt reductions at this stage could impair the learning trajectory. Thus, the guide steps' reduction follows an S-shaped curve, ensuring a smooth transition from guided to learned policy execution.

To construct the S-shaped curve for the JSRL regulator, we select two odd functions as the foundation: $\tanh(x)$ and x^3 . Since $\tanh(x)$ inherently has an S-shaped profile, the x^3 term is utilized to adjust the function value at the endpoints. To modulate the steepness of the S-curve, a scaling factor k_s is introduced into the variable x . Next, map both the horizontal and vertical coordinates to the range of 0–1 through function transformations. The final regulator function, which indicates the rate at which the guide policy steps are reduced, is formulated as follows:

$$f(x) = \frac{1}{2} \left(-\tanh(k_s X) - \varepsilon X^3 + 1 \right) \quad (11)$$

where $X = 2x - 1$, $\varepsilon = 1 + \tanh(-k_s)$, and $f(\cdot)$ represents the JSRL regulator function.

Let E denote the total training episodes, $c_g \in (0, 1)$ is a ratio coefficient, $E_g = c_g E$ is defined as the training episodes of JSRL, and E_t denotes the current training episodes that have been completed. The ratio $E_t/E_g \in [0, 1]$ serves as an indicator of the progress made in the JSRL phase. Let M indicate the total steps in an episode, ρ_0 indicate the initial proportion of steps executed by guide policy, then the current number of guide-steps M^g is calculated by $M^g = M \times \rho_0 \times f(E_t/E_g)$. The remaining steps M^θ executed by the learned policy is given by $M^\theta = M - M^g$. When E_t/E_g reaches 1, the JSRL process is completed, and the training proceeds with the learned policy taking over all steps for the remainder of the training episodes.

B. Training Through Multiple Curriculum Regulators

Since EMOCA is a complex task, relying solely on JSRL to achieve encirclement is challenging. Given this, we implement a two-stage training strategy with a *Base* model trained in the first stage, and train a more complicated scenario in the *Fine Tune* stage. We divide the *Base* training process into multiple subtasks according to different goals. Each subtask is trained through CL during a certain process, from easy to hard. After that, we consider increased target speeds and various escape policies in the *Fine Tune* phase. The evolution of the entire training progress with multiple regulators is shown in Fig. 4.

There are four CL regulators in the training of *Base*, which are the JSRL regulator, angle regulator, distance regulator, and obstacle avoidance regulator. The JSRL regulator helps to reduce the exploration space, as mentioned previously. The angle regulator and distance regulator control the difficulty of terminal encirclement conditions by adjusting the angle bias and distance bias from large to small. This adjustment allows the pursuers to earn rewards early in the training process, even if they have not achieved the exact encirclement conditions, thereby mitigating the issue of sparse rewards. The training episodes of these two regulators are $E_e = c_e E$, and both

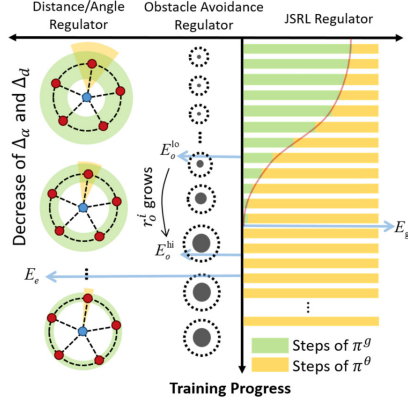


Fig. 4. Illustration of multiregulators that assist training. From left to right is the evolution of angle and distance bias, obstacle radius, and guide policy steps.

regulators are defined in linear form

$$\Delta_\alpha = \begin{cases} \Delta_\alpha^{sta} - (\Delta_\alpha^{sta} - \Delta_\alpha^{exp}) E_t / E_e, & E_t \leq E_e \\ \Delta_\alpha^{exp}, & E_t > E_e \end{cases} \quad (12)$$

$$\Delta_d = \begin{cases} \Delta_d^{sta} - (\Delta_d^{sta} - \Delta_d^{exp}) E_t / E_e, & E_t \leq E_e \\ \Delta_d^{exp}, & E_t > E_e \end{cases} \quad (13)$$

where the superscript “sta” indicates the start value, while “exp” indicates the expected value. Similar to c_e , the parameter c_e is a ratio that defines the proportion of episodes influenced by the regulator. The same applies to the later-mentioned parameters c^{lo} and c^{hi} .

As for the obstacle avoidance regulator, it modulates the sizes of obstacles within a specific training phase. Given that obstacles significantly influence the encirclement process, we introduce a mechanism where obstacles are not immediately presented at full size during training. Instead, they emerge incrementally, starting as mere points and progressively expanding to reach their original sizes over time. This process occurs between the training phase E_o^{lo} and E_o^{hi} , where $E_o^{lo} = c^{lo}E$ and $E_o^{hi} = c^{hi}E$. For $i \in \mathcal{O}$, the radius of obstacle i during this period is defined as follows:

$$r_o^i = \begin{cases} 0, & E_t < E_o^{lo} \\ r_o^{exp} \times \frac{E_t - E_o^{lo}}{E_o^{hi} - E_o^{lo}}, & E_o^{lo} \leq E_t \leq E_o^{hi} \\ r_o^{exp}, & E_t > E_o^{hi} \end{cases} \quad (14)$$

where r_o^{exp} is the original radius of obstacle i .

In the *Fine Tune* phase, the target escapes in different policies, with its speed gradually increasing through a linear curriculum regulator. Besides, the penalty for collision is raised, and the learning rate is set smaller. The comprehensive MRA-RLEC methodology is presented in Algorithm 1.

Algorithm 1 MRA-RLEC

Input: Total training episodes E , maximum episode steps M , learning rate l_r , PPO gradient update times K

Output: Actor network θ and critic network ϕ

- 1: Initialize actor parameter θ and critic parameter ϕ , define E_t as the current training episodes
- 2: **for** $E_t = 1$ to E **do**
- 3: Compute E_t/E and select relevant regulators. Compute Δ_α and Δ_d by equation (12) and (13). Calculate r_o^i through (14) and set up encirclement scenario.
- 4: Compute E_t/E_g and obtain guide step M^g via JSRL regulator function (11)
- 5: **for** step = 1 to M **do**
- 6: Execute the joint action $a_{1:N}^t = \pi(o_1, o_2, \dots, o_N)$. If step < M^g , then set $\pi = \pi^g$, else set $\pi = \pi^\theta$
- 7: Run the mixed policy of π^g and π^θ for M timesteps in total, collecting $\{o, a, r\}$ for each agent and obtain a trajectory τ . Store τ into replay buffer
- 8: **end for**
- 9: Estimate advantage function A_t from τ in replay buffer
- 10: **for** $k = 1$ to K **do**
- 11: Calculate policy loss $L^\pi(\theta)$ through equation (6)
- 12: Update network parameters for each agent:
- 13: $\theta' \leftarrow \theta - l_r \nabla_\theta L^\pi(\theta)$, $\phi' \leftarrow \phi - l_r \nabla_\phi L^V(\phi)$
- 14: **end for**
- 15: **end for**

C. Design of MARL Framework

1) *Reward Function*: As for the EMOCA problem, it is difficult to describe the system reward only considering the final state, because the final state is hard to trigger which generally leads to sparse reward issues. To counter this, we propose the GERD method to allow rewards for partial encirclement, and the reward is designed as follows: before encirclement is finished, each pursuer receives a step reward R_{step} as an individual reward. To motivate local cooperation, pursuer i will receive a local surrounding reward R_{local} as long as it satisfies the condition in (2), at which point we consider pursuer i to be finished. Once finished, the pursuer keeps adjusting its position to maintain the partial encirclement and get continual rewards. This process continues until all pursuers are finished, ultimately leading to a complete encirclement. Upon the successful encirclement by all pursuers, each pursuer will receive a global capture reward R_{global} . The reward of pursuer i is defined as

$$R_t = \begin{cases} R_{global}, & \text{all pursuers are finished} \\ R_{local}, & \text{pursuer } i \text{ is finished} \\ R_{step}, & \text{otherwise.} \end{cases} \quad (15)$$

In the course of encirclement, the pursuer team needs to implement both surrounding and chasing actions simultaneously. Considering the symmetrical nature of the encirclement task, the sum of all the relative positional vectors between the pursuers and the target offers a natural representation of the encirclement state. Ideally, upon achieving a perfect encirclement, the magnitude of this aggregate vector should

converge to zero. Denote this as the formation reward R_f , which encourages both surrounding and chasing actions at the same time. To normalize this reward, R_f is defined as

$$R_f = \exp\left(-k_1 \cdot \left\| \sum_{i=1}^N (p^T - p^i) \right\|\right) - 1. \quad (16)$$

To ensure that the pursuers achieve the designated capture distance d_c , we introduce a distance reward R_d defined as follows:

$$R_d = \exp\left(-k_2 \cdot \sum_{i=1}^N (d_{i,T} - d_c)^2\right) - 1 \quad (17)$$

where k_1 and k_2 are tunable coefficients. Finally, we establish a fixed penalty for collisions, denoted as R_l . The step reward R_{step} is then formulated as a weighted sum of the formation reward R_f , the distance reward R_d , and the collision penalty R_l

$$R_{\text{step}} = w_1 R_f + w_2 R_d + w_3 R_l \quad (18)$$

where w_1 , w_2 , and w_3 are weights that determine the relative importance of each component in the overall reward. Specifically, we set $R_l = -3$ as the penalty for collisions, $R_{\text{local}} = 5$ as the reward for local cooperation, and $R_{\text{global}} = 10$ as the reward for successfully completing the encirclement. These values are chosen to create a balanced incentive structure that guides the learning process toward effective encirclement policies while avoiding collisions.

2) *State Space and Communication Protocol*: In most multiagent cooperative tasks, it is ideal for agents to communicate with all other agents to obtain complete information. However, in dynamic multiagent tasks such as encirclement, communicating with all agents is resource-consuming and puts forward higher requirements for communication bandwidth. Considering that the terminal encirclement state is a standard circle, we propose a distributed bidirectional communication protocol, so that each pursuer only exchanges information with its left neighbor and right neighbor.

Define S_t^i as the state vector for pursuer i at time t , and its neighbors are pursuer $j, k \in \mathcal{N}(i)$. S_t^i consists of four distinct components, that are $S_t^{i,j}$, $S_t^{i,k}$, $S_t^{i,T}$, and $S_t^{i,o}$. Specifically, $S_t^{i,j}$ and $S_t^{i,k}$ represent the relative states of pursuer i to its two neighbors. For instance, the state relative to its left neighbor P_j is given by $S_t^{i,j} = [\alpha_{i,j}, d_{i,j}, v_{i,j}, Q_{i,j}, Q_{j,i}]^T$, where $v_{i,j}$ is the relative velocity's magnitude, and $Q_{i,j}$ is the relative bearing angles from P_i 's velocity to the position vector pointing from P_i to P_j , as illustrated in Fig. 5. The right neighbor state $S_t^{i,k}$ has the same components. The term $S_t^{i,T}$ represents the observed state of target, defined as $S_t^{i,T} = [d_{i,T}, d_{i,T}, Q_{i,T}, \omega, \dot{\omega}]^T$, where $Q_{i,T}$ is the bearing angle between pursuer's velocity and the relative position vector, while ω denotes the target's rotational velocity from the perspective of pursuer i . The observation of the closest obstacle o is captured by $S_t^{i,o} = [x_{i,o}, y_{i,o}, d_{i,o}, d_{j,o}, d_{k,o}, r_o, v_o^x, v_o^y]^T$, with $x_{i,o}$ and $y_{i,o}$ reflecting the positional differences along the x - and y -axes between pursuer i and

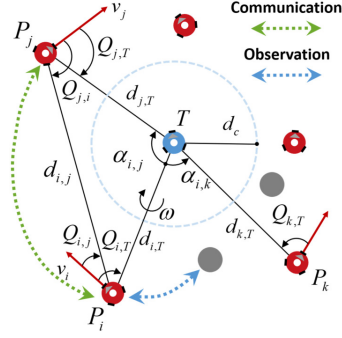


Fig. 5. Schematic of encirclement features.

obstacle o , respectively. Finally, a concatenated state S_t^i is derived

$$S_t^i = \left[(S_t^{i,j})^T, (S_t^{i,k})^T, (S_t^{i,T})^T, (S_t^{i,o})^T \right]^T \quad (19)$$

and S_t^i is the immediate input to the actor network.

3) *Network Structure*: Given that MAPPO is under actor-critic architecture, each pursuer is equipped with an actor network that maps the observed states into its actions, and the whole system has a centralized critic network that evaluates the overall state. Both actor network and critic network are based on fully connected and recurrent neural network (RNN) structure, where hidden states $h^{x,v}$ in RNN restore historical information and promote decision-making. The fully connected structure is constructed using a multilayer perceptron (MLP), and the gate recurrent unit is employed as the RNN layer to capture temporal dependencies. Moreover, $\tanh(\cdot)$ is selected as the activation function, and all the actor networks have the same structure and share the model parameters.

V. SIMULATIONS AND RESULTS

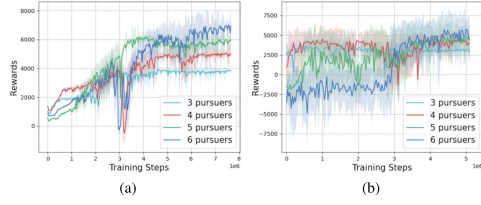
In this section, we apply the MRA-RLEC to different MARL algorithms to verify its robustness and effectiveness, and we train the model with different numbers of agents to show the scalability of the proposed method. Without loss of generality, we choose the default number of agents as five, i.e., $N = 5$. Besides, we compare MRA-RLEC with other encirclement algorithms to validate its efficacy. We also conduct ablation tests and demonstrate the practicality of our method through experiments in Gazebo using self-designed omnidirectional vehicles.

A. Simulation Settings

In the training process of *Base*, we set the regulator parameters c_g , c_e , c^{lo} , and c^{hi} to 0.6, 0.75, 0.45, and 0.75, respectively. The discount factor γ is set to 0.985, and the clip coefficient ϵ is set to 0.15. The weights of step reward are set to $\omega_1 = 0.3$, $\omega_2 = 0.3$, and $\omega_3 = 0.4$. In terms of pursuit dynamics, the maximum velocity of pursuer v_{max} is set

TABLE II
NETWORK AND TRAINING RELATED PARAMETERS

Name	Parameter
Actor network MLP layers	2
Actor network hidden units	128, 64
Actor network RNN hidden units	32
Critic network MLP layers	3
Critic network hidden units	256, 128, 64
Critic network RNN hidden units	32
Length of RNN data chunk	10
Updating batch size	100
Learning rate in <i>Base</i>	0.0002
Learning rate in <i>Fine Tune</i>	0.00015
PPO update epoch	15


 Fig. 6. Training rewards of different numbers of pursuers. (a) Reward of *Base*. (b) Reward of *Fine Tune*.

to 0.5 m/s, while $v_{\max}^T$ is set to 0.15 m/s in the *Base* and 0.25 m/s in *Fine Tune*. The acceleration limits, represented by $u_{\max}$, are set to 0.5 m/s² for both sides.

For encirclement condition parameters, we set $d_c = 1$ m, $\Delta_\alpha = 0.3$ rad, and $\Delta_d = 0.1$ m. We also set the parameters in the guide policy as $k_r = 4.5, k_o = 4.0, k_a = 2.0$, and $k_b = 1.5$, and the collision distance threshold is $L_s = 0.3$ m. Furthermore, the maximum steps of each episode are set to 200, and each simulation time interval dt represents 0.1 s. The *Base* phase is designed to undergo a total of 7.68×10^6 training steps, while the *Fine Tune* phase comprises 5.12×10^6 steps. In addition, the experiment is applied on a computer with an Intel Xeon E5-2678 v3 CPU, 128 GB of memory, and Nvidia GTX 1080Ti GPU. Other network and training-related parameters are illustrated in Table II.

B. Scalability and Robustness of MRA-RLEC

To verify the scalability of MRA-RLEC, we implement our algorithm on the environments presented in Section II with numbers of pursuers varying from 3 to 6. During the training of *Base*, we set the JSRL parameter $k_s = 2.0$. In the *Fine Tune* stage, we increase the maximum target speed and use a linear CL regulator to gradually increase the speed. The potential field policy (PFP) [37] is selected as the escape policy for the target, and experiments are conducted under the rMAPPO scheme.

The visualizations of rewards in the two stages are shown in Fig. 6. As seen from Fig. 6(a), in *Base* the reward curves usually experience a drop before going upward. This phenomenon can be attributed to the retreat of the guide policy, which can

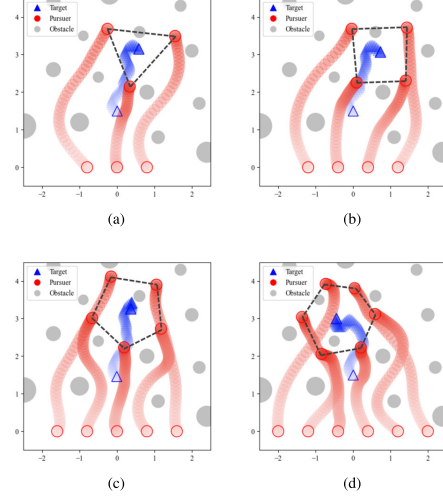


Fig. 7. Trajectories of different numbers of pursuers via MRA-RLEC, trained with rMAPPO. The axis unit is meters (m). (a) 3 pursuers encirclement. (b) 4 pursuers encirclement. (c) 5 pursuers encirclement. (d) 6 pursuers encirclement.

lead to a decrease in training rewards. While in *Fine Tune*, there is no guide policy, and the reward steadily increases. The convergence of reward in Fig. 6(a) and (b) indicates that MRA-RLEC is scalable to different numbers of pursuers.

The model is further validated, with the trajectories of the encircling process visualized in Fig. 7. The radius of all the entities is consistent with the simulation parameters, and the trajectory colors vary in depth to indicate the chronological sequence of movements. Fig. 7 illustrates the successful encirclement formation, where the pursuers form a regular polygon centered by the target, satisfying $|a_{i,j} - a_{\text{expl}}| < \Delta_\alpha, j \in \mathcal{N}(i)$ and $|d_{i,T} - d_c| < \Delta_d$. Besides, any apparent overlap in the trajectories of the pursuers is a result of differences in timing and not due to physical collisions. The figure also confirms that there is no intersection between the trajectories and the obstacles, indicating that no collisions occur between the agents and the obstacles during the encircling process.

To verify the robustness of MRA-RLEC, we apply it to various MARL algorithms to train a five-pursuer encirclement scenario. Among these algorithms, the independent proximal policy optimization (IPPO) method provides a scheme where each agent estimates its local value function independently [38]. MAPPO has no RNN layers compared with rMAPPO, and in multiagent transformer (MATransformer) the networks use transformer structure rather than convolutional neural networks [39]. In addition, heterogeneous-agent PPO (HAPPO) [40], MADDPG [32], and multiagent twin delayed DDPG (MATD3) [41] are all prominent algorithms in the domain of MARL. The reward visualization is shown in Fig. 8, and by all methods the training rewards converge properly under our framework.

TABLE III
 REWARD INDICES WITH MRA-RLEC BASED ON DIFFERENT MARL ALGORITHMS

Training method	Average reward ($\uparrow$)	Training steps to achieve average reward ($\downarrow$)	Maximum reward ($\uparrow$)	Average epoch standard deviation ($\downarrow$)	Standard deviation of average reward ($\downarrow$)
rMAPPO	4345.177	2630400	6390.158	599.2639	1908.99
MAPPO	2870.418	2598400	5941.362	567.4647	1942.78
IPPO	4141.887	2342400	6372.418	727.4509	1790.40
HAPPO	3366.252	1862400	5518.821	612.6576	1533.79
MATransformer	4290.802	2374400	6312.289	864.2967	1304.26
MADDPG	4017.395	2438400	6836.482	405.1290	2414.49
MATD3	3977.205	2790400	6848.849	252.7119	2658.52

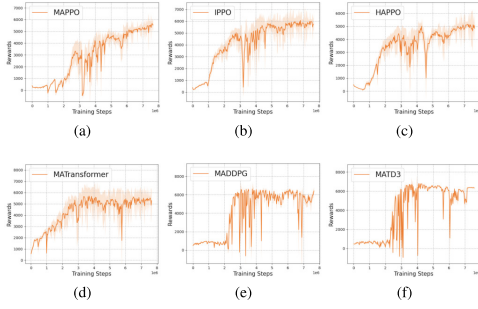


Fig. 8. Training rewards of different MARL algorithms trained with MRA-RLEC. (a) MAPPO. (b) IPPO. (c) HAPPO. (d) MATransformer. (e) MADDPG. (f) MATD3.

Table III compares training metrics of different algorithms, utilizing $\uparrow$ for higher desirable values and $\downarrow$ for the opposite. The epoch standard deviation, indicating reward variance in an epoch, is shown in the shaded area under the reward curve. The standard deviation of the average reward across all epochs is a measurement of reward fluctuation, which reflects the policy stability during training. Among these methodologies, off-policy algorithms (MADDPG and MATD3) distinguish between target policy and behavior policy, and are more likely to explore the global optimal policy. This can lead to more reward oscillation since they explore more policies. Therefore, off-policy algorithms have higher maximum reward and standard deviation in average reward. In comparison, PPO-based algorithms are on-policy methods, which may settle for local optima. Furthermore, the presence of RNN layers, as seen in the contrast between MAPPO and rMAPPO, significantly affects average rewards.

C. Ablation Test

We first conduct an ablation test to demonstrate the effectiveness of the JSRL regulator. In this experiment, a five-pursuers encirclement scenario is focused, and the JSRL regulator is introduced in the training process. The initial ratio of guide policy steps ρ_0 is set to 0.8. By varying the JSRL parameter k_s , we generate a spectrum of regulator functions, each of which is utilized to train the model.

As seen from Fig. 9, the training reward achieves convergence when the proportion curve exhibits a convex shape in

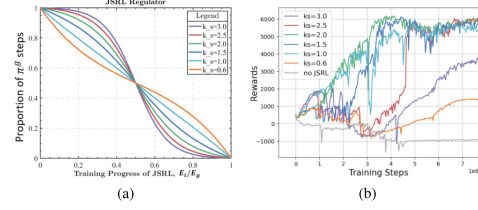
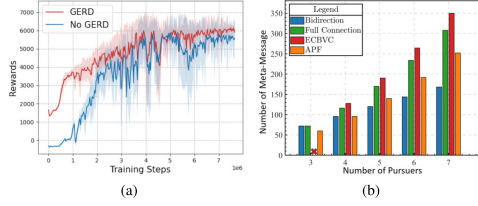

 Fig. 9. Ablation test of JSRL regulators. (a) Proportion curves of guide policy steps in different k_s . (b) Training rewards under different JSRL regulators.


Fig. 10. Comparative results in reward function and communication design. (a) Training reward with and without GERD. (b) Meta-messages cost in different encirclement algorithms.

the initial stage, where $k_s > 1$. Nevertheless, the proportion curve cannot be overly convex, as this could cause an abrupt withdrawal of the guide policy in the middle, thereby affecting the exploration. This is well illustrated by the reward curves $k_s = 2.5$ and $k_s = 3.0$. Conversely, when the proportion curve is concave, with $k_s < 1$, the model's convergence performance deteriorates. This aligns with the analysis that the model requires additional time to explore the terminal condition in the initial stage. Besides, we thoroughly withdraw the guide policy after a few episodes of training to demonstrate the influence of JSRL, and the gray curve shows that the rewards cannot converge in such a complex scenario.

Next, we test the effectiveness of GERD. To simplify the scenario, we remove obstacles and assist in training with the JSRL regulator. The first experiment incorporated GERD, granting local rewards for partial encirclement. In the comparative experiment, rewards are given only when complete encirclement is formed. As illustrated from Fig. 10(a), the introduction of GERD accelerates the increase in rewards and leads to a higher convergence value. In contrast, the model

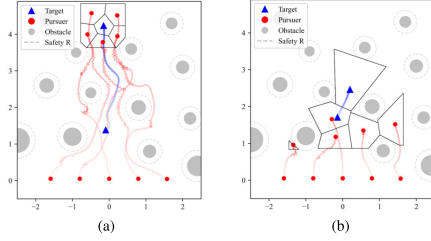


Fig. 11. Encirclement trajectories via ECBVC. (a) Successful encirclement. (b) Unsuccessful encirclement.

trained without GERD spends more time exploring for full encirclement, yielding lower initial rewards. This comparison highlights GERD's role in enhancing training efficiency and convergence.

We further evaluate the communication efficiency of our bidirectional protocol using the meta-message [20], which could measure the algorithms' communication load complexity in a single timestep. Define N_p , N_T , and N_o as the observation dimensions for pursuers, the target, and obstacles, respectively. The total communication meta-message under the bidirectional protocol is calculated as $(2N_p + N_T + N_o)N$, the load complexity is $O(N)$. While in the fully connected situation, the communication load is $((N-1)N_p + N_T + N_o)N$, and the complexity is $O(N^2)$. The ECBVC and APF methods require each pursuer to have full knowledge of all entities, resulting in meta-message scales with the square of N , the load complexity is also $O(N^2)$. The comparison result is shown in Fig. 10(b), with the ECBVC algorithm lacking support for scenarios $N = 3$. This analysis underscores the efficiency of our bidirectional protocol for conserving communication resources in encirclement tasks.

D. Comparison With Other Encirclement Algorithms

In this section, we compare MRA-RLEC with other encirclement algorithms. We utilize rMAPPO, MADDPG, and MATD3 as fundamental RL frameworks and integrate MRA-RLEC on these training bases, with APF serving as the guide policy. In comparison, we have selected three typical traditional approaches: ECBVC [25], pseudo-force-based encirclement control (PFEC) [23], and formation-based encirclement control (FBEC) [11]. The performance is evaluated by the following key metrics: average step reward, encirclement time, danger rate, and successful rate. Danger rate represents the frequency of situations when two entities are too close, and such situation is determined by safety buffer, specifically when $\|p^i - p^j\| < r^i + r^j + k_\delta \times (\delta^i + \delta^j)$.

An encirclement test is deemed unsuccessful by either failing to meet the encirclement criteria or colliding with allies or obstacles. Fig. 11 illustrates the instances of successful and unsuccessful encirclements under ECBVC, where the cells in black lines represent Voronoi cells. The trajectory indicates the center of the pursuer and target, while the buffer distance around the obstacle corresponds to the radius of the pursuer. In Fig. 11(b), due to the close distance between the obstacle

and pursuer, the left pursuer is trapped in the local extremum within its Voronoi cell. This leads to an oscillation in its control, ultimately resulting in a failed encirclement attempt.

To further evaluate the generality of different encirclement algorithms, we have designated different evasion policies for the target, as outlined below.

- 1) The PFP considers the coalition threats of the pursuer team and helps the target act responsively. The corresponding control law v_1^T is

$$v_1^T = \|v_{\max}^T\| \frac{\sum_i \frac{\hat{d}_{i,T}}{\|p^T - p^i\|} (p^T - p^i)}{\left\| \sum_i \frac{\hat{d}_{i,T}}{\|p^T - p^i\|} (p^T - p^i) \right\|} \quad (20)$$

where $\hat{d}_{i,T} = d_{i,T} - (r^T + r^i)$, and the entity radius are involved to compute the potential forces.

- 2) The greedy policy (GP) drives the target away from the nearest pursuer g . The corresponding control law v_2^T is

$$v_2^T = \|v_{\max}^T\| \frac{p^T - p^g}{\|p^T - p^g\|} \quad (21)$$

- 3) The Apollonius policy (AP) [42] helps the target stay within its dominance region and maximizes the capture time for pursuers. The corresponding control law v_3^T is

$$v_3^T = \|v_{\max}^T\| \frac{p^A - p^T}{\|p^A - p^T\|} \quad (22)$$

where p^A is the farthest boundary point of the closest Apollonius circles among all target-pursuer pairs.

- 4) The move-to-centroid policy (MCP) [25] keeps the target away from neighboring pursuers by moving toward its Voronoi centroid. The corresponding control law v_4^T is

$$v_4^T = \|v_{\max}^T\| \frac{C_{V_T^{V^v}} - p^T}{\|C_{V_T^{V^v}} - p^T\|} \quad (23)$$

where $C_{V_T^{V^v}}$ is the center of target's Voronoi cell.

We performed a Monte Carlo simulation, running each algorithm 100 times, with the results presented in Table IV. The first column represents different escape policies, while the second corresponds to different encirclement algorithms. As indicated in Table IV, MADDPG and MATD3 combined with MRA-RLEC have the best performance in encirclement time, owing to their capability in finding globally optimal solutions. For RL-based methods, the faster pursuers form encirclement formation, the greater terminal rewards they accumulate, resulting in a smaller encircle time. In terms of danger rate and successful rate, RL-based methods generally have better performance than traditional ones. Because in traditional methods, the history information is merely considered, and the control laws greatly rely on the positions of obstacles, which could be less effective in a complex multiobstacle environment. As a result, much time is spent on avoiding collisions, leading to a poor performance in encirclement.

Another comparative dimension is the computational complexity. With our bidirectional protocol and a fixed network structure, each pursuer only processes information from two neighbors and the most threatening obstacle. Hence, the forward inference complexity remains constant from the view

TABLE IV
 ENCIRCLEMENT INDICES WITH DIFFERENT METHODS
 UNDER VARIOUS ESCAPE POLICIES

Escape Policy	Encircle Method	Step Reward	Encircle Time (s)	Danger Rate (%)	Success Rate (%)
PFP	rMAPPO	21.44	9.79±0.43	7	100
	MADDPG	19.57	10.61±0.98	16	99
	MATD3	21.28	9.65±0.66	12	100
	ECBVC	8.73	13.07±1.23	42	28
	PFEC	9.04	13.58±2.21	79	41
	FBEC	10.69	14.33±1.51	60	47
GP	rMAPPO	24.48	8.89±0.11	11	100
	MADDPG	23.65	9.21±0.59	15	100
	MATD3	22.97	8.74±0.32	9	100
	ECBVC	7.29	12.38±1.12	49	26
	PFEC	13.25	12.20±2.58	58	67
	FBEC	11.99	13.51±3.97	70	32
AP	rMAPPO	21.85	9.81±0.17	12	99
	MADDPG	22.29	9.52±0.38	7	100
	MATD3	23.91	8.66±0.49	4	100
	ECBVC	8.83	13.21±1.96	45	43
	PFEC	9.82	13.54±3.08	79	52
	FBEC	8.80	14.57±1.86	62	42
MCP	rMAPPO	25.43	8.13±0.11	8	100
	MADDPG	26.78	7.45±0.20	6	100
	MATD3	26.39	7.93±0.09	5	100
	ECBVC	7.87	11.58±1.15	39	37
	PFEC	10.77	11.96±2.45	58	74
	FBEC	13.79	13.23±1.68	34	72

of information. In contrast, for other methods that require fully connected communication, each pursuer has to integrate information from all other entities, resulting in a computational complexity of $\mathcal{O}(NM)$. From this perspective, our approach demonstrates a significant advantage in terms of efficiency.

E. Performance Analysis

In addition, we have conducted experiments to validate the performance of our algorithm under different levels of task complexity, and analyzed the effectiveness of balancing between encircle a mobile target and avoid obstacles. The task complexity is determined by the maximum target speed, and we select a series of typical velocities to perform the Monte Carlo test, with each parameter conducting 1000 trials. Denoted by S and R, the success rate and the average step reward are documented, as detailed in Table V.

As observed from Table V, there is a decline in performance when $v_{\max}^T$ exceeds a critical threshold. This decrement is attributed to the increased difficulty for pursuers to catch up with a faster-escaping target. Consequently, the model prioritizes maximizing the encirclement reward, potentially at the expense of obstacle avoidance, to obtain a higher overall return. Such behavior reflects the intrinsic tradeoffs in RL-based methodologies, where the learned policy adapts to maximize overall task performance. Therefore, the algorithm achieves its best effectiveness when the task is simple, and could give poor performance under highly complicated conditions. In extreme scenarios, particularly when $v_{\max}^T$ is nearly equal to $v_{\max}$, pursuers are unable to catch up with the target, making encircle it even more unattainable.

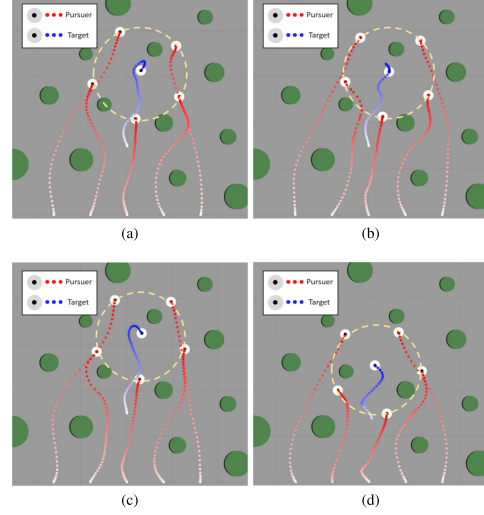


Fig. 12. Encirclement trajectories under different escape policies running in ROS2, trained via MRA-RLEC framework and rMAPPO. (a) Escape with PFP. (b) Escape with GP. (c) Escape with AP. (d) Escape with MCP.

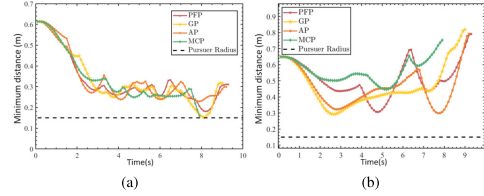


Fig. 13. Distance to evaluate collisions when encircling the target. (a) Minimum distance between pursuers and obstacles $D_o(t)$. (b) Minimum distance among all pursuers $D_p(t)$.

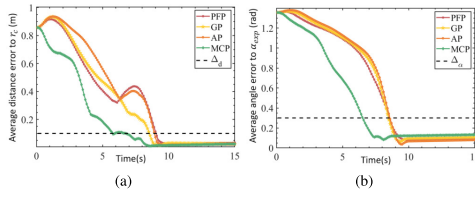
F. Experiments Under Various Escape Policies in ROS2

For future real-world vehicle deployment, we conducted additional experiments in Gazebo with ROS2 managing vehicle control and communication. The model, trained using MRA-RLEC combined with rMAPPO, generates accelerations from the actor network. The wheel velocities are further calculated through kinematics and the velocity commands are published via the ROS2 system. Vehicles communicate by publishing and subscribing to states through a shared message pool. We performed a five-pursuer encirclement test with varied target escape policies, as depicted in Fig. 12, where trajectory lines in red and blue indicate the sequence of movements over time.

In addition, visualizing the minimum distances helps evaluate collision avoidance effectiveness. D_o is defined as the minimum distance from any pursuer to an obstacle, calculated

TABLE V
 PERFORMANCE UNDER DIFFERENT ENCIRCLEMENT COMPLEXITY

Method	$v_{\max}^T = 0.10$ m/s		$v_{\max}^T = 0.20$ m/s		$v_{\max}^T = 0.25$ m/s		$v_{\max}^T = 0.27$ m/s		$v_{\max}^T = 0.30$ m/s	
	S (%)	R	S (%)	R	S (%)	R	S (%)	R	S (%)	R
MADDPG	99.7	36.74	99.2	23.53	99.1	20.24	59.8	9.59	31.4	3.87
MATD3	99.8	35.97	99.5	25.08	99.3	21.17	63.2	11.82	29.8	3.07
RMAPP	99.8	39.48	99.3	25.76	98.7	20.65	65.7	13.37	34.2	4.16


 Fig. 14. Distance error and angle error under four escape policies over time. (a) Distance error e_d , calculated by average distance minus capture radius. (b) Angle error e_α , calculated by average included angle minus expected encirclement angle.

as $D_o = \min_{i \in \mathcal{I}, j \in \mathcal{O}} (d_{i,j} - r_o^j)$, where r_o^j is the radius of the obstacle j . Fig. 13(a) illustrates that D_o is always above the pursuer's radius, signifying effective obstacle avoidance. Fig. 13(b) presents the minimum distances between pursuers as $D_p = \min_{i,j \in \mathcal{I}, i \neq j} (d_{i,j} - r^j)$, with r^j as P_j 's radius, confirming no collisions occur among pursuers during encirclement.

Fig. 14 illustrates the errors associated with the encirclement condition. The average distance error of pursuers, denoted by $e_d = \frac{1}{N} \sum_{i \in \mathcal{I}} |d_{i,T} - d_c|$, gradually descends below Δ_d and converges to zero, as shown in Fig. 14(a). Similarly, the average angle error of expected encirclement formation, denoted by $e_\alpha = \frac{1}{N} \sum_{i=1}^N |\alpha_{i,j+1} - \alpha_{\text{exp}}|$, also descends below Δ_α and converges properly.

VI. CONCLUSION AND FUTURE WORK

This article presents a novel policy-guided RL framework, MRA-RLEC, to tackle the problem of mobile target encirclement in a multiobstacle environment. In MRA-RLEC, multiple regulators are devised to assist training, and the GERD method is proposed to alleviate reward sparsity. Moreover, we designed a resource-saving bidirectional protocol which has a linear communication complexity as the number of pursuers increases. We tested the scalability and robustness of MRA-RLEC with different pursuer numbers and MARL structures, and compared our method with other encirclement algorithms, demonstrating our superiority. Along with this, we also conducted experiments under various escape policies in the physical simulation engine Gazebo to demonstrate MRA-RLEC's potential for real-world application.

While our experimental outcomes are favorable within the defined settings, certain limitations still exist. Some factors present in real-world systems are not fully accounted for in our model such as communication range, and the target moves

at a relatively low speed. In future work, we intend to address these challenges by conducting real-world experiments with highly maneuverable targets that have broader sensing ranges. Besides, it is worth noting that the concept of guide policy and curriculum regulators is extendable to other multiagent tasks. This framework could potentially improve training efficiency in any multiagent system that adopts the RL approach.

REFERENCES

- [1] M. Ashish, J. Vilela, G. Nejat, and B. Benhabib, "A multirobot path-planning strategy for autonomous wilderness search and rescue," *IEEE Trans. Cybern.*, vol. 45, no. 9, pp. 1784–1797, Sep. 2014.
- [2] Z. Sui, Z. Pu, J. Yi, and S. Wu, "Formation control with collision avoidance through deep reinforcement learning using model-guided demonstration," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 6, pp. 2358–2372, Jun. 2021.
- [3] R. Zhang, Q. Zong, X. Zhang, L. Dou, and B. Tian, "Game of drones: Multi-UAV pursuit-evasion game with online motion planning by deep reinforcement learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 10, pp. 7900–7909, Sep. 2022.
- [4] A. T. Hafez, M. Iskandarani, S. N. Givigi, S. Yousefi, and A. Beaulieu, "UAVs in formation and dynamic encirclement via model predictive control," *IFAC Proc. Volumes*, vol. 47, no. 3, pp. 1241–1246, 2014.
- [5] X. Qu, W. Gan, D. Song, and L. Zhou, "Pursuit-evasion game strategy of USV based on deep reinforcement learning in complex multi-obstacle environment," *Ocean Eng.*, vol. 273, Apr. 2023, Art. no. 114016.
- [6] A. Sarwal, D. Agrawal, and S. Chaudhary, "Surveillance in an open environment by co-operative tracking amongst sensor enabled robots," in *Proc. Int. Conf. Inf. Acquisition*, Jul. 2007, pp. 345–349.
- [7] G. Antonelli, F. Arrichiello, and S. Chiaverini, "The Entrapment/Escorting mission," *IEEE Robot. Autom. Mag.*, vol. 15, no. 1, pp. 22–29, Mar. 2008.
- [8] A. Khan, B. Rinner, and A. Cavallaro, "Cooperative robots to observe moving targets: Review," *IEEE Trans. Cybern.*, vol. 48, no. 1, pp. 187–198, Jan. 2018.
- [9] F. Chen, W. Ren, and Y. Cao, "Surrounding control in cooperative agent networks," *Syst. Control Lett.*, vol. 59, no. 11, pp. 704–712, Nov. 2010.
- [10] Z. Sun, H. Sun, P. Li, and J. Zou, "Cooperative strategy for pursuit-evasion problem with collision avoidance," *Ocean Eng.*, vol. 266, Dec. 2022, Art. no. 112742.
- [11] N. Huang, J. Zhuge, Z. Sun, D. Huang, Y. Kong, and Q. Lu, "Dynamic encirclement for anonymous agents with arbitrary formations," *IEEE Trans. Control Netw. Syst.*, vol. 11, no. 3, pp. 1644–1654, Sep. 2024.
- [12] B. Xu, H.-T. Zhang, Y. Ding, and W. Ren, "Event-triggered surrounding formation control of multiagent systems for multiple dynamic targets," *IEEE Trans. Control Netw. Syst.*, vol. 10, no. 2, pp. 752–764, Jun. 2023.
- [13] F. Liu, S. Yuan, W. Meng, R. Su, and L. Xie, "Multiple noncooperative targets encirclement by relative distance-based positioning and neural antisynchronization control," *IEEE Trans. Ind. Electron.*, vol. 71, no. 2, pp. 1675–1685, Feb. 2024.
- [14] A. T. Hafez, A. J. Marasco, S. N. Givigi, M. Iskandarani, S. Yousefi, and C. A. Rabbath, "Solving multi-UAV dynamic encirclement via model predictive control," *IEEE Trans. Control Syst. Technol.*, vol. 23, no. 6, pp. 2251–2265, Nov. 2015.
- [15] J. Liao, C. Liu, and H. H. T. Liu, "Model predictive control for cooperative hunting in obstacle rich and dynamic environments," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2021, pp. 5089–5095.
- [16] T. T. Nguyen, N. D. Nguyen, and S. Nahavandi, "Deep reinforcement learning for multiagent systems: A review of challenges, solutions, and applications," *IEEE Trans. Cybern.*, vol. 50, no. 9, pp. 3826–3839, Sep. 2020.

- [17] J. Ma, H. Lu, J. Xiao, Z. Zeng, and Z. Zheng, "Multi-robot target encirclement control with collision avoidance via deep reinforcement learning," *J. Intell. Robotic Syst.*, vol. 99, no. 2, pp. 371–386, Aug. 2020.
- [18] T. Zhang, Z. Liu, Z. Pu, and J. Yi, "Multi-target encirclement with collision avoidance via deep reinforcement learning using relational graphs," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, May 2022, pp. 8794–8800.
- [19] T. Zhang, Z. Liu, S. Wu, Z. Pu, and J. Yi, "Multi-robot cooperative target encirclement through learning distributed transferable policy," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2020, pp. 1–8.
- [20] Y. Wang, L. Dong, and C. Sun, "Cooperative control for multi-player pursuit-evasion games with reinforcement learning," *Neurocomputing*, vol. 412, pp. 101–114, Oct. 2020.
- [21] C. de Souza, R. Newbury, A. Cosgun, P. Castillo, B. Vidolov, and D. Kulic, "Decentralized multi-agent pursuit using deep reinforcement learning," *IEEE Robot. Autom. Lett.*, vol. 6, no. 3, pp. 4552–4559, Jul. 2021.
- [22] W. M. Spears, D. F. Spears, J. C. Hamann, and R. Heil, "Distributed, physics-based control of swarms of vehicles," *Auton. Robots*, vol. 17, nos. 2–3, pp. 137–162, 2004.
- [23] F. Liu, S. Yuan, K. Cao, W. Meng, and L. Xie, "Distance-based multiple noncooperative ground target encirclement for complex environments," *IEEE Trans. Control Syst. Technol.*, vol. 33, no. 1, pp. 261–273, Jan. 2025.
- [24] B. Tian, P. Li, H. Lu, Q. Zong, and L. He, "Distributed pursuit of an evader with collision and obstacle avoidance," *IEEE Trans. Cybern.*, vol. 52, no. 12, pp. 13512–13520, Dec. 2021.
- [25] X. Wang, L. Xi, Y. Ding, and B. M. Chen, "Distributed encirclement and capture of multiple pursuers with collision avoidance," *IEEE Trans. Ind. Electron.*, vol. 71, no. 7, pp. 7520–7530, Jul. 2024.
- [26] A. Richards and J. P. How, "Aircraft trajectory planning with collision avoidance using mixed integer linear programming," in *Proc. Amer. Control Conf.*, vol. 3, May 2002, pp. 1936–1941.
- [27] Y. F. Chen, M. Liu, M. Everett, and J. P. How, "Decentralized non-communicating multiagent collision avoidance with deep reinforcement learning," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2017, pp. 285–292.
- [28] Z. Pu, T. Zhang, X. Ai, T. Qiu, and J. Yi, "A deep reinforcement learning approach combined with model-based paradigms for multiagent formation control with collision avoidance," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 53, no. 7, pp. 4189–4204, Jul. 2023.
- [29] I. Uchendu et al., "Jump-start reinforcement learning," in *Proc. 40th Int. Conf. Mach. Learn.*, Jan. 2022, pp. 34556–34583.
- [30] S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, "Robot operating system 2: Design, architecture, and uses in the wild," *Sci. Robot.*, vol. 7, no. 66, p. 6074, May 2022.
- [31] C. Yu, A. Velu, E. Vinitsky, Y. Wang, A. M. Bayen, and Y. Wu, "The surprising effectiveness of PPO in cooperative, multi-agent games," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, Jan. 2021, pp. 24611–24624.
- [32] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch, "Multi-agent actor-critic for mixed cooperative-competitive environments," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, Jan. 2017, pp. 1–12.
- [33] X. Wang, Y. Chen, and W. Zhu, "A survey on curriculum learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 4555–4576, Sep. 2022.
- [34] S. Narvekar, B. Peng, M. Leonetti, J. Sinapov, M. E. Taylor, and P. Stone, "Curriculum learning for reinforcement learning domains: A framework and survey," *J. Mach. Learn. Res.*, vol. 21, no. 1, pp. 1–50, Jul. 2020.
- [35] R. Portelas, C. Colas, L. Weng, K. Hofmann, and P.-Y. Oudeyer, "Automatic curriculum learning for deep RL: A short survey," 2020, *arXiv:2003.04664*.
- [36] G. Wang, M. Xin, W. Wu, Z. Liu, and H. Wang, "Learning of long-horizon sparse-reward robotic manipulator tasks with base controllers," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 3, pp. 4072–4081, Mar. 2022.
- [37] X. Fang, C. Wang, L. Xie, and J. Chen, "Cooperative pursuit with multi-pursuer and one faster free-moving evader," *IEEE Trans. Cybern.*, vol. 52, no. 3, pp. 1405–1414, Mar. 2020.
- [38] C. Schroeder de Witt et al., "Is independent learning all you need in the StarCraft multi-agent challenge?" 2020, *arXiv:2011.09533*.
- [39] M. Wen et al., "Multi-agent reinforcement learning is a sequence modeling problem," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, Jan. 2022, pp. 16509–16521.
- [40] Y. Zhong, J. G. Kuba, S. Hu, J. Ji, and Y. Yang, "Heterogeneous-agent reinforcement learning," *J. Mach. Learn. Res.*, vol. 25, pp. 1–67, Jan. 2023.
- [41] J. Ackermann, V. Gabler, T. Osa, and M. Sugiyama, "Reducing overestimation bias in multi-agent domains using double centralized critics," 2019, *arXiv:1910.01465*.
- [42] A. Von Moll, D. W. Casbeer, E. Garcia, and D. Milutinovic, "Pursuit-evasion of an evader by multiple pursuers," in *Proc. Int. Conf. Unmanned Aircr. Syst. (ICUAS)*, Jun. 2018, pp. 133–142.



Fandi Gou received the B.Eng. degree in automation from Shanghai Jiao Tong University, Shanghai, China, in 2023, where he is currently pursuing the Ph.D. degree in control science and engineering with the Department of Automation.

He is the first Chair-Elect of the Chinese Association of Automation Student Chapter with Shanghai Jiao Tong University. His research interests include multiagent reinforcement learning, encirclement control, and cooperative coverage.



Haikuo Du received the B.Eng. degree in automation from Shanghai Jiao Tong University, Shanghai, China, in 2023, where he is currently pursuing the master's degree in control science and engineering with the Department of Automation.

His current research interests include path planning, obstacle avoidance control, and safe reinforcement learning.



Chenyu Zhao is currently pursuing the B.Eng. degree with the Department of Electronic Engineering, Shanghai Jiao Tong University, Shanghai, China.

Her current research interests include multiagent cooperation and reinforcement learning.



Yunze Cai received the B.S. degree from the Department of Applied Mathematics, Northwestern Polytechnical University, Xi'an, China, in 1997, the M.S. degree from the Department of Automation, Northwestern Polytechnical University, in 2000, and the Ph.D. degree from the Department of Automation, Shanghai Jiao Tong University, Shanghai, China, in 2003.

She is currently a Professor with the Department of Automation, Shanghai Jiao Tong University, and a Visiting Researcher at the National Key Laboratory of Air-based Information Perception and Fusion. She has been engaged in the research of information processing of complex systems, and the application of intelligent decision-making and control in multi-agent systems.

Prof. Cai is a member of the Chinese Association of Automation, the Information Fusion branch of the Chinese Society of Aeronautics and Astronautics, and the Chinese Artificial Intelligence Association.

毕业论文（设计）文献综述和开题报告考核

对文献综述、外文翻译和开题报告评语及成绩评定：

成绩比例	文献综述 占（10%）	开题报告 占（15%）	外文翻译 占（5%）
分值			

开题报告答辩小组负责人（签名）_____

年 月 日